Class 9

Mathematics

Volume 2

Solutions

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8. Binomial Theorem

WORKSHEET – 1
Level – I

1) B 2) B 3) B 4) D 5) C 6) A 7) D
8) D 9) B 10) C

1. [1] Sol: $(x+a)^5 = x^5 + 5x^4a + 10x^3a^2 + 10x^2a^3 + 5xa^4 + a^5$

   $\therefore$ The sum of the coefficients $= 1 + 5 + 10 + 10 + 5 + 1 = 32$

2. [2] $T_{r+1} = {}^nC_r x^{n-r} y^r$ in $(x+y)^n$

3. [4] Sol: The middle term in the expansion $\left(x - \frac{1}{2x}\right)^{10}$

   $= T_{\frac{10}{2}+1} = T_6 = {}^{10}C_5 x^{10-5}\left(\frac{-1}{2x}\right)^5 = -{}^{10}C_5 \frac{1}{2^5} = \frac{-63}{8}$

4. [1] Hint : use $r = \frac{np}{p+q}, n = 9, p = 2, q = 1$

5. [4] Sol: In the expansion of $\left(y + \frac{c^3}{y^2}\right)^{10}$, the rth term is given by

   $T_r = {}^{10}C_{r-1} y^{10-r+1}\left(\frac{c^3}{y^2}\right)^{r-1} = {}^{10}C_{r-1} y^{11-r}\frac{c^{3r-3}}{y^{2r-2}} = {}^{10}C_{r-1} y^{11-r-2r+2} c^{3r-3}$

   Thus, $T_r = {}^{10}C_{r-1} y^{13-3r} c^{3r-3}$

   Let the rth term contains y^{-2} $\therefore 13 - 3r = -2 \Rightarrow r = \frac{15}{3} = 5.$

   Therefore, 5th term will contain y^{-2}

   From (1), $T_5 = {}^{10}C_{5-1} c^{15-3} y^{-2} = {}^{10}C_4 c^{12} y^{-2}$

   $\therefore$ Coefficient of $y^{-2} = {}^{10}C_4 c^{12} = \frac{10! c^{12}}{4!(10-4)!} = \frac{10! c^{12}}{4!6!} = \frac{10\times9\times8\times7}{4\times3\times2\times1} \times c^{12} = 210c^{12}$

6. (1) $(3n - (n+1) + 2)^{th}$ term from beginning i.e $(2n+1)^{th}$ term from beginning

7. [21] In$(3 + 7x)^{29}$, $T_{r+1} = {}^{29}C_r 3^{29-r}(7x)^r$

   Coefficient of T_{r+1} = coefficient of T_r

   ${}^{29}C_r 3^{29-r} \cdot 7^r = {}^{29}C_{r-1} 3^{29-(r-1)} \cdot 7^{r-1}$

   $$\frac{29!}{(29-r)!\,r!} \cdot 3^{29-r} \cdot 7^r = \frac{29!}{(30-r)!\,(r-1)!} \cdot 3^{30-r} \cdot 7^{r-1}$$

   $$\frac{(30-r)(29-r)!\,(r-1)!}{(29-r)!\,r(r-1)!} = \frac{3}{7}$$

   $$\frac{30-r}{r} = \frac{3}{7} \Rightarrow 210 - 7r = 3r \Rightarrow r = 21$$

8. [3] ${}^{18}C_{2r+3} = {}^{18}C_{r-3} \Rightarrow (15-2r)!\,(2r+3)! = (21-r)!\,(r-3)!$

9. [4] $= \frac{T_6}{T_6} = \frac{(-1)}{(-2)^5} = \frac{1}{32}$

10. [3] Sol: We know that $(x-1)^6 = x^6 - 6x^5 + 15x^4 - 20x^3 + 15x^2 - 6x + 1$

$$(x+1)^6 = x^6 + 6x^5 + 15x^4 + 20x^3 + 15x^2 + 6x + 1$$

$$(x-1)^6 + (x+1)^6 = 2[x^6 + 15x^4 + 15x^2 + 1] = 2x^6 + 30x^4 + 30x^2 + 2$$

11. [5] $T_{17} = T_{18}$ simplify

12. [3] $T_5 = 1120 \Rightarrow p \cdot q = 4 \therefore p + \frac{4}{P} = 5 \Rightarrow p = 1$ or 4

13. [3] $T_1 = 1, T_2 = 8x, T_3 = 24x^2 \Rightarrow na = 8; \frac{n(n-1)}{2}a^2 = 24 \ \therefore a = 4$

14. [1]; Given ${}^{10}C_{2r} = {}^{10}C_{4r+4} \Rightarrow (10-2r)! + (6-4r)!\,(4r+4)! \Rightarrow r = 1$ (Verification)

15. [3] $T_3 = 10^6 \Rightarrow 5_{C_2} x^3 \left(x^{\log_{10}^x}\right)^2 = 10^6$, x = 10

16. [3] $n^2 - (4r+1)n + 4r^2 - 2 = 0$ where $r = 5$ (or)

$2({}^nC_5) = {}^nC_4 + {}^nC_6 \Rightarrow \frac{2n-13}{5(n-4)(n-5)} = \frac{1}{30}$; Verify for n = 7 or 14

17. [2] Coefficient of $x^{17} = (1 + 2 + 1 \ldots\ldots + 18) = -171$.

18. [2] Expand $(x + ai)^n$ and $(x - ai)^n$ then multiply

19. [4] $(1-x)(1-x^4)^4$

20. [2] ${}^nC_{r-1}$, ${}^nC_r$, ${}^nC_{r+1}$ are in A.P $\Rightarrow 2 = \frac{{}^nC_{r-1}}{{}^nC_r} + \frac{{}^nC_{r+1}}{{}^nC_r} \Rightarrow 2 = \frac{r}{n-r+1} + \frac{n-r}{r+1}$

21. [2,3,4]

22. [1]

23. [2]

24. [1]

25. [3] Sol: $(m-n)^7$

The total number of terms in the expansion = 8

The 1st term = $m^{7;}$ The 2nd term = $-7\ m^6n$

The 3rd term = $\frac{7\times6}{2}m^5n^2 = 21m^5n^2$ The 4th term = $-\frac{21\times5}{3}m^4n^3 = -35m^4n^3$

26. a-r; b-p; c-s; d-q

a) Since $n = 10$ is an even number. Then the given expansion has only one middle

term. i.e., $\frac{T_n}{2} + 1 = T_{5+1} = T_6; T_6 = 10_{c5}\left(\frac{3x}{7}\right)^{10-5} \cdot (-2y)^5 = -10_{c5}\left(\frac{6}{7}\right)^5 \cdot x^5 \cdot y^5$

b) Since n = 8 is an even number. Then they given expansion has only one middle term. i.e., $T_5 = {}^8C_4(7p)^4.(2q)^4$

c) Since n = 17 is an odd number. Then the expansion has two middle terms i.e., T_9& T_{10}

$T_9 = {}^{17}C_8(4x^2)^{17-8} \cdot (5x^3)^8 = {}^{17}C_8 4^9 \cdot 5^8 \cdot x^{42}; T_{10} = {}^{17}C_9(4x^2)^{17-9} \cdot (5x^3)^9$

$= {}^{17}C_9 4^8 \cdot 5^9 \cdot x^{43}$

d) Since $n = 20$ is even number. Then the expansion has only one middle term

i.e., $T_{11} = {}^{20}C_{10}\left(\frac{3}{a^3}\right)^{20-10} \cdot (5a^4)10 = {}^{20}C_{10}(15)^{10} \cdot a^{10}$

27. Ans : 5 ; Since $n = 9$ is odd the expansion of $(1 + 5\sqrt{2}x)^9 + (1 - 5\sqrt{2}x)^9$ has $\left(\frac{n+1}{2}\right)$

i.e., $\frac{9+1}{2} = 5$ terms

28. [2] Sol: $(x + a)^n = x^n + {}^nC_1x^{n-1}a + {}^nC_2x^{n-2} + {}^nC_3x^{n-3}a^3 + \cdots . + {}^nC_na^n =$ $(x^n + {}^nC_2x^{n-2}a^2 + \cdots . + ({}^nC_1x^{n-1}a + {}^nC_3x^{n-3}a^3 + \cdots .) = P + Q$ and $(x - a)^n = x^n -$ ${}^nC_1x^{n-1}a + {}^nC_2x^{n-2}a^2 - {}^nC_3x^{n-3}a^3 + \cdots \ldots . + (-1)^n\,{}^nC_na^n = (x^n + {}^nC_2x^{n-2}a^2 +$ $\cdots \ldots)({}^nC_1x^{n-1}a + {}^nC_3x^{n-3}a^3 + \cdots \ldots) = P - Q.\ P^2 - Q^2 = (P + Q)(P - Q) = (x +$ $a)^n(x - a)^n = (x^2 - a^2)^n$

29. [1] Sol: $4PQ = (P + Q)^2 - (P - Q)^2 = (x + a)^{2n} - (x - a)^{2n}$

30. [2] Sol: $p^2 + q^2 = (p + q)^2 - 2pq$

$$p^2 + q^2 = (x + a)^{2n} - \left(\frac{(x + a)^{2n} - (x - a)^{2n}}{2}\right) = \frac{(x + a)^{2n} + (x - a)^{2n}}{2}$$

31. [3,4] Given

$$T_5 = 24\ T_3$$
$${}^{11}C_4x^4 = 24\ {}^{11}C_2x^2$$
$$\frac{11!}{(11-4)!\,4!}x^2 = 24\frac{11!}{(11-2)!\,2!}$$
$$x^2 = 4$$
$$x = \pm 2$$

32. [1,2,3,4] Sum of coefficients in $(a\alpha^2x^2 + 2ba\alpha x + c)^n$ is $(a\alpha^2 + 2ba\alpha + c)^n$.

Let $f(\alpha) = a\alpha^2 + 2ba\alpha + c$

Such that sum $= [f(\alpha)]^n$. Now, $f(\alpha) > 0 \forall n$, and $\alpha \in R$ then $a > 0$.

Also, $f(\alpha) < 0 \forall n \in$ odd and $\alpha \in R$ if $a < 0$. Similarly, options (b) and (d) are also true.

33. [1] Sol: $T_2 = 135,\ T_3 = 30,\ T_4 = 10/3$

${}^nC_1 \cdot a^{n-1} \cdot b^1 = 135 \ldots \ldots (1)\ {}^nC_2 \cdot a^{n-2} \cdot b^2 = 30 \ldots \ldots (2);\ {}^nC_3 \cdot a^{n-2} \cdot b^3 = 10/3 \ldots \ldots \cdot (3)$

$$\Rightarrow \frac{{}^nC_2 \cdot a^{n-2}\ b^2}{{}^nC_1a^{n-1}\ b} = \frac{30}{135};\ \frac{(n-1)(a^{-1}) \cdot b}{2} = \frac{2}{9}; 9(n-1)b = 4a \rightarrow (4) \Rightarrow \frac{{}^nC_3a^{n-3}\ b^3}{{}^nC_2a^{n-2}\ b^2} = \frac{\frac{10}{3}}{30}$$

$$(n-2)(a^{-1})b = \frac{1}{3}; 3\ b(n-2) = a \rightarrow (5) \Rightarrow \frac{(4)}{(5)} \Rightarrow 3(n-1) = 4(n-2)$$

$$4n - 3n - 8 + 3 = 0 \Rightarrow n = 5$$

34. [2] $T_{21} = {}^{44}C_{20} \cdot x^{20}; T_{22} = {}^{44}C_{21} \cdot x^{21}$. Given that $T_{21} = T_{22} \Rightarrow {}^{44}C_{20}x^{20} = {}^{44}C_{21}x^{21}$
$\Rightarrow \frac{44!}{(24!)20!} = \frac{44!}{(23)!(21)!} \cdot x \Rightarrow \frac{21}{24} = x \Rightarrow x = 7/8$

35. [2] Sol: The rth term in the given expansion is $\left(\frac{n+1}{2}\right) = \frac{50}{2} = 25$
Thus the coefficient of rth term $= 18_{c_{r-1}}$
Given that the coefficient of $(2r+4)^{th}$ term = the coefficient of $(r-2)^{nd}$ term.
i.e., $18_{c_{2r+3}} = 18_{c_{r-3}} \Rightarrow 2r+3 = r-3 \Rightarrow r = 6$ or -6 Since r is + ve, $\therefore r = 6$

WORKSHEET – 2
Level – I

1) B	2) D	3) B	4) D	5) B	6) C
7) B	8) B	9) A			

1. [3] Sol: We have, $(2x+y+3z)^{10} = \sum_{r_1+r_2+r_3=10} \frac{10!}{r_1!r_2!r_3!}(2x)^{r_1}(y)^{r_2}(3z)^{r_3}$
The general term in the above expansion is
$\frac{10!}{r_1!\,r_2!\,r_3!}(2x)^{r_1}(y)^{r_2}(3z)^{r_3} = \frac{10!}{r_1!\,r_2!\,r_3!}2^{r_1}3^{r_3}x^{r_1}y^{r_2}z^{r_3}$
For the coefficient of $x^2y^5z^3$, we must have $r_1 = 2, r_2 = 5$ and $r_3 = 3$.
So, coefficient of $x^2y^5z^3 = \frac{10!}{2!5!3!}2^2 \times 3^3$

2. [1] $\frac{f(1)+f(-1)}{2}$

3. [1] Sol: In $(a+b+c)^{18}$, general term is given by $\frac{18!}{(r_1)!(r_2)!(r_3)!}(a)^{r_1}(b)^{r_2}(c)^{r_3}$
Coefficient of $a^8\,b^6c^4$ is obtained by putting $r_1 = 8, r_2 = 6, r_3 = 4$ in (1)
i.e., $\frac{(18)!}{8!\cdot6!\cdot4!}a^8\,b^6c^4$. Thus required coefficient $= \frac{(18)!}{8!.6!\cdot4!} \times \frac{14!}{14!} = {}^{18}C_{14} \times {}^{14}C_8$

4. [3] Use the formula $\frac{(n+1)|x|}{1+|x|}$

5. [3] Use the formula $\frac{(n+1)|x|}{1+|x|}$

6. [1] Sol: The general term in the expansion of $(1+x^2-x^3)^8$ is
$\frac{8!}{r!s!t!}(1)^r(x^2)^s(-x^3)^t = \frac{8!}{r!s!t!}(-1)^t x^{2s+3t}$, where $r+s+t = 8$.
For the coefficient of x^6, we must have $2s+3t = 6$
Now, $r+s+t = 8$ and $2s+3t = 6 \Rightarrow r = \frac{10+t}{2}$, $s = \frac{6-3t}{2}$, where $0 \le t \le 8$.
$\therefore t = 0 \Rightarrow r = 5, s = 3$ and $t = 2 \Rightarrow r = 6, s = 0 \therefore$ Coefficient of x^6
$= \frac{8!}{5!\,3!\,0!}(-1)^0 + \frac{8!}{6!\,0!\,2!}(-1)^2 = \frac{8!}{5!\,3!} + \frac{8!}{6!\,2!} = 56 + 28 = 84$

7. [1] Sol: $(2-3x)^7 = 2^7\left(1-\frac{3}{2}x\right)^7$, where $n = 7$ and $|\alpha| = \frac{3}{2}\cdot(1) = \frac{3}{2}$

   Now, $\frac{(n+1)|\alpha|}{|\alpha|+1} = \frac{24}{5} = 4+\frac{4}{5}$ (which is not an integer)

   $\therefore$ The greatest term in the expansion of $(2-3x)^7$ is $T_5 = {}^7C_4(2)^{7-4}(3x)^4 = 22680$

8. [7] $(n+1)(n+2) = 72, n = 7.$

9. [1] Sol: The general term in the expansion of $(2-x+3x^2)^6$ is

   $\frac{6!}{r!s!t!}2^r(-x)^s(3x^2)^t$, where $r+s+t = 6 = \frac{6!}{r!s!t!}2^r\times(-1)^s\times 3^t\times x^{s+2t}$,

   where $r+s+t=6$. For the coefficient of x^5, we must have $s+2t=5$

   But, $r+s+t=6$ $\therefore s = 5-2t$ and $r = 1+t$, where $0\le r,s,t\le 6$

   Now, $t=0\Rightarrow r=1, s=5; t=1\Rightarrow r=2, s=3; t=2\Rightarrow r=3, s=1$

   Thus, there are three terms containing x^5 and coefficient of x^5

   $$= \frac{6!}{1!\,5!\,0!}\times 2^1\times(-1)^5\times 3^0 + \frac{6!}{2!\,3!\,1!}\times 2^2\times(-1)^3\times 3^1 + \frac{6!}{3!\,1!\,2!}\times 2^3\times(-1)^1\times 3^2$$
   $$= -12-720-4320 = -5052$$

10. [4] $\frac{f(1)-f(-1)}{2}$ 11. [4] Put $x=1; x=-1$, and then add

11. [4] Use the formula $\frac{(n+1)|x|}{1+|x|}$

12. [3] Put $x = 1, x = \omega, x = \omega^2$ and add those then it is $\frac{f(1)}{3} = 3^{n-1}$

13. [4] let x =

    let x =

$$\log_e 10 = \sum_{r=0}^{n}(-1)^r\cdot {}^nC_r\frac{(1+rx)}{(1+nx)^r} = \sum_{r=0}^{n}(-1)^r\cdot {}^nC_r\frac{1}{(1+nx)^r} + \sum_{r=0}^{n}(-1)^r\cdot\frac{n}{r}\cdot {}^{n-1}C_{r-1}\frac{rx}{(1+nx)^r}$$

    Simplify ẩove

15. [4] $(2+\sqrt{3})^n = [x]+f.\,\text{Let}(2-\sqrt{3})^n = f^1, 0<f^1<1$

    $[x]+f+f^1 = 2[{}^nC_0 2^n + {}^nC_2 2^{n-2}3+\cdots]$ an even integer

    $\therefore f+f^1 = 1$ $\therefore [x]$ is an odd integer, $f^1 = 1-f$

    But $([x]+f)f^1 = (2+\sqrt{3})^n(2-\sqrt{3})^n = 1$

    $\therefore [x] = \frac{1}{f^1}-f = \frac{1}{1-f}-f = \frac{1-f+f^2}{1-f}$ $\therefore \frac{f^2}{1-f} = [x]-1$, an even integer

    The number of terms in the above expansion is equal to the number of non-

    18 .

16. [7] use multinomial theorem.

    General term of given expansion is $\frac{6!}{a!b!c!}1^a 2^{\frac{b}{2}}3^{\frac{c}{3}}$ where $a+b+c=6$ and

a	b	c
6	0	0
4	2	0
2	4	0
0	6	0
3	0	3
1	2	3
0	0	6

17. [1] $\frac{6!}{3!2!1!} = 60$
18. [3] Coefficient of $\boldsymbol{a^8b^4c^9d^9}$ in $(\boldsymbol{abc + abd + acd + bcd})^{10}$ is

    coefficient of $\boldsymbol{a^8b^4c^9d^9}$ in $(\boldsymbol{abcd})^{10}\left(\frac{1}{\boldsymbol{a}}+\frac{1}{\boldsymbol{b}}+\frac{1}{\boldsymbol{c}}+\frac{1}{\boldsymbol{d}}\right)^{10}$

    $\Rightarrow$ coefficient of $a^{-2}b^{-6}c^{-1}d^{-1}$ in $\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right)^{10} = \frac{10!}{2!6!1!1!}$
19. [2] Given that $\boldsymbol{J + f} = (2+\sqrt{3})^n; 0 \le \boldsymbol{f} < 1\ \boldsymbol{F} = (2-\sqrt{3})^n, 0 < \boldsymbol{F} < 1$

    $\boldsymbol{J + f + F} = 2[\ {}^n\boldsymbol{C}_6 2^n + {}^n\boldsymbol{C}_2 2^{n-2}\cdot 3 + \cdots..] = 2 \times$ integer

    since $\boldsymbol{o < f < F < 2 \Rightarrow f + F} = 1$ i.e $\boldsymbol{F} = 1 - \boldsymbol{f}(\boldsymbol{I + f})(\boldsymbol{J - f}) = (\boldsymbol{J + f})\boldsymbol{F} = (2+\sqrt{3})^{\boldsymbol{n}}(2-\sqrt{3})^{\boldsymbol{n}} = 1$
20. [3] The sum of coefficients of terms not containing $y = 3^6$

    The sum of coefficients of terms not containing both x&y $= 2^6$

    So the required number $= 3^6 - 2^6 = 665$.
21. [1,2] Sum of coefficient of $= (1+3-5)^{2n} = (-1)^{2n} = 1(\because$ 2n is even)
22. [1] Let T_r and T_{r+1} denote the rth and (r + 1) th terms in the expansion of $(1 + x)^{10}$.

    Then, $T_r = {}^{10}C_{r-1}x^{r-1}$ and $T_{r+1} = {}^{10}C_r x^r \therefore \frac{T_{r+1}}{T_r} = \frac{{}^{10}C_r x^r}{{}^{10}C_{r-1}x^{r-1}} = \frac{{}^{10}C_r}{{}^{10}C_{r-1}}$

    $\therefore \frac{T_{r+1}}{T_r} = \frac{11-r}{r}x \Rightarrow \frac{T_{r+1}}{T_r} = \left(\frac{11-r}{r}\right)\times\frac{2}{3}[\because x = 2/3]$

    Now, $\therefore \frac{T_{r+1}}{T_r} > 1 \Rightarrow \left(\frac{11-r}{r}\right)\times\frac{2}{3} > 1 \Rightarrow 22 > 5r \Rightarrow r < 4\frac{2}{5}$

    $\therefore (4 + 1)$ th i.e 5th term is the greatest term.

    Putting r = 4 in T_{r+1}, we get $T_5 = {}^{10}C_5x^4[\because x = 2/3] \Rightarrow T_5 = 210\left(\frac{2}{3}\right)^4$

23. [2] Sol: Here, n = 15 and m = 4 ∴ q = 3 and r = 3

Hence, greatest coefficient $= \frac{n!}{(q!)^{m-r}[(q+1)!]^r} = \frac{15!}{3!(4!)^3}$

24. [4] Sol: We have, $(xy + yz + zx)^6 = \sum_{r+s+t=6} \frac{6!}{r!s!t!}(xy)^r(yz)^s(zx)^t =$

$\sum_{r+s+t=6} \frac{6!}{r!s!t!}x^{r+t}y^{r+s}z^{s+t}$

If the general term in the above expansion contains $x^3y^4z^5$, then r + t = 3, r + s = 4 and s + t = 5

Also, r + s + t = 6. Solving these equations, we get r = 1, s = 3, t = 2

∴ Coefficient of $x^3y^4z^5 = \frac{6!}{1!3!2!} = \frac{6!}{2!3!} = 60$

25. [4] Sol: General term in $(1 + a - b + c)^9 = \sum \frac{9!}{r_1!r_2!r_3!r_4!} \cdot (1)^{r_1} \cdot (a)^{r_2}(-b)^{r_3}(c)^{r_4} \Rightarrow$

Coefficient of $a^3\ b^4c = \frac{9!}{1!3!4!1!} = \frac{9!}{3!4!}$

26. a-s; b-r; c-q; d-p

a) We know that the number of terms in the expansion of $(x_1 + x_2 + \cdots .. + x_r)^n$ is The number of terms $= (n + 1) = 12 + 1 = 13$

b) The number of terms $= {}^{(n+r-1)}C_{r-1} = {}^{(5+3-1)}C_{3-1} = 21$

c) The number of terms $= {}^{(3+4-1)}C_{4-1} = 20$

d) The number of terms $= {}^{(5+4-1)}C_{4-1} = 56$

27. Ans : 3 Sol. $(3 - 2x)^5 = 3^5\left[1 - \frac{2x}{3}\right]^5$. Numerically greatest term is $\frac{(5+1)\left|\frac{-2x}{3}\right|}{\left|\frac{-2x}{3}\right|+1} = \frac{6.\frac{2}{3}}{\frac{2}{3}+1} =$

2 + (int eger) ∴ 3rd term is the numerically greatest term

28. [1,2,3,4] Differentiate w.r.t. x and put x = 0 we get $a_1 = 20$

again differentiate w.r.t. x and x = 0 we get $a_2 = 210$

replace x by 1/x and put x = 0 we get $a_{20} = 3^{10}$

again differentiate w.r.t. x and put = 0 we get $a_{19} = 20.3^9$

29. [1] Sol: Here n = 19, an odd integer.

∴ The largest binomial coefficients are 19_{c_9}&$19_{c_{10}}$.

30. [2] Here n = 24 is an even integer ∴ The largest binomial coefficient is $24_{c_{12}}$.

31. [3] Here n = 22 is an even integer. ∴ The largest binomial coefficient is $n_{c_{\left(\frac{n}{2}\right)}}$

$= 22_{c_{11}} = 22_{c_r} \Rightarrow r = 11;\ 13_{c_r} = 13_{c_{11}} = 13_{c_2} = \frac{13 \times 12}{1 \times 2} = 78$

32. [1,2,4] Sum of coefficients vanish $\Rightarrow a^2 - a\alpha + 1 = 0 \Rightarrow \alpha^2 - 4 \geq 0 (\because a \in n)$

$\Rightarrow \alpha \in (-\alpha, -2] \cup [2, \infty)$

33. [4] $a_0 + a_2 + a_4 + \cdots + a_{4030} = \frac{(1+1+1)^{2015}+(1-1+1)^{2015}}{2} = \frac{3^{2015}+1}{2}$

$= \frac{(3+1)(3^{2014} - 3^{2013} + \cdots + 3^2 - 3 + 1)}{2} = 2[3(3^{2013} - 1) + \cdots + 3(3-1) + 1]$

$= 2[3(\text{ even }) + \cdots + 3(\text{ even }) + 1] = 12k + 2^\circ 12k - 10$

34. [4] $a_1 - a_3 + a_5 - a_7 + \cdots =$ Image. part of $(1 + i + i^2)^{2015} = I_m(i^{2015}) = I_m - (i) = -1$

Again, a_1 = coefficient of x in $(1 + x + x^2)^{2015}$ = coefficient xx in $(1 + x)^{2015} = 2015$

So, $a_3 - a_5 + a_7 - \cdots = 2015 - (-1) = 2016$.

35. [1] $\sum_{r=0}^{4030} a_r x^r = (1 + x + x^2)^{2030}$.

Replacing x by $\left(-\frac{1}{x}\right)$ we get, $\sum_{r=0}^{4030}(-1)^r a_r \left(\frac{1}{x}\right)^r = \frac{(1-x+x^2)^{2015}}{x^{4030}}$.

Now, $\sum_{r=0}^{4018}(-1)^r a_r^2$ = coefficient of x^0 in $\left(\sum_{r=0}^{4030} a_r x^r\right)\left(\sum_{r=0}^{4030}(-1)^r a_r \left(\frac{1}{x}\right)^r\right)$

= coefficient of x^0 in $(1 + x + x^2)^{2015} \times \frac{(1-x+x^2)^{2015}}{x^{4030}}$

= coefficient of x^{4030} in $(1 + x^2 + x^4)^{2015}$ = coefficient of x^{4030} in $\sum_{r=0}^{4030} a_r (x^2)^r = a_{2015}$.

WORKSHEET – 3
Level – I

1) B 2) D 3) A 4) D 5) (4) Hint: $n_{C_r} = \frac{n!}{(n-r)!r!}$

6) (3) 7) (3) 8) (1) 9) (1) Sol: $C_0 + C_2 + C_4 + C_8 = 2^{9-1} = 2^8 = 256$

10. (3) $\sum_{r=1}^{n} r \cdot {}^nC_r = \sum_{r=1}^{n} r \cdot \frac{n}{r} {}^{n-1}C_{r-1} = \sum_{r=1}^{n} n \cdot {}^{n-1}C_{r-1} = n\{{}^{n-1}C_0 + {}^{n-1}C_1 + {}^{n-1}C_2 + \cdots . + {}^{n-1}C_{n-1}\} = n \cdot 2^{n-1}$

1. [4] In the expansion of $(1 + x)^{2n}$ sum of coefficients is ${}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + \cdots + {}^{2n}C_{2n} = 2^{2n}$

${}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + \cdots + {}^{2n}C_{2n} = 2^{2n-1} \therefore {}^{2n}C_2 + {}^{2n}C_4 + \cdots + {}^{2n}C_{2n} = 2^{2n-1} - 1$

2. [2] sum of coefficients ${}^nC_0 + {}^nC_1 + \cdots + {}^nC_n = 2^n$; ${}^nC_1 + {}^nC_3 + \cdots + {}^nC_{n-1} = 2^{n-1}$

3. [4] $C_0 + 2C_1 + 3C_2 + \cdots . + (n+1)C_n$

$$\sum_{r=0}^{n}(r+1)C_r = \sum_{r=0}^{n} r \cdot C_r + \sum_{r=0}^{n} C_r = n2^{n-1} + 2^n = n \cdot 2^{n-1} + 2 \cdot 2^{n-1} = (n+2)2^{n-1}$$

4. [2] $3C_1 - 4C_2 + 5C_3 + \cdots + (-1)^{n-1}(n+2)C_n$

$$\sum (-1)^{r-1}(r+2)C_r = \sum (-1)^{r-1} r\, {}^nC_r + \sum (-1)^{r-1} 2C_r$$

$= \sum(-1)^{r-1} n^{(n-1)} C_{r-1} + 2 \cdot \sum(-1)^{r-1} C_r = 0 - 2[0 - 1] = 2$

5. [2] $\frac{{}^{n}C_r}{{}^{n}C_{r-1}} = \frac{n-r+1}{r}$

6. [4] $\frac{C_0}{2} - \frac{C_1}{3} + \frac{C_2}{4} - \frac{C_3}{5} + \cdots$

$$= \frac{1}{(n+1)(n+2)}\left[\frac{(n+2)(n+1)}{2!} - \frac{2\cdot(n+2)(n+1)\cdot n}{3!} + \frac{3\cdot(n+2)(n+1)n(n-1)}{4!} - \cdots\right]$$

$$= \frac{1}{(n+1)(n+2)}\left[\ {}^{(n+2)}C_2 - {}^{2(n+2)}C_3 + {}^{3(n+2)}C_4 - \cdots.\right.$$

7. [2] Differentiate with respect x both sides and then put $x = 1$

8. [3] Standard Result

9. [4] ${}^{n}C_r + {}^{n}C_{r-1} = {}^{(n+1)}C_r$

10. [2] ${}^{(n+1)}C_0 + {}^{(n+1)}C_1 + \cdots \ldots \ldots + {}^{(n+1)}C_n + {}^{(n+1)}C_{n+1} = 2^{n+1}$

11. [4] Given expression contains $3n$ factors Choose r brackets out of $3n$ brackets to get aterm of degree r. Hence the coefficient is ${}^{3n}C_r$. Let n = 3 verify

12. [1] $(1+x)^{30} = C_0 + C_1x + C_2x^2 + \cdots . + C_{30}x^{30}$

Set x = 1, −1, i and −i then add $C_0 + C_4 + C_8 + \cdots . + C_{28} = \frac{1}{4}[2^{30} + (1+i)^{30} + (1-i)^{30}] = 2^{28}$.

13. [2] $2s = 2^{20} + {}^{20}C_{10}$

14. [2] assign n value

$$\sum\sum 1_1 = \frac{1}{2}\left[\sum_{i=1}^{n}\sum_{j=1}^{n} - \sum_{i=j} 1^2\right] = \frac{1}{2}[\sum 1 \sum 1 = \sum 1] = \frac{1}{2}[n\pi n - n]$$

$$\frac{n^2-n}{2} = \frac{n(n-1)}{2}$$

15. [1] assign n value

16. [3] all are independent

17. [2] assign n value

18. [1] $\sum_{r=0}^{n-1}\left(\frac{{}^{n}C_r}{{}^{n}C_r + {}^{n}C_{r+1}}\right)^3 = \sum_{r=0}^{n-1}\left(\frac{{}^{n}C_r}{{}^{n}C_r\left(1+\frac{n-r}{r+1}\right)}\right)^3 = \sum_{=0}^{n-i}\left(\frac{r+1}{n+1}\right)^3 = \frac{n^2}{4(n+1)} = \frac{4}{5}$ gives n = 4

19. [1] The given series is equal to $\frac{1}{2}\int((1+x)^{50} + (1-x)^{50})dx$

20. [4] $\sum_{r=0}^{20} r(20-r)({}^{20}C_r)^2 = \sum_{r=0}^{20}(20-r)\,{}^{20}C_r \times r \times {}^{20}C_r$

$$= \sum_{r=0}^{19} 20 \times {}^{19}C_r \times 20 \times {}^{19}C_{r-1} = 400\sum_{r=0}^{19} {}^{19}C_r \times {}^{19}C_{r-1} = 400 \times {}^{38}C_{20}$$

21. (1, 2) k = $\frac{(C_0+C_1)(C_1+C_2)\ldots(C_{n+1}+C_n)}{C_0C_1\ldots..C_0} = \frac{{}^{(n+1)}C_1\,{}^{(n+1)}C_2\ldots.\,{}^{(n+1)}C_n}{{}^{n}C_0\,{}^{n}C_1\ldots\,{}^{n}C_n} = \frac{(n+1)^n}{n!}$

22. (1)

23. (2) L.H.S $= 15_{c_2} + 2.15_{c_3} + \cdots . + 14.15_{c_{15}}$

$= 1 + (-1)^{15} \cdot c_0 + 0.15_{c_1} + 1 \cdot 15_{c_2} + 2 \cdot 15_{c_3} + \cdots . + 14 \cdot 15_{c_{15}} = 1 + (-1 + 14) \cdot 2^{15-1}$

$= 13 \cdot 2^{14} + 1$

24. (3)

Sol: $C_0 + C_1 + C_2 + \cdots . + C_n = 2^n = 128 \Rightarrow n = 7$. Now $C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \frac{C_3}{4} +$

$\cdots . (-1)^n \frac{C_n}{n+1} = \frac{1}{1+7} = \frac{1}{8}$

25. (1) Sol: $2 \cdot C_0 + 7C_1 + 12 \cdot C_2 + \cdots . + (5n + 2) \cdot C_n = (2a + nd) \cdot 2^{n-1} = (4 + 5n)2^{n-1}$

26. Key : a-p; b-r; c-q; d-s

a) Given $(1 + x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \cdots . + a_{2n} x^{2n}$

Put x = 1, then, $3^n = a_0 + a_1 + a_2 + \cdots . + a_{2n}$

b) Then $(1 + x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \cdots . + a_{2n} x^{2n}$

Put $x = 1$, then $3^n = a_0 + a_1 + a_2 + \cdots . + a_{2n} \rightarrow (1)$

Put x = −1, then $1 = a_0 - a_1 + a_2 \ldots . + a_{2n} \rightarrow (2)$

adding (1) and (2) equations.

$\Rightarrow 3^n + 1 = (a_0 + a_1 + a_2 + \cdots . + a_{2n}) + (a_0 - a_1 + a_2 + \cdots . + a_{2n})$

$\Rightarrow (2a_0 + 2a_2 + \cdots . + 2a_{2n}) = 3^n + 1 \Rightarrow a_0 + a_2 + \cdots . + a_{2n} = \dfrac{3^n + 1}{2}$

c) Subtracting above (1) and (2) equations

$\Rightarrow 3^n - 1 = (a_0 + a_1 + a_2 + \cdots . + a_{2n}) - (a_0 - a_1 + a_2 + \cdots . + a_{2n})$

$\Rightarrow a_1 + a_3 + \cdots . + a_{2n-1} = \dfrac{3^n - 1}{2}$

d) Given expansion $(1 + x + x^2)^n = a_0 + a_1 x + a_2 x + \cdots . + a_{2n} x^{en}$

Put x = 1, then $3^n = a_0 + a_1 + \cdots . + a_{2n} \rightarrow (1)$

Put x = ω, then, $(1 + \omega + \omega^2) = 0 = a_0 + a_1 \omega + a_2 \omega^2 + \cdots . + a_{2n} (\omega)^{2n} \rightarrow (2)$

Put x = ω^2, then, $a_0 + a_1 \omega^2 + a_2 \omega^4 + \cdots . + a_{2n} \omega^{4n} = 1 + \omega^2 + \omega^4 = 0 \rightarrow (3)$

adding (1), (2) and (3)

$3a_0 + a_1(1 + \omega^2 + \omega^4) + a_3(1 + \omega^3 + \omega^6) + \cdots + a_{2n}(1 + \omega^{2n} + \omega^{4n}) = 3^n$

27. Ans : 0 Sol. sum $= \sum(-1)^n \cdot (n + 2) \cdot C_n = \sum(-1)^n \cdot {}^nC_n + 2\sum(-1)^n \cdot C_n = 0 + 0 = 0$

28. [2,3]

29. Key : a-p; b-q; c-r; d-s

30. [1] Sol: $(1 + x)^{2n-1} = {}^{2n+1}C_0 + {}^{2n+1}C_1 x + {}^{2n+1}C_2 x^2 + \cdots . + {}^{2n+1}C_{2n} x^n + {}^{2n+1}C_{2n+1} \cdot$ x^{2n+1} If x = −1, then G.E. = 1

31. [1,2,3] $S = C_0^2 + \sum_{r=1}^{n} \frac{C_r^2}{C_{r-1}} = 1 + \sum[(n + 1)C_r - {}^rC_r] = 1 + (n + 1)(2^n - 1) - n \cdot 2^{n-1}$

$= 1 + n \cdot 2^n + 2^n - -n \cdot 2^{n-1} = n \cdot 2^{n-1} + 2^n - n\,S + n = 2^{n-1}(n + 2)$

32. [2] $x^3 - 1 = 0$; Cube roots of unity are $1, \omega, \omega^2$

33. [3]

$$x = 1 \Rightarrow C_0 + C_1 + C_2 + C_3 + \cdots \ldots + C_n$$

$$= 2^n - \cdots \pm \cdots + C_0 + C_1\omega + C_2\omega^2 + C_3\omega^3 + \cdots \ldots + C_n\omega^n = (1+\omega)^n - - - - - - - - ($$

$$x = \omega \Rightarrow C_0 + C_1\omega^2 + C_2(\omega^2)^2 + C_3(\omega^2)^3 + \cdots \ldots + C_n(\omega^2)^n = (1+\omega^2)^n$$

$$x = \omega^2 \Rightarrow C_0$$

$$(1) + (2) + (3)$$

$$3[C_0 + C_3 + C_6 + C_9 + \cdots..]$$

$$= \frac{2^n + 2(-1)^n}{3}$$

34. [4] $x^4 - 1 = 0 \Rightarrow x = \pm 1, \pm i \Rightarrow$ Sum of the roots is zero.

WORKSHEET – 4

Level – I

1) D	2) C	3) C	4) B	5) B	6) C	7) B
8) C	9) C	10) C				

8. $(627)^{\frac{1}{4}} = (625+2)^{\frac{1}{4}} = (625)^{\frac{1}{4}}\left[1 + \frac{2}{625}\right]^{\frac{1}{4}}$

$$= 5\left[1 + \frac{1}{4}\left[\frac{2}{625}\right] + \frac{\frac{1}{4}\left(\frac{1}{4}-1\right)}{2!}\left(\frac{2}{625}\right)^2 \ldots \ldots.\right]$$

$$= 5\left[1 + \frac{1}{1250} + \frac{\frac{1}{4}\left(\frac{3}{4}\right)}{2}\left(\frac{4}{625}\right) + \cdots..\right] = 5\left[1 + \frac{1}{1250} + \frac{3}{5000}\right] = 5[1.0014] = 5.007$$

Level – II

1) B	2) C	3) A	4) B	5) B	6) A	7) 6
8) B	9) B	10) C	11) A	12)C	13)C	14)B
15) A	16)B	17)C	18)D	19)A,D	20)A	21)B
22) A	23)C	24)	25)12	26)B,C,D	27)A	28)C
29) D	30)B	31)A				

1. Key : B

$$(2+3x)^{-2/3} = 2^{-2/3}\left(1 + \frac{3}{2}x\right)^{-2/3} \Rightarrow \left|\frac{3}{2}x\right| < 1 \Rightarrow -1 < \frac{3}{2}x < 1 \Rightarrow \frac{-2}{3} < x < \frac{2}{3}$$

2. [C] $(1+x)^n\ T_{r+1} = \frac{n(n-1).....(n-r+1)}{r!}x^r$

$$\left(1-\frac{5}{2}x\right)^{-3/5} T_8 = T_{7+1} = \frac{\left(\frac{-3}{5}\right)\left(\frac{-3}{5}-1\right)\ldots\ldots\left(\frac{-3}{5}-6\right)\left(\frac{-5}{2}x\right)^7}{7!}$$

$$= \frac{3\times 8\times 13\times \ldots\ldots\times 33}{5^7\times 7!}\cdot\frac{5^7}{2^7}\cdot x^7 = \frac{3\times 8\times 13\times \ldots\ldots\times 33}{7!}\cdot\left(\frac{x}{2}\right)^7$$

3. Key : A $\left(1-\frac{5x}{3}\right)^{-3} \Rightarrow (1-x)^{-n}$ General term $T_{r+1} = \frac{n(n-1).....(n-r+1)}{r!}x^r$

$$T_{r+1} = \frac{(-3)(-3-1)\ldots\ldots(-3-r+1)}{r!}\cdot\left(\frac{-5}{3}x\right)^2 = \frac{3.4\ldots\ldots\ldots(r+2)}{r!}\left(\frac{5}{3}x\right)^r$$

$$= \frac{(r+1)(r+2)}{1-2}\left(\frac{5x}{3}\right)^r = \frac{1.2.3\ldots\ldots\ldots.r(r+1)(r+2)}{1.2r!}\left(\frac{5}{3}x\right)^r$$

4. Key : C

$$(1-4x)^{-3/5} \Rightarrow T_5 = \frac{\left(\frac{-3}{5}\right)\left(\frac{-3}{5}-1\right)\left(\frac{-3}{5}-2\right)\left(\frac{-3}{5}-3\right)\left(\frac{-3}{5}-4\right)}{4!}(-4x)^4$$

$$= \frac{\left(\frac{-3}{5}\right)\left(\frac{-8}{5}\right)\left(\frac{-13}{5}\right)\left(\frac{-18}{5}\right)(-4x)^4}{4!} = \frac{3\times 8\times 13\times 18}{24\times 5^4}\times 4^4\cdot x^4 = 234\times\frac{4^4}{5^4}\times x^4$$

x^4 Coefficient $= 234\times\frac{4^4}{5^4}$

5. Key : B

$$11^9 + 9^{11} = (10+1)^9 + (10-1)^{11} = 10^9 + 9c_1 10^8 + \cdots\ldots. + 1 + 10^{11} - 11C_1 10^{10} +$$
$$= 10[10^8 + 9c_1 10^7 + \cdots.. + 10^{10} - 11c_1 10^9 \ldots..] = 10\ \text{K}$$

6. Key : A

$$(1.01)^{1000000} = (1+0.1)^{1000000} \Rightarrow 1 + 1000000\times 0.1 = 1 + 100000 = 100000$$

7. Ans: 6

$$9^7 + 7^9 = -(1-8)^9 + (1+8)^7$$
$$= -[1 - 9c_1 8 + 9c_2 8^2 + \cdots..9c_9 8^9] + [1 + 7c_1 8 + 7c_2 8^2 + \cdots..]$$
$$= 8[9c_1 + 7c_1] + 8^2[7c_2 - 9c_2 + \cdots..] = 8[9+7] + 8^2[K] = 8^2(P) = 64P \Rightarrow 2^n = 64 \Rightarrow n = 6$$

8. Key : B

$$(4+5x)^{-7/3} = 4^{-7/3}\left(1+\frac{5x}{4}\right)^{-7/3} \Rightarrow \left|\frac{5x}{5}\right| < 1 \Rightarrow -1 < \frac{5x}{4} < 1 \Rightarrow -\frac{4}{5} < x < \frac{4}{5} \Rightarrow x$$
$$\in \left(\frac{-4}{5}, \frac{4}{5}\right)$$

9. Key : B

$$1+\frac{2}{3}\cdot\frac{1}{2}+\frac{2}{3}\cdot\frac{5}{6}\cdot\frac{1}{2^2}+\cdots\ldots\Rightarrow 1+\frac{\left(\frac{p}{q}\right)}{1}x+1+\frac{\frac{p}{q}\left(\frac{p}{q}+1\right)}{2!}(x)^2+\cdots\ldots=(1-x)^{-p/q}$$

$$\frac{p}{q}=\frac{2}{3}\Rightarrow\frac{p}{q}=\frac{-2}{3}\ x=\frac{1}{2}$$

$$\therefore\ 1+\frac{2}{3}\cdot\frac{1}{2}+\frac{2}{3}\cdot\frac{5}{6}\cdot\frac{1}{2^2}+\cdots\ldots=\left(1-\frac{1}{2}\right)^{-2/3}\Rightarrow=\left(\frac{1}{2}\right)^{-2/3}=2^{2/3}\Rightarrow\sqrt[3]{2^2}=\sqrt[3]{4}$$

10. Key : C

$$(2a-3b)^{-1/2}\cdot(2a)^{-1/2}\left[1-\frac{3b}{2a}\right]^{-1/2}\Rightarrow T_{r+1}=\frac{1}{\sqrt{2a}}\left[\frac{\left(1\frac{1}{2}\right)\left(1\frac{1}{2}+1\right)\ldots\left(\frac{t!}{2}\right)+r-1}{r!}\right]\left(\frac{+3b}{2a}\right)$$

$$=\frac{1}{\sqrt{2a}}\frac{1.3.5\ldots\ldots(2r+1)\frac{1}{2^r}}{r!}\left(\frac{3b}{2a}\right)=\frac{1}{\sqrt{2a}}\frac{1.3.5\ldots\ldots(2r+1)}{r!}\left(\frac{3b}{4a}\right)^r$$

11. Key: $1;1+\frac{1}{4}+\frac{1.3}{4.8}+\frac{1\cdot3\cdot5}{4\cdot8.12}+\cdots$

$$1+\frac{1}{2}-\frac{1}{2}+\frac{\frac{1}{2}\left(\frac{1}{2}+1\right)}{1.2}\cdot\left(\frac{1}{2}\right)^2+\frac{1\cdot\left(\frac{1}{2}+1\right)\left(\frac{1}{2}+2\right)}{1.2\cdot3}\left(\frac{1}{2}\right)^3\ldots\ldots..\Rightarrow\left(1-\frac{1}{2}\right)^{-\frac{1}{2}}=\left(\frac{1}{2}\right)^{\frac{-1}{2}}$$

$$=2^{1/2}=\sqrt{2}$$

12. Key : C

$$(1.03)^{19}=(1+0.03)^{19}=1+19\times0.03+\frac{19\times8}{2}\times(0.03)^2+\cdots..\Rightarrow\approx1.7250$$

13. Key : C

$$(1+x)^2(8-x)^{-1/3}\Rightarrow8^{-1/3}\left(1-\frac{x}{8}\right)^{-1/3}(1+x)^2$$

$$\frac{1}{2}\left(1+\frac{1}{3}\left(\frac{x}{8}\right)+\frac{\frac{1}{3}\left(\frac{1}{3}+1\right)}{2!}\left(\frac{x}{8}\right)^2+\cdots\ldots..\right)(1+2x+x^2)\Rightarrow\frac{1}{2}\left(1+\frac{x}{24}+\frac{x^2}{288}+\cdots\ldots\right)(1+2x+x^2)$$

x^2 coeff... $\frac{1}{2}\left[1+\frac{1}{288}+\frac{1}{12}\right]=\frac{1}{2}\times\frac{288+1+24}{288}=\frac{313}{576}$

14. Key: B

$$1+\frac{K}{3}+\frac{K(K+1)}{3.6}+\frac{K(K+1)(K+2)}{3.6.9}+\cdots.$$

$$1+K\left(\frac{1}{3}\right)+\frac{K(K+1)}{1.2}\left(\frac{1}{3}\right)^2+\frac{K(K+1)(K+2)}{1.2.3}\left(\frac{1}{3}\right)^3+\cdots..\left(1-\frac{1}{3}\right)^K=\left(\frac{2}{3}\right)^K$$

15. Key : A

$$8^{2n}-(62)^{2n+1}=(9-1)^{2n}-(63-1)^{2n+1}=(9^{2n}-{}^{2n}C_1 9^{2n-1}+\cdots..+1)$$

$$=(63^{2n+1}-{}^{2n+1}C.63^{2n}+\cdots\ldots..)=9K+2$$

16. Key : B

$$\text{In}(1+x)^{\frac{27}{5}}\Rightarrow T_{r+1}=\frac{27}{5}\left(\frac{27}{5}-1\right)\left(\frac{27}{5}-2\right)\ldots\ldots.\left(\frac{27}{5}-r+1\right)\frac{x^2}{r!}$$

If $r = 6$ coefficients of x^6 is $\frac{\frac{27}{5}\left(\frac{27}{5}-1\right)\left(\frac{27}{5}-2\right)\left(\frac{27}{5}-3\right)\left(\frac{27}{5}-4\right)\left(\frac{27}{5}-5\right)}{5!}$ is r =? $\Rightarrow x^7$ coefficents is $-ve$.

17. Key : C

x^{10} coefficient in $(1+x)^{15}$ is $15C_{10} = 15C_5$

x^5 coefficient in $(1-x)^{-n}$ is ${}^{n+4}C_5$

$\Rightarrow 15C_5 = {}^{n+4}C_5 \Rightarrow n+4 = 15 \Rightarrow n = 11$

18. Key : D; $\frac{2^{403}}{15} = \frac{2^3 \cdot 2^{400}}{15} = \frac{8\left[\left(2^4\right)^{100}\right]}{15} = \frac{8(16)^{100}}{15} = \frac{8(15+1)^{100}}{15}$

$= \frac{8[15^{100} + 100C_1 15^{99} + \cdots .. + \cdots]}{15}$

$= \frac{8 \times (15P+1)}{15} = \frac{8 \times 15P}{15} + \frac{8}{15} \Rightarrow$ K = 8

19. Key: A,D

$$(1-4x)^{1/2} \Rightarrow T_{3+1} = \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)}{3!}(-4x)^3 \Rightarrow = \frac{\frac{1}{2}\left(\frac{-1}{2}\right)\left(\frac{-3}{2}\right)}{3 \times 2}(-4)^3 x^3$$

$$= \frac{3}{8 \times 3 \times 2}(-4^3) = -4$$

20. Key : A

21. Key : B

$$(3-4x)^{3/4} = 3^{3/4}\left(1 - \frac{4}{3}x\right)^{3/4} \Rightarrow -1 < \frac{4}{3}x < 1 \Rightarrow \frac{-3}{4} < x < \frac{3}{4}$$

22. Key : A

$$(2+5x)^{-1/2} = 2^{-1/2}\left(1 + \frac{5}{2}x\right)^{-1/2} \Rightarrow -1 < \frac{5}{2}x < 1 \Rightarrow -\frac{2}{5} < x < \frac{2}{5}$$

23. Key : C

$$(7+3x)^{-5} = 7^{-5}\left(1 + \frac{3}{7}x\right)^{-5} = -1 < \frac{3}{7}x < 1 \Rightarrow \frac{-7}{3} < x < \frac{7}{3}$$

Change the option 3 in question.

24. $a-r, b-q, c-s, d-p$

25. Ans: 12

$$(a+bx)^{-3} = \frac{1}{27} + \frac{1}{3}x + \cdots \ldots \Rightarrow a^{-3}\left(1 + \frac{b}{a}x\right)^{-3} = \frac{1}{27}(1 + 9x \ldots \ldots)$$

$$\frac{1}{a^3}\left(1 + \frac{b}{a}\right)^{-3} = \frac{1}{27}(1 + 9x \ldots \ldots) \Rightarrow \frac{1}{a^3} = \frac{1}{27} \Rightarrow a = 3$$

$$\left(1 + \frac{b}{3}x\right)^{-3} = 1 + bx + \cdots \ldots \Rightarrow = 1 + 9x + \cdots . \Rightarrow b = -9 \Rightarrow a - b = 3 - (-9) = 12$$

26. Key : 2,3,4

$$\left(\frac{1-t^6}{1-t}\right) = (1-t^6)^3(1-t)^{-3}$$

$$\Rightarrow (1-3t^6+3t^{12}-t^{18})(1+3t+6t^2+10t^3+15t^4+\cdots.)$$

$$t^4 \text{ coefficient } = 15$$

27. Key: 1

$$\left(1+\frac{2}{3}x\right)^{3/2}(32+5x)^{-1/5} = \left(1+\frac{2}{3}\left(\frac{3}{2}\right)x\right)(32)^{-1/5}\left(1+\frac{5}{32}x\right)^{-1/5}$$

$$= \frac{1}{32^{1/5}}(1+x)\left(1-\frac{1}{5}\left(\frac{5}{3^2}\right)x\right) = \frac{1}{2}(1+x)\left(\frac{32-x}{32}\right) = \frac{1}{64}(32+32x-x) = \frac{32+31x}{64}$$

28. Key : 3; $\frac{(1-2x)^{-1}(1-3x)^{-2}}{(1-4x)^{-3}} = (1-2x)^{-1}(1-3x)^{-2}(1-4x)^3 = (1+2x)(1+6x)(1-12x)$

$$= (1+2x+6x-12x) = 1-4x$$

29. Key: 4

$$\frac{\left(1+\frac{2}{3}x\right)^3(1-15x)^{-1/5}}{(2-3x)^4} = \frac{\left(1+\frac{2}{3}x\right)^{-3}(1-15x)^{-1/5}}{2^4\left(1-\frac{3}{2}x\right)^{-4}}$$

$$= \frac{1}{16}\left(1+\frac{2}{3}x\right)^{-3}(1-15x)^{-1/5}\left(1-\frac{3}{2}x\right)^4 = \frac{1}{16}\left(1+(-3)\left(\frac{2}{3}x\right)\right)\left(1-15\left(\frac{-1}{5}\right)x\right)\left(1+\frac{3}{2}x\right)$$

$$= \frac{1}{16}[(1-2x)(1+3x)(1+6x)] = \frac{1}{16}[1-2x+3x+6x] = \frac{1}{16}(1+7x)$$

30. Key : 2

$1+\frac{2}{4}+\frac{2.5}{4.8}+\frac{2.5.8}{4.8.12}+\cdots\ldots$

Compare with

$$1+\frac{p}{1}\left(\frac{x}{q}\right)+\frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2+\cdots\ldots \Rightarrow 1+\frac{2}{1}\left(\frac{1}{4}\right)+\frac{2.5}{1.2}\left(\frac{1}{4}\right)^2+\cdots\ldots..$$

$$P=2, \frac{x}{q}=\frac{1}{4} \Rightarrow p+Q=5 \Rightarrow q=3 \Rightarrow x=\frac{3}{4}$$

$$\therefore\ 1+\frac{2}{4}+\frac{2.5}{4.8}+\frac{2.5.8}{4.8.12}+\cdots\ldots = \left(1-\frac{3}{4}\right)^{\frac{-2}{3}} = \left(\frac{1}{4}\right)^{-2/3} = 4^{2/3} = \sqrt[3]{16}$$

31. Key : 1

$$1+\frac{1}{4}+\frac{1.3}{4.8}+\frac{1.3.5}{4\cdot 8\cdot 12}+\cdots \Rightarrow 1+\left(\frac{1}{2}\right)\frac{1}{2}+\frac{\frac{1}{2}\left(\frac{1}{2}+1\right)}{2!}\left(\frac{1}{2}\right)^2+\cdots\ldots$$

$$\left(1-\frac{1}{2}\right)^{-1/2} = \left(\frac{1}{2}\right)^{-1/2} = 2^{1/2} = \sqrt{2}$$

32. Key: 1

$$1+\frac{2}{3}\left(\frac{1}{8}\right)+\frac{2\times 5}{3\times 6}\left(\frac{1}{8}\right)^2+\cdots\ldots\Rightarrow 1+\frac{2}{3}\left(\frac{1}{8}\right)+\frac{\frac{2}{3}\left(\frac{2}{3}+1\right)}{2!}\left(\frac{1}{8}\right)^2+\cdots\ldots$$

$$\frac{p}{q}=\frac{2}{3},x=\frac{1}{8}\Rightarrow 1+\frac{2}{3}\left(\frac{1}{8}\right)+\frac{2\times 5}{3\times 6}\left(\frac{1}{8}\right)^2+\cdots\ldots=\left(1-\frac{1}{8}\right)^{-2/3}$$

$$=\left(\frac{7}{8}\right)^{-2/3}=\left(\frac{8}{7}\right)^{2/3}=\sqrt[3]{\frac{64}{49}}=\frac{4}{\sqrt[3]{49}}$$

9. Permutations and Combinations

WORKSHEET – 1
Level – I

1) B	2) A	3) C	4) D	5) D	6) B
7) A	8) B	9) A	10) A	11) D	12) D
13) A					

1. (3) $1! + 2! + 3! + 4! + 5! = 1 + 2 + 6 + 24 + 120 = 153$
2. (1) We have, $1 \times 3 \times 5 \times 7 \times \ldots \times (2n-1)$

   $= \dfrac{1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \ldots \times (2n-2) \times (2n-1) \times 2n}{2 \times 4 \times 6 \times 8 \times \ldots \times (2n-2) \times 2n} = \dfrac{(2n)!}{2^n n!}$
3. (4) We have

   ${}^nP_r + r \cdot {}^nP_{r-1} = {}^{(n+1)}P_r \Rightarrow {}^9P_5 + 5 \cdot {}^9P_4 = {}^{10}P_5 \therefore r = 5$
4. (4) Clearly a person can enter the hall through any of the three entrance doors. So, there are 3 ways of entering the hall. After entering the hall, the person can come out through any of the two exit doors. So, there are 2 ways of coming out. Hence, the number of ways in which a person can enter the hall and then come out = $3 \times 2 = 6$ ways
5. (3) By option verification $= n^{(n-1)}P_{(r-1)} = \frac{n \cdot (n-1)!}{(n-1-r+1)!} = \frac{n!}{(n-r)!} = n_{p_r}$
6. (1) Given word 'KRISHNA'

   No.of letter = 7; The no. of vowels = 2

   Treat two vowels are one unit.

   The no.of units = 5 different units +1 unit of vowels = 6 units

   (2 vowels arrange among themselves in 2 ! ways)

   The no.of required arrangements = $6! \times 2!$
7. Given word CHEMISTRY

   No. of letters = 9

   C_____ _____Y

   No. of 4 letter words begin with ' C ' and ends with ' Y ' = $7p_2 = 42$ ways
8. (1) $1! + 2! + 3! + 4! + 5! = 153$
9. (4) No. of boys = 4

   No. of girls = 3

   The first place with boy in ${}^4p_1$ ways

   The last place with girls in ${}^3p_1$ ways

   Remaining 5 places are filled by 3 boys and 2 girls in 5 ! ways.

The required no. of ways = $4 \times 3 \times 5$!

10. (3) Given

The no.of diff. Maths books = 5

The no.of diff. Physics books = 4

The no.of diff. Chemistry books = 3

Since the books of the same subject are together.

The no.of units = 1 unit of maths +1 unit of physics +1 unit of chemistry = units

These can be arranged in 3 ! ways

Same subject books can be arranged among themselves in $5! x 4$!x 3 ! ways.

The required no.of arrangements = $3! \times 5! \times 4! \times 3$!

11. The man can travel from X to Y in 10 ways and as he is not to return by the same bus that he took while going, corresponding to each of the 10 ways of going, there are 9 ways of returning. Hence the total number of ways in which he can go to Y and return back is $10 \times 9 = 90$ ways.

12. (1) Given word "TRIANGLE"

No.of vowels = $3(\ A, E, I)$

No.of consonants = $5(\ T, R, N, G, L)$

The required no. of ways = $5! \times 3$!

13. (1) The first square can be coloured with any one of ten colours. So, there are 10 ways to colour it.

1st	2nd	3rd	4th	5th	6 th

Since, the adjacent squares should not have the same colour, so the 2 nd can be coloured in $10 - 1 = 9$ ways (i.e., by using any of the remaining colours).

Again, the 3rd can be coloured in $10 - 1 = 9$ ways leaving the colour used in the second square and so on.

Hence, the total number of ways of colouring the given strip = $10 \times 9 \times 9 \times 9 \times 9 \times 9 = 590{,}490$ ways

14. (2) Given word 'EQUATION'

No.of vowels = 5

No.of consonants = 3

No.of 4 letter words that can begin and end with vowel = ${}^5P_2 \times {}^3P_2 = 120$

15. Under the restriction, first digit can be selected 5 ways.
    After the selection of first digit, four digits are left.
    Second digit can be selected in 4 ways and the third digit can be selected in 3 ways.
    According to fundamental principal of counting total number of ways = $5 \times 4 \times 3 = 60$ ways.
16. (3) Given word 'MIXTURE'
    The no.of 4 letters word which contains letter ' x ' = ${}^{6}P_{3} = 120$
17. (1)
    First item in column-A can be matched with any of the 6 items in column-B.
    This can be done in 6 ways. When this has been done, the second item in column-A can be matched with any of the remaining 5 items of column-B.
    i.e., Here are 5 ways of matching second item of column-A.
    Similarly the number of ways of matching the 3rd , 4th , 5th , and 6th item of A are 4,3,2 and 1 respectively.
18. (1) Given word ORGANIC
    No.of vowels = 3
    No.of consonants = 4
    No two vowels come together
    First 4 consonants in 4 places can be arranged in 4 ! ways
    In b/w first and last 4 consonants there are 5 gaps.
    In 5 gaps 3 vowels can be arrangement = $4 \times {}^{5}P_{3}$ ways
    ∴ The required no.of arrangements = $4!\, x^{5}P_{3}$
19.
    (2) $\sum_{r=1}^{10} r \cdot r! = \sum_{r=1}^{10} (r+1-1) \cdot r! = \sum_{r=1}^{10} [(r+1)! - r!] = 11! - 1$
20. (2) No.of boys = 6
    No.of girls = 6
    Boys and girls sit alternately
    First arrange all the boys in 6 ! ways
    girls can be arranged alternatively in 6 ! ways
    No. of ways of arrangements = $2 \times 6! \times 6!$.
21. (2) Fix 5 at middle place, 1, 2, 3, 4 on left side and 6,7,8,9 on right side can be arranged in $(4!)^2$ ways.

$a = (x+2)!\ b = {}^{x}P_{11}\ c = (x-11)!$

$a = 182bc$

$(x+2)! = 182\dfrac{x!}{(x-11)!}(x-11)!$

$(x+2)(x+1) = 182 \Rightarrow x^2 + 3x - 180 = 0 \Rightarrow (x+15)(x-12) = 0$

$x \neq -15;\ \therefore x = 12$

23. (1) ${}^{9}P_5 + 5 \cdot {}^{9}P_4 \Rightarrow {}^{10}P_5 = {}^{10}P_r \Rightarrow r = 5$

24. (1) Let E_p denotes the exponent of p in n

Then, $E_p(n!) = \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \cdots\ldots + \left[\frac{n}{p^s}\right]$,

[Where s is the largest + ve integer such that $p^s \leq n \leq p^{s+1}$]

Here $n = 100, p = 3 \therefore 3^4 < 100 < 3^5$

So, $E_3(100!) = \left[\frac{100}{3}\right] + \left[\frac{100}{3^2}\right] + \left[\frac{100}{3^3}\right] + \left[\frac{100}{3^4}\right] = 33 + 11 + 3 + 1 = 48$

25. (1) No.of boys = 5

No.of girls = 5

girls together = 1 unit

5 different units [boys]+1unit (girls) = 6 units

6units arranged in 6!ways

5 girls themselves arranged 5 ! ways

26. (2) No two girls sit together arrange 5 boys in 5 ! ways $-\mathrm{B_B} - \mathrm{B_B_B_}$

there are 6 places for 5 girls arranged in ${}^{6}P_5$ ways

$\therefore$ The required arrangement is ${}^{6}P_5 \times 5$! ways = $6! \times 5$! ways.

27. (3) Boys and girls sit alternately

First arrange all the boys in 5 ! ways

girls can be arranged alternatively in 5 ! way

similarly, girls first and then boys

$\therefore$ no.of ways of arrangements are 2×5 ! $x5$! ways

28. a-s; b-r; c-q; d-p

(a) $\frac{n(n-1)(n-2)(n-3)(n-4)}{n(n-1)(n-2)} = 2;$

$n^2 - 7n + 10 = 0 \Rightarrow n^2 - 2n - 5n + 10 = 0 \Rightarrow n(n-2) - 5(n-2) = 0 \Rightarrow n = 2,5$

(b) Since ${}^{n}p_r = {}^{n-1}p_r + r^{n-1}p_{r-1} \Rightarrow n = 6$

(c) ${}^{n}p_4 = 10 \times 168 = 10 \times 8 \times 21 = 10 \times 8 \times 7 \times 3 = 8 \times 7 \times 6 \times 5$

${}^{n}p_4 = {}^{8}p_4 \Rightarrow n = 8$

(d) $\frac{n(n-1)(n-2)(n-3)(n-4)}{n(n-1)(n-2)(n-3)} = 12 \Rightarrow n - 4 = 12 \Rightarrow n = 16$

29. (2,3,4) Sol:Product of first ' n ' natural numbers $= 1 \times 2 \times 3 \times \ldots \times n = n\,! = n_{p_n}$

Also, $\mathrm{n}_{\mathrm{p}_{\mathrm{n}-1}} = \frac{\mathrm{n!}}{(\mathrm{n}-(\mathrm{n}-1))!} = \frac{\mathrm{n!}}{1!} = \mathrm{n}\,!$

30. a-s; b-r; c-p; d-q

a) $7^{(n+1)}P_5 = 2 \cdot {}^{n}P_6 \Rightarrow 7(n+1) = 2(n-4)(n-5) \Rightarrow 7n + 7 = 2n^2 - 18n + 40$

$2n^2 - 25n + 33 = 0 \Rightarrow n = 11\; n = \frac{3}{2} \notin N$

b) $11({}^{n}P_4) = (20)({}^{n-2}P_4) \Rightarrow 11\frac{n!}{(n-4)!} = 20\frac{(n-2)!}{(n-6)!} \Rightarrow 11\frac{n(n-1)}{(n-4)(n-5)} = 20$

$\Rightarrow 11(n^2 - n) = 20(n^2 - 9n + 20) \Rightarrow n = \frac{23}{9} \notin N$ or $n = 16$.

c) $7({}^{18}P_{r-1}) = 9({}^{17}P_{r-1}) \Rightarrow 7\frac{18!}{(18-r+1)!} = 9\frac{17!}{(17-r+1)!} \Rightarrow \frac{14}{(19-r)} = 1 \Rightarrow r = 5.$

d) ${}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800 : 1 \Rightarrow \frac{56!}{(50-r)!} \times \frac{(51-r)!}{54!} = \frac{30800}{1}$

$56 \times 55 \times (51 - r) = 30800 \Rightarrow 51 - r = 10 \Rightarrow r = 41$

31. (1) Total number of letters in the word 'MONDAY' is 6 and all are distinct.

Words of 4 letters

Number of words of 4 letters $= {}^{6}\mathrm{P}_4 = 6 \times 5 \times 4 \times 3 = 360.$

32. (2) Number of words by using all the letters $= {}^{6}\mathrm{P}_6 = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720.$

33. (3) Words beginning with a vowel

In the letters $\mathrm{M, O, N, D, A, Y}$, only $\mathrm{O, A}$ are vowels.

$\therefore$ In the required words, the first letter can be filled in ${}^{2}\mathrm{P}_1$

= 2 ways (any one of $\mathrm{O, A}$) and the remaining 5 letters in ${}^{5}\mathrm{P}_5 = 5\,!$

= 120 ways (4 consonants and 1 vowels).

$\therefore$ By F.P.C., total number of words $= 2 \times 120 = 240.$

34. (2,3)p $= \mathrm{E}_2(10!) = \left[\frac{10}{2}\right] + \left[\frac{10}{2^2}\right] + \left[\frac{10}{2^3}\right] = 5 + 2 + 1 = 8$

$\mathrm{q} = \mathrm{E}_3(10!) = \left[\frac{10}{3}\right] + \left[\frac{10}{3^2}\right] = 3 + 1 = 4$

$\mathrm{r} = \mathrm{E}_5(10!) = \left[\frac{10}{5}\right] = 2;\; \mathrm{s} = \mathrm{E}_7(10!) = \left[\frac{10}{7}\right] = 1$

$\therefore q = 4, r = 2$

35. (4)

WORKSHEET – 2
Level – I

1) A	2) C	3) A	4) A	5) B	6) A	7) C
8) B	9) D	10) A				

Level – II

1) A	2) C	3) B	4) B	5) D	6) B	7)
8) C	9) A	10) C	11) C	12) B	13) B	14) A
15) A	16) A	17) D	18) A	19)	20) B	21) B
22) C	23) A,D	24) C	25) A	26) A	27) a-q; b-r; c-s; d-p	
28)	29) A,B	30) C	31)			

1. (1) Required number of words = $5^3 = 125$

2. (3) Given word is MONKEY

   No.of 4 letter words that can be formed in which at least one letter repeated = $6^4 - 6p_4$

3. (2) Total number of molecules = 12.

   In a chain, each molecule A, C, G, T occur thrice,

   ∴ Total number of arrangements = $\frac{12!}{3!3!3!3!} = 369600$.

4. (2) Given word MATHS

   Dictionary order of letters A,H,M,S,T

   No.of words begin with

   A = 4 !

   H = 4 !

   M = 4 !

   S = 4 !

   TAH = 2 !

   TAMH = 1!

   TAMSH = 1 !

   Rank of TAMSH = $4 \times 4! + 2 + 1 + 1 = 96 + 4 = 100$.

5. (4) Given digits are 2,4,5,7,8

   If unit digit is 5 then only number is divisible by 5

   4 digit numbers:

   unit place has one chance and remaining three places are filled with $5 \times 5 \times 5$ ways.

   ∴ required no.of numbers divisible by 5 = $5^3 \times 1 = 5^3$ ways.

6. (2) Total no.of persons = 12(8 m + 4w)

   Treat 4 women as one unit.

   then we have 8 different units (men) +1 unit of (women) = 9 units.

   Circular arrangements = (9 – 1) ! ways = 8 !ways

   Now 4 women arrange among themselves in 4 ! ways.

The required no.of arrangements = $8!\,4!$

7. Given word is BANANA

Given word contains 6 letters in which 3A's are alike, 2 N's are alike and one B all A's together:

Treat 3A's as one unit then we have $3 + 1 = 4$ units

in which 2 N 's are a like and rest are different they can arranged in $\frac{4!}{2!}$ ways.

The 3A's among themselves can be arranged in $\frac{3!}{3!} = 1$ way.

The number of required arrangements $= \frac{4!}{2!} \times 1 = 12$ ways

8. (3) Since repetition is allowed, each of the 3 places in 3 digits number can be filled in 5 ways.

H	T	O
5 ways	5 ways	5 ways

9. (1) Given word PACIFIC

Treat 2 C'S as 1 unit

total units = 5 units +1 unit = 6 units

these units can be arranged in $\frac{6!}{2!}$ ways

The two C's arranged among them selves in $\frac{2!}{2!} = 1$ way

$\therefore$ The two C's come together is $\frac{6!}{2!} \times 1 = 360$ ways

10. (3) No.of Indians = 5

No. of Americans = 4

No. of Russians = 3

Treat each nationalist as 1 unit

these 3 units arranged in round table in 2!ways

5 Indians among themselves is 5 ! ways.

4 Americans among themselves is 4 ! ways.

3 Russians among themselves is 3 ! ways. persons same nationality sit together is $2!x3! \times 4!x5$! ways.

11. (3) ABLORU ; $2 \times 5! + 0 \times 4! + 0 \times 3! + 0 \times 2! + 1 \times 1! + 1 = 242$

12. (2) Required number of numbers $= 4 + 4^2 + 4^3 + 1 \times 4^3 = 148$

13. (2) Non-zero at ten's place:

Case-1 : Same number at ten's and hundred's places,

Number of numbers $= 1 \times 9 \times 9 = 81$

Case-II : Same number at ten's and unit's places.

Number of numbers $= 8 \times 9 \times 1 = 72$($\because$ 0 cannot be at hundred's place)

Zero at ten's place:

Case-I: Same number at ten's and hundred's place. This is impossible.

Case-II : Same number at ten's and unit's place. Number of numbers $=$

$9 \times 1 \times 1 = 9.$

$\therefore$ Total number of numbers $= 81 + 72 + 9 = 162.$

14. (1) First arrange vowels in the beginning, then arrange remaining letters.

15. [1] Let all the yellow roses be taken as one flower then the no. of flowers becomes $(6 + 1) = 7$ flowers.

They can be arranged in a garland in $\frac{1}{2}(7-1)! = \frac{1}{2}(6!)$

But the 3 yellows can be rearranged in 3 ! ways.

The arrangements of roses so that the yellow roses may be together $= \frac{1}{2}(6!)(3!)$

16. (1) M I R R O R I M ORRR1. $\frac{5!}{3!} + 0.4! + 1 \cdot \frac{3!}{3!} + 1 \cdot \frac{2!}{2!} + 0.1! + 1 = 23$

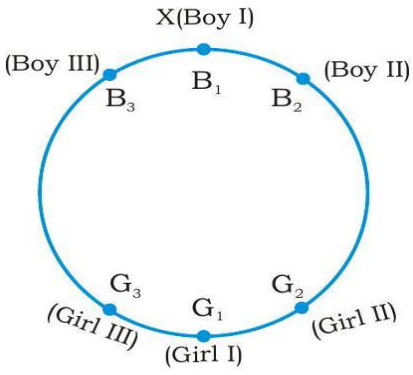
17. [4] Sol: The 6 red roses can be arranged in a garland in $\frac{1}{2}(6-1)! = \frac{1}{2}(5!)$ ways.

The 3 yellow roses are to be arranged in 6 places in between red roses. i.e. in $^6P_3$ ways.

The no. of arrangements so that no two yellow roses come together $= \frac{1}{2}(5!)\ {}^6P_3$

18. (1) $\frac{n(n^r-1)}{n-1} = \frac{3(3^6-1)}{3-1} = \frac{728\times3}{2} = 1092.$

19. As shown in the figure, the boy X will have neighbours as boys B_2 and B_3. The girl Y will have neighbours as G_1 and G_3.



The two girls G_2 and G_3 can be arranged at their own places in 2 ways.

Similarly, the two boys B_2 and B_3 can be arranged at their own places in two ways. Hence total number of arrangements $= 2 \times 2 = 4$.

20. (2) 2.18 !

21. (2)

$$\frac{\lfloor 40}{\lfloor 6(\lfloor 4)^2(\lfloor 6)^3}$$

22. [3] Let the 4 particular stones be taken as one stone then we have 5 stones. They can be strung to form a necklace in 4 ! ways.

But 4 particular stones can be arranged in 4 ! ways.

Hence the required no. of ways $= \frac{1}{2}(4!)(4!) = 288$

23. (1,4) Number of words $= \frac{(11)!}{2!2!2!} = \frac{(11)!}{(2!)^3} = 4989600$

24. (3) Treat two $R's$ as one unit, total number of units = 1 unit +5 different units = 6 units Then we have 6 units in which A's are two

Hence they can be arranged in $\frac{6!}{2!} = 360$ way

Now the two $R's$ can be arranged themselves $\frac{2!}{2!} = 1$ way

Hence the number of required permutations is 360 .

25. (1)

First fill the first and last place with A's in $\frac{2!}{2!} = 1$ way

Now the remaining 5 letters can be arranged in $\frac{5!}{2!}$ ways.

∴ The number of required permutations is 60 .

26. (1) First arranged the vowels in $\frac{3!}{2!}$ ways and consonants can be arranged in $\frac{4!}{2!} = 12$ ways.

V			V			V
A	R	R	A	N	G	E
	C	C		C	C	

∴ total number of arrangements $= \frac{3!}{2!} \times \frac{4!}{2!} = 3 \times 12 = 36$

27. a-q; b-r; c-s; d-p

28. The word of "PATALIPUTRA' has eleven letters, in which $2P's, 3A's, 2T's, 1L, 1U, 1R, 1I$. Vowels are AAIUA.

These vowels can be arranged themselves in $\frac{5!}{3!} = 20$ ways.

The consonants are P, T, L, P, T, R can be arranged themselves in $\frac{6!}{2!2!} = 180$ ways.

$\therefore$ Required number of words $= 20 \times 180 = 3600$ ways.

$\therefore$ The number of required permutations is 36 .

29. (1,2) The rank of TOJCBE $= 5.5! + 4.4! + 3.3! + 2.2! + 1 = 717$

The rank of OBJECT $= 4.5! + 2.3! + 1.2! + 1 = 495$

30. (3) Required number of permutations $= \frac{6!}{3!}$

31. (2) Required numbers of permutations $= \frac{5!}{3!}$

32. (3) Required number of permutations $= 5$!

33. a-q; b-t; c-(p,s); d-r

a) $\frac{{}^8P_4}{4} = \frac{8\times7\times6\times5}{4} = 420$

b) $\frac{(7-1)!}{2} = \frac{720}{2} = 360$

c) $4! \times {}^5P_4 = 5! \times 4! = 2880$

d) $(6-1)! \times {}^6P_2 = 5! \times 30 = 3600$

34. (1,2,3,4) The number of permutations of r a's and $(2n-r)$ b's is ${}^{2n}C_r$. The maximum of ${}^{2n}C_r = {}^{2n}C_n$

WORKSHEET – 3

Level – I

1) D 2) A 3) A 4) B 5) B 6) B 7) A

Level – II

1) D	2)C	3)D	4) 31	5)C	6) D
7)C	8)A	9)301	10)C	11)C	12)B
13)C	14)C	15)B	16)C	17)A	18)6
19)B	20)B	21)C	22)A,C	23)C	24)A
25)D	26)C	27) a-q, b-r, c-s, d-p		28)A,C	29)A,B,D
30) D	31)C	32)A	33)A,B,C	34) D	35)C
36)A					

1. men out of 6 and 2 ladies out of 5 can be selected in ${}^6C_3 \times {}^5C_2 = 200$ ways

2. Following are the ways of selecting the team

a) 4 boys and one girl: This can be done in ${}^7C_4 \times {}^4C_1$ ways.

b) 3 boys and two girls: This can be done in ${}^7C_3 \times {}^4C_2$ ways.

c) 2 boys and three girls: This can be done in ${}^7C_2 \times {}^4C_3$ ways.

d) 1 boy and 4 girls: This can be done in ${}^7C_1 \times {}^4C_4$ ways.

If p is the total number of ways then p = ${}^{7}C_4 \times {}^{4}C_1 + {}^{7}C_3 \times {}^{4}C_2 + {}^{7}C_2 \times {}^{4}C_3 + {}^{7}C_1 \times {}^{4}C_4$

$= \frac{7!}{4!\,3!} \times \frac{4!}{1!\,3!} + \frac{7!}{3!\,4!} \times \frac{4!}{2!\,2!} + \frac{7!}{2!\,5!} \times \frac{4!}{3!\,1!} + \frac{7!}{1!\,6!} \times \frac{4!}{4!\,0!}$

$= 35 \times 4 + 35 \times 6 + 21 \times 4 + 7 = 441$ ways.

3. ${}^{n}C_r + \sum_{i=1}^{m} {}^{(n+m-i)}C_{r-1} = {}^{(n+m)}C_r$

4. The candidate can fail by failing in 1 or 2 or 3 or 4 or 5 subjects out of 5 in each case

∴ Total number of ways in which they can fail = ${}^{5}C_1 + {}^{5}C_2 + {}^{5}C_3 + {}^{5}C_4 + {}^{5}C_5$

$= {}^{5}C_1 + {}^{5}C_2 + {}^{5}C_2 + {}^{5}C_1 + {}^{5}C_5 = 31$ ways

5. No. of arrangement of N. I, N, T, N = $\underline{\underline{5}}$

No. of ways of selecting 3 places out of 6 places for $E = {}^{6}C_3$

Total no. of arrangements = $\frac{5!}{3!} \times {}^{6}C_3$

6. ${}^{7}C_3 = 35$

7. Since each set must contain a_3, other two elements taken from remaining (n-1) can be selected in ${}^{n-1}C_2$ ways

8. ${}^{n+1}C_7 > {}^{n+1}C_6 \Rightarrow n + 1 > 7 + 6$

$\Rightarrow n > 12$

$\Rightarrow n = 13,14, \ldots$. ∴ $n = 13$

9. ${}^{4}C_2\,{}^{7}C_3 + {}^{4}C_3\,{}^{7}C_2 + {}^{4}C_4\,{}^{7}C_1$

10. ${}^{6}C_4 \times {}^{8}C_7 + {}^{6}C_5 \times {}^{8}C_6 + {}^{6}C_6 \times {}^{8}C_5 = 344$

11. ${}^{10}C_3 \times 3!$

12. The number of required selections is ${}^{9}C_5$

13. ${}^{8}C_5$

14. $l = 3$

15. $3!\,(3 + 4 + 5 + 6) = 108$

16. $3! \times 1111 \times 10 - 2! \times 111 \times 10 = 64440$

17. $12(2 + 3 + 5 + 4 + 4)(1 \quad 1 \quad 1 \quad 1 \quad 1) = 2399976$

18. $(1 + 3 + 5 + 7 + 9)[{}^{5}P_3 \cdot (1111) - {}^{4}P_2(111)] = 16{,}33{,}200$

19. 3 alike = 3 O's = 11 different from {P, R, T, I, N} = ${}^{5}C_1 \cdot 1 \cdot \frac{4!}{3!} = 20$

20. All different = ${}^{6}C_5 \cdot 5!$ 2 similar +3 diff. = ${}^{3}C_1 \cdot {}^{5}C_3 \cdot \frac{5!}{2!}$ 3 same +2 diff. = ${}^{2}C_1 \cdot {}^{5}C_2 \cdot \frac{5!}{3!}$

2 same +2 same +1 diff. = ${}^{3}C_2 \cdot {}^{4}C_1 \cdot \frac{5!}{2!2!}$

3 same +2 same = ${}^{2}C_1 \cdot {}^{2}C_1 \cdot \frac{5!}{3!2!}$

∴ Total selections = 72 Total arrangements = 3320

21. $31 \times (2 + 3 + 4 + 5) \times 1111 = 93324$

22. The total number of ways = ${}^{10}\mathrm{C}_5 = \frac{10\times9\times8\times7\times6}{5\times4\times3\times2\times1} = 252$ ways

23. Number of required ways = (1 women, 2 men) or (2 women, 3 men) or (3 women, 2 men) or (4 women, 1 men)
$= {}^{4}C_1 \times {}^{8}C_4 \times {}^{8}C_2 \times {}^{8}C_3 + {}^{4}C_3 \times {}^{8}C_2 + {}^{4}C_4 \times {}^{8}C_1 = 736$

24. The number of ways in which 11 players can be chosen out of 12 = ${}^{15}\mathrm{C}_{11}$ = 1365 ways

25. When a particular player is always chosen, we will have to choose 10 players out of 14 ∴ required number of ways = ${}^{14}\mathrm{C}_{10} = 1001$

26. When a particular player is never chosen, we will have to choose 11 players out of 14 . ∴ required number of ways = ${}^{14}\mathrm{C}_{11} = 364$ ways

27. a-q, b-r, c-s, d-p

28. The number of ways of selecting = (3 consonants out of 7) and (2 vowels out of 4) = ${}^{7}\mathrm{C}_3 \times {}^{4}\mathrm{C}_2 = 210$
∴ The number of groups each containing 3 consonants and 2 vowels = 210
∴ Each group contains 5 letters which can be arranged amongst themselves in 5! = 120 ways.
∴ Required number of words = 210 × 120 = 25200 ways

29. (1,2,4)

30. ${}^{12}C_4$

31. $2^{12} - ({}^{12}C_0 + {}^{12}C_1 + {}^{12}C_2) = 4017$

32. ${}^{12}C_0 + {}^{12}C_1 + {}^{12}C_2 + {}^{12}C_3 = 299$

33. (1,2,3) Required number of ways = ${}^{11}\mathrm{C}_5 - {}^{9}\mathrm{C}_3 = {}^{9}\mathrm{C}_5 + {}^{9}\mathrm{C}_4 + {}^{9}\mathrm{C}_4 = 3\,{}^{9}\mathrm{C}_4$

34. Number of ways = ${}^{39}\mathrm{C}_1 = 39$

35. Number of ways = $2\mathrm{C}_1({}^{20}\mathrm{C}_1 \times {}^{15}\mathrm{C}_1) = 600$

36. Number of ways = (3 !) $({}^{72}\mathrm{C}_1 \times {}^{108}\mathrm{C}_2 + {}^{72}\mathrm{C}_2 \times {}^{108}\mathrm{C}_1 + {}^{72}\mathrm{C}_3 \times {}^{108}\mathrm{C}_0) = 4510224$
Alternate: Number of ways = $(3!)({}^{180}\mathrm{C}_3 - {}^{108}\mathrm{C}_3) = 4510224$

WORKSHEET – 4

Level – I

1) D	2) A	3) D	4) B	5) B	6) C	7) D
8) D						

Level – II

1) B	2) A	3)B	4) A	5)A	6) D	7)448
8) A	9) B	10)B	11)12	12)A	13)C	14)112
15)B	16)A	17)A	18)A,C	19)A,B,C,D	20)C	21)A
22) B	23)C	24)a-q,b-s,c-r,d-p		25)C	26)B	27)C
28) D	29)A	30)B	31)A	32)A,B,C,D	33)A	34)A
35) A	36)a-s,b-r,c-q,d-p					

1. (2) $900 = 2^2 \times 3^2 \times 5^2$ No. of ways $= \frac{(2+1)(2+1)(2+1)+1}{2} = 14$
2. (1) In a village

   No. of families = 600

   No. of families having no children = 100

   No. of families having only boys = 200

   No. of families having only girls = 200
3. (2) Given

   No. of families having boys and girls $= 600 - (100 + 2 \times 200) = 600 - 500 = 100$

   No. of students wrote exam = 105

   No. of students passed in English = 80

   No. of students passed in Maths = 75

   No. of students passed in both subjects = 60

   No. of students passed in only in English $= 80 - 60 = 20$

   No. of students passed in only in Maths $= 75 - 60 = 15$

   No. of students failed $= 105 - (20 + 15 + 60) = 105 - 15 - 80 = 10$
4. (1) No. of integers between 1 and 567 $= 567 - 1 - 1 = 565$

   No. of integers divisible by 3 $= \left[\frac{565}{3}\right] = 188$

   No. of integers divisible by 5 $= \left[\frac{565}{3}\right] = 113$

   No. of integers divisible by 7 $= \left[\frac{565}{7}\right] = 80$

   No. of integers divisible by 15 $= \left[\frac{565}{15}\right] = 37$

   No. of integers divisible by 35 $= \left[\frac{565}{35}\right] = 16$

No. of integers divisible by $21 = \left[\frac{565}{21}\right] = 26$

No. of integers divisible by 3 x $5 \times 7 = \left[\frac{565}{105}\right] = 5$

$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$
$= 188 + 113 + 80 - 37 - 16 - 26 + 5 = 386 - 79 = 307$

5. (1) Integers between 999 and 9999

Number of 3 digited numbers except 9999

Number of integers begin with $3\overset{3}{\underset{1}{\downarrow}}$

$n(A) = 1000$

No. of integers end with $3, n(B) = 900$

No. of integers begin with 3 and end with $3n(A \cap B) = 10^2 = 100$

$n(A \cup B) = n(A) + n(B) - n(A \cap B) = 1000 + 900 - 100 = 1800$

6. (4) Given digits are $0,1,2, \ldots .,9$

No. of arrangements end with 8 = 9 !

No. of arrangements begin with 3 = 9 !

No. of arrangements that begin with 3 and end with 8 = 8 !

No. of arrangements that do not end with 8 and do not begin with

$3 = 10! - (9! + 9! - 8!) = 10! - 2 \times 9! + 8! = 8!\,(90 - 18 + 1) = 8! \times 73$

7. (448)Sum of even divisors = sum of divisors - sum of odd divisors = $480 - 32 = 448$

8. (1) The numbers of ways of a person make selections of fruits from the basket is $(4 + 1)(5 + 1)(6 + 1) - 1 = 209$

9. (2) The different sums is $(1 + 1)(1 + 1)(1 + 1)(1 + 1)(1 + 1) - 1 = 31$.

10. (2) Any divisor of $2^5 \cdot 3^4 \cdot 5^2$ is of the form $2^a \cdot 3^b \cdot 5^c$ where $0 \le a \le 5, 0 \le b \le 4$ and $0 \le c \le 2$.

Hence the sum of the divisors = $\sum_{a=0}^{5} \sum_{b=0}^{4} \sum_{c=0}^{2} 2^a 3^b 5^c$

$= (1 + 2 + \cdots 2^5)(1 + 3 + \cdots + 3^4)(1 + 5 + 5^2) = \left(\frac{2^6 - 1}{2 - 1}\right)\left(\frac{3^5 - 1}{3 - 1}\right)\left(\frac{5^3 - 1}{5 - 1}\right)$

$= 63 \times 121 \times 31 = 3^2 \cdot 7^1 \cdot 11^2 . 31.$

11. (12) For the number $2^a 3^b 5^c 7^d \ldots$ the number of positive integral divisors which are multiples of 6 is $ab(c + 1)(d + 1) \ldots \ldots$

12. (1) We have $9! = 2^7 \times 3^4 \times 5 \times 7$

Odd factors of the form $3m + 2$ are neither multiples of 2 nor multiples of 3 .

So the factors may be 1,5,7,35 of which just 5 and 35 are of the form 3 m + 2. Their sum is 40 .

13. (3) According to question 1 cm$(p,q) = r^2t^4s^2$. For 1 cm, (p,q) to contain r^2 following are the possible powers of r in p and q. $(0,2);(1,2);(2,2)$. Out of these all except (2,2) can be interchanged for p and q.

$\therefore$ Total no. of ways for p, q to contain different powers of r are $= 2 \times 3 - 1 = 5$

Similarly different powers of t that can be given to p and q are $(0,4),(1,4),(2,4)$ $(3,4),(4,4)$ out of which except (4,4) all others can be interchanged for p and q. $\therefore$ Total no. of ways for p, q to contains different powers of t are $= 2 \times 5 - 1 = 9$

Similarly total no. of ways for p, q to contains different powers of s are $= 2 \times 3 - 1 = 5$ So total no. of ordered pairs $= 5 \times 9 \times 5 = 225$.

14. (112) $(2008)^8 = 2^{24} \times 251^8$ has $25 \times 9 = 225$ positive divisors, including $(2008)^4 = \sqrt{(2008)^8}$. There is a one to one correspondence between the positive divisors less than $(2008)^4$ and those larger than $(2008)^4$. It follows that there are $\frac{1}{2}(225-1) = 112$ positive divisors less than $(2008)^4$

15. (2) Number of letters = 5

All letters are in wrong envelopes $= D_5 = 5!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right)$

16. (1) The total number without any restrictions containing digits 2,3,4,5,6,7 is $n(S) = 6^n$

The total number of numbers that contain 3,4,5,6,7 is $n(A) = 5^n$ The total number of numbers that contain 2,3,4,5,6 is $n(B) = 5^n$ The total number of numbers that contain 3,4,5,6 is $n(A \cap B) = 4^n$ The tortal number of numbers that do not contain digits 2 and 7 is $5^n + 5^n - 4^n$. The total number of numbers that contain 2 and 7 is $6^n - 5^n - 5^n + 4^n$

17. (1) k = 1 18. (1,3) Conceptual

18. (1,2,3,4) Exponent of ' 2 ' is $= \left[\frac{10}{2}\right] + \left[\frac{10}{2^2}\right] + \left[\frac{10}{2^3}\right] = 8$

Exponent of ' 3 ' is $= \left[\frac{10}{3}\right] + \left[\frac{10}{3^2}\right] = 3 + 1 = 4$

exponent of ' 5 ' is $= \left[\frac{10}{5}\right] = 2$; exponent of ' 7 ' is $= \left[\frac{10}{7}\right] = 1$

No of divisors of $10! = (8+1)(4+1)(2+1)(1+1) = 270$.

No of ways of putting ' N ' as a product of two natural numbers $= 270/2 = 135$

20. (3) Sol: In statement1, total number of selections

$= 21_{C_0} + 21_{C_1} + 21_{C_2} + 21_{C_3} + \cdots . + 21_{C_{10}} = 2^{20}$

but statement 2 is false, as given series is not exact half series.

21. (1) ${}^3C_1 \times {}^4C_1 \times {}^5C_1 = 3 \times 4 \times 5 = 60$

22. (2) $(2^3 - 1)(2^4 - 1)(2^5 - 1) = 7 \times 15 \times 31 = 3255$

23. (3) $(2^5 - 1) \times 2^7 = 31 \times 128 = 3968$

24. a-q; b-s; c-r; d-p ; a) $1331 = (11)^3$; No. of divisors $= 3 + 1 = 4$

b) $1024 = 2^{10}$(which is a perfect square) ; No. of ways $= \frac{1}{2}[(10 + 1) + 1] = 6$

c) $2484 = 2^2 \times 3^3 \times (23)^1$; No of ways $= \frac{1}{2}[(2 + 1)(3 + 1)(1 + 1)] = 12$

d) $70200 = 2^3 \times 3^3 \times 5^2 \times (13)^1$

Sum of proper divisors $= \left(\frac{2^4-1}{1} \times \frac{3^7-1}{2} \times \frac{5^3-1}{4} \times \frac{13^2-1}{12}\right) - 70200 - 1$

$= 260400 - 70201 = 190199.$

25. (3) The required number is $4 \times 3 + 3 \times 2 + 2 \times 4 = 26.$

26. (2) Total no of positive factors $= (2 + 1)(2 + 1)(1 + 1)(1 + 1)(1 + 1) = 72$

27. (3) Sum of all factors $= \frac{5^3-1}{5-1} \times \frac{2^3-1}{2-1} \times \frac{3^2-1}{3-1} \times \frac{7^2-1}{7-1} \times \frac{17^2-1}{17-1} = 124992$

28. (4) The sum of all odd proper divisors $= \left(\frac{5^3-1}{5-1} \times \frac{3^2-1}{3-1} \times \frac{7^2-1}{7-1} \times \frac{17^2-1}{17-1}\right) - 1 = 17855$

29. (1) A + B = 48 + 60 = 108

30. (1) A + E = 48 + 240 = 288

31. (2) C + D = 144 + 120 = 264

32. (1,2,3,4)

33. (1) ${}^6C_4D_2 + {}^6C_3D_3 + {}^6C_2D_4 + {}^6C_1D_5 + {}^6C_0D_6$

$15\left[2!\left(\frac{1}{2!}\right)\right] + 20\left[3!\left(\frac{1}{2!} - \frac{1}{3!}\right)\right] + 15\left[4!\left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right)\right] + 6 \times 5!\left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right] +$

$6! \times \left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!}\right] = 719$

34.

(1) $6! \times \left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!}\right] = 265$

35. (1) $6 \times 5!\left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right] + 6! \times \left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!}\right] = 529$

36. a-s, b-r, c-q, d-p

a) $5! - 1 = 119$

b) ${}^{10}C_6 \cdot D_4 = 210 \times 9 = 1890$

c) $5!\left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right) = 44$

d) ${}^5C_2 \cdot D_3 + {}^5C_3 \cdot D_2 + {}^5C_4 \cdot D_1 + {}^5C_5 \cdot D_0 = 31$

10. Probability

WORKSHEET – 1

Level – I

1) A	2) C	3) C	4) B	5) A	6) B
7) B	8) B				

Level – II

1) A	2) D	3) A	4) B	5) D	6) 31
7) D	8) B	9) C	10) B	11) D	12) D
13) C	14) A	15) 7	16) A	17) C	18) A
19) D	20) B	21) D	22) D	23) C	24) B
25) 6	26) A,B	27) D	28) B	29) a-q, b-s, c-p, d-r	
30) a-r,b-q,c-s d-p	31) B	32) C	33) A	34) a-s, b-p, c-t, d-q	

1. Leap year has 366 days.

$$
\begin{array}{r|l}
7 & 366 \\
\cline{2-2}
 & 35 \\
\cline{2-2}
 & 16 \\
 & 14 \\
\cline{2-2}
 & 2
\end{array}
\quad (52
$$

366 = 52 weeks +2 days $\Rightarrow \therefore n(S) = 7$

Let E be the event of getting 53 Mondays and 53 Tuesdays $\Rightarrow E = \{M \cdot T\} \Rightarrow n(E) = 1$

Now, $P(E) = \frac{n(E)}{n(S)} = \frac{1}{7}$.

2. Non leap year has 365 days.

7. 365(52

$$
\begin{array}{r}
35 \\
\hline
15 \\
14 \\
\hline
1
\end{array}
$$

365 days = 52 weeks +1 day

Let E be the event of having exactly 52 Mondays

$$E = \{S(\text{ or })T(\text{ or })W(\text{ or })Th(\text{ or })F(\text{ or })Sat\}$$
$$n(E) = 6 \Rightarrow P(E) = \frac{6}{7}$$

3. Let S be the event of rolling a die twice $\Rightarrow n(S) = 6^2 = 36$

   Let E be the event of getting a doublet. Then,

   $E = \{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\} \Rightarrow n(E) = 6.$

   Now, $P(E) = \frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}$.

4. Let S be the event of drawing two cards from a pack of 52 cards. Then, $n(S) = 52C_2$. Let E be the event of getting card such that one is black and other is red. Then, $n(E) = 26_{C_1} \cdot {}^{26}C_1$.

   Now, $P(E) = \frac{n(E)}{n(S)} = \frac{{}^{26}C_1 \cdot {}^{26}C_1}{{}^{52}C_2} = \frac{26\times26}{\frac{52\times51}{2}} = \frac{26}{51}$.

5. Let S be the event of drawing two balls from 10 balls containing ' 3 ' red balls.

   Then, $n(S) = 10C_2 = \frac{10.9}{1.2} = 45$.

   Let E be the event of drawing at least one red ball. Then,

$$n(E) = 3C_1.7C_1 + 3C_2 = 3.(7) + 3 = 24 \Rightarrow P(E) = \frac{n(E)}{n(S)} = \frac{24}{45} = \frac{8}{15}$$

6. Given, A, B are disjoint $\Rightarrow A \cap B = \emptyset \Rightarrow P(A \cap B) = 0$

   We have, $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{2}{7} + \frac{3}{5} - 0 = \frac{10+21}{35} = \frac{31}{35}$.

7. Let S be the event of drawing two cards from a well shuffled pack. Then,

   $n(S) = {}^{52}C_2 = \frac{52\times51}{1.2} = 26.\times 51.$

   Let E be the event of getting at least one of the two is a heart card. Then,

   $n(E) = {}^{13}C_1 \cdot {}^{39}C_1 + {}^{13}C_2 \Rightarrow 13.39 + \frac{13.12}{1.2} \Rightarrow 13(39+6) = 13(45)$

   $\Rightarrow P(E) = \frac{n(E)}{n(S)} = \frac{13.45}{26.51} = \frac{15}{34}$.

8. The total number of coins in the bag = 12 + 7 + 4 = 23

   Let S be the event of selecting ' 3 ' coins randomly $\Rightarrow n(S) = {}^{23}C_3$

   Let E be the event of the selecting 3 coins such that sum is maximum. Then,

   $n(E) = {}^{12}C_3$.

   $\therefore P(E) = \frac{n(E)}{n(S)} = \frac{{}^{12}C_3}{{}^{23}C_3}$.

9. Let S be the event of tossing n coins simultaneously. Then, $n(S) = 2^n$.

   Let E be the event of getting head an odd number of times. Then, $n(E) = \frac{2^n}{2}$.

   Now, $P(E) = \frac{n(E)}{n(S)} = \frac{2^n}{2^n}\left(\frac{1}{2}\right) = \frac{1}{2}$.

10. Let S be the event of selecting a single letter randomly from the word "PROB ABILITY". Then, $n(S) = 11$.

    Let E be the event of selecting a letter which is a vowel. Then, E $= \{A, O, I, I\}$

    Now, $P(E) = \frac{4}{11}$.

11. Let S be the event of throwing a die thrice. Then, $n(S) = 6^3 = 216$.

    Let E be the event of selecting the ' 3 ' numbers whose sum is 6 . Then, $n(E) = 10$.

    Now, $P(E) = \frac{10}{216} = \frac{10}{6^3} = \frac{a}{b^3} \Rightarrow$ a $= 10$, b $= 6 \Rightarrow$ a $-$ b $= 10 - 6 = 4$.

12. Let S be the event of selecting a child (boy (or) girl) out of 4-children. Then, $n(S) = 2^4 = 16$.

    Let E be the event of that at least one child is boy

    $\Rightarrow n(E) = {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 = 15$.

    Now, $P(E) = \frac{n(E)}{n(S)} = \frac{15}{16}$.

13. Let S be the event of arranging 7 persons in a row. Then, $n(S) = 7$!.

    Let E be the event of arranging persons in a row containing ' 3 ' persons A, B, C sit together in a particular order.

    Then, $n(E) = 5! \Rightarrow P(E) = \frac{n(E)}{n(S)} = \frac{5!}{7!}$.

14. Given digits are 1,2,3,4,5,6,7

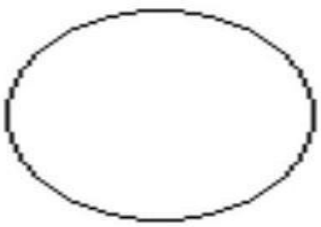
    Let S be the event of arranging 4-digited numbers $\Rightarrow n(S) = 7.6.5.4$.

    Let E be the event of getting a 4 - digited odd number

    $\Rightarrow n(E) = 6p_3 \times 4p_1 = (6.5.4) \times 4$.

    Now, $P(E) = \frac{n(E)}{n(S)} = \frac{(6 \cdot 5 \cdot 4) \cdot (4)}{(7 \cdot 6 \cdot 5 \cdot 4)} = \frac{4}{7}$.

15. Let S be the event of arranging ten boys in a circle.



    Then, $n(S) = (10 - 1)! = 9$!.

    If E is the event that two specific boys want to sit together, the number of arrangements = 8! 2!

$\therefore n(E) = 8!\,2\,!$. Now, $P(E) = \frac{n(E)}{n(S)} = \frac{8!2!}{9!} = \frac{2}{9} = \frac{k}{l} \Rightarrow$ k = 2,1 = 9 $\Rightarrow |k - l| = |2 - 9| = |-7| = 7$

16. Let S be the event of selecting of a card from a pack of 100 cards. Then, $n(S) = 100$. Let E be the event of drawing a perfect square. Then, $n(E) = 10$

$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{10}{100} = \frac{1}{10}$.

17. Let S be the event of selecting 5 tickets randomly from 50 tickets. Then, $n(S) = 50_{C_5}$.

But, given condition is that $x_1 < x_2 < x_3 < x_4 < x_5 (x_1, x_2, x_3, x_4, x_5$ are the drawn tickets) $x_1, x_2 \in \{1,2,3 \ldots\ldots..29\}; x_4, x_5 \in \{31,32, \ldots\ldots\ldots\ldots 50\}$, $x_3 = 30$

Let E be the event of that the tickets are drawn with the given condition.

$$n(E) = {}^{29}C_2 \times 20_{C_2} \times {}^{1}C_1 \Rightarrow \therefore P(E) = \frac{n(E)}{n(S)} = \frac{{}^{29}C_2 \times 20_{C_2}}{{}^{50}C_5}$$

18. Let S be the event of Selecting 3 balls randomly from the bag. Then, $n(S) = {}^{9}C_3 = 84$.

Let E be the event of selecting at least one white ball.

Then, $n(E) = 2C_1 \cdot {}^{7}C_2 + 2C_2 \cdot {}^{7}C_1 = 49 \Rightarrow \therefore P(E) = \frac{n(E)}{n(S)} = \frac{49}{84} = \frac{7}{12}$.

19. Let S be the event of selecting two numbers from 1 to 100 $\Rightarrow n(S) = 100^2$ Let E be the event of selecting the two numbers which are different $\Rightarrow n(E) = 100C_1 \times 99C_1$.

Now, $P(E) = \frac{{}^{100}C_1 \times {}^{99}C_1}{{}^{100}C_2} = \frac{100 \times 99}{{}^{100}C_2}$.

20. Let S be the event of selecting ' 3 ' numbers randomly from the first 20 natural numbers. Then, $n(S) = {}^{20}C_3$.

Let E be the event of selecting ' 3 ' numbers whose product is odd.

Then, $\bar{E}$ is the event of selecting of 3 numbers from even/odd whose product is even.

$$P(\bar{E}) = \frac{{}^{10}C_3}{{}^{20}C_3} \Rightarrow P(E) = 1 - P(\bar{E}) = 1 - \frac{{}^{10}C_3}{{}^{20}C_3} = 1 - \frac{{}^{p}C_3}{{}^{q}C_3}$$

$$p = 10, q = 20, \Rightarrow \frac{q}{p} = \frac{20}{10} = 10$$

21. Given $x \in [1,5]$

Let S be the sample space. Then, S = {1,2,3,4,5} and $n(S) = 5$.

Let E be the event of selecting ' x ' values satisfying $x^2 - 3x + 2 > 0$.

$\Rightarrow x^2 - 2x - x + 2 > 0 \Rightarrow x(x-2) - 1(x-2) > 0 \Rightarrow (x-1)(x-2) > 0$

$\therefore x < 1$ and $x > 2$

$\therefore E = \{3,4,5\} \Rightarrow p(E) = \frac{n(E)}{n(S)} = \frac{3}{5}$.

22. Given 2×2 matrix $\begin{bmatrix} - & - \\ - & - \end{bmatrix}_{2\times2}$

Let S be the event of selecting total number of 2×2 matrices with entries $\{0,1\} \Rightarrow n(S) = 8$

Let E be the event of selecting a 2×2 matrix whose determinant is 0 .

Then, $E = \left\{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right\} \Rightarrow n(E) = 5$

Now, $P(E) = \frac{n(E)}{n(S)} = \frac{5}{8}$.

23. Given that ' 2 ' cards are accidentally dropped from a pack of 52 playing cards. then, $n(S) = 52C_2$.

Let E be the event of missing cards are of different colours.

Then, $n(E) = 26_{C_1} \times 26_{C_1}$.

Now, $P(E) = \frac{n(E)}{n(S)} = \frac{26\times26}{\frac{52\times51}{2}} = \frac{26}{51}$.

24. A box conatins [5 white socks +5 blue socks] 10 socks

Let S be the event of selecting 2 socks randomly. Then, $n(S) = 10C_2$.

Let E be the event of both sock will match. Then, $n(E) = 5C_2 + 5C_2$.

Now, $P(E) = \frac{n(E)}{n(S)} = \frac{5C_2+5C_2}{10C_2} = \frac{\frac{5.4}{1.2}+\frac{5.4}{1.2}}{\frac{10\times9}{1.2}} = \frac{4}{9}$.

25. Given digits are 0,1,2,3,4

Let S be the event of forming a 5 digited number. Then, $n(S) = 4P_1 \times 4! = 4(4!)$.

Let E be the event that 5 -digited number contains ' 2 ' m ten's place and ' 0 ' in unit place. Then, $n(E) = 3! \times 1p_1 \times 1p_1 = 3$!.

Now, $P(E) = \frac{n(E)}{n(S)} == \frac{3!}{4!\cdot4} = \frac{3!}{3!4\cdot4} = \frac{1}{16} = \frac{1}{a^b} = \frac{1}{4^2}$.

$\Rightarrow a = 4, b = 2 \Rightarrow a + b = 4 + 2 = 6$.

26. Let S be the event of drawing two cards from a pack of cards. Then, $n(S) = {}^{52}C_2$. Let E be the event of drawing the cards such that one is a club and another is not a club. Then, $n(E) = {}^{13}C_1 \times {}^{39}C_1$.

Now, $P(E) = \frac{n(E)}{n(S)} = \frac{{}^{13}C_1 \cdot {}^{39}C_1}{{}^{52}C_2} = \frac{\frac{13.39}{52.51}}{1.4} = \frac{13}{2(17)} = \frac{13}{34}$.

27. A box contains 15 bulbs.

5 bulbs are detective $\Rightarrow$ 10 bulbs are non-detective

Let S be the event of selecting 5 bulbs. Then, $n(S) = 15C_5$.

I. Let E_1 be the event of selection contains none of them is defective. Then, $n(E_1) = 10C_5$.

Now, $P(E_1) = \frac{n(E_1)}{n(S)} = \frac{10C_5}{15C_5} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6}{14 \cdot 15 \cdot 13 \cdot 12.11} = \frac{4.3}{13.11} = \frac{12}{143}$.

II. Let E_2 be the event of selection containing only one of them is detective. Then, $n(E_2) = 5C_1 \times 10C_4$.

Now, $P(E_2) = \frac{n(E_2)}{n(S)} = \frac{5C_1 \times 10C_4}{15C_5} = \frac{5 \cdot \left(\frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1}\right)}{\frac{15 \cdot 14 \cdot 13 \cdot 12.11}{5 \cdot 4 \cdot 3 \cdot 2.1}} = \frac{5 \cdot 10}{13.11} = \frac{50}{143}$.

III. Let E_3 be the event of at least one of them is defective $= P(E_1^C)$.

$P(E_3) = 1 - P(E_1) = 1 - \frac{12}{143} = \frac{131}{143}$.

28. Let S be the event of tossing two unbiased dice. Then, $n(S) = 6^2 = 36$. Let E be the event of getting the sum on the faces as 8 . Then, E = $\{(2,6), (3,5), (4,4), (5,3), (6,2)\} \Rightarrow n(E) = 5$.

Now, $P(E) = \frac{n(E)}{n(S)} = \frac{5}{36}$.

Let F be the event of getting even numbers on both the dice. Then, F = $\{(2,2), (2,4), (2,6), (4,2), (4,4), (4,6), (6,2), (6,4), (6,6)\} \Rightarrow n(F) = 9$.

Now, $P(F) = \frac{n(P)}{n(S)} = \frac{9}{36} = \frac{1}{4}$.

a) Leap year has 366 days = 52 weeks +2 days

$S = \{SM, MT, TW, WTh, ThF, FSat, \text{Sat}S\} \Rightarrow n(S) = 7$

Let E be the event of getting 53 Sundays. Then, E = $\{SM$, SatS $\}$.

$\Rightarrow n(E) = 2$. Now, $P(E) = \frac{n(E)}{n(S)} = \frac{2}{7}$.

b) In a leap year, $n(S) = 7$

Let E be the event of getting exactly 52 Sundays. Then,

$E = \{MT, TW, WTh, ThF, FSat\} \Rightarrow n(E) = 5 \Rightarrow$ Now, $P(E) = \frac{n(E)}{n(S)} = \frac{5}{7}$.

c) A non- leap year has 365 days = 52 weeks +1 day.

Then, $S = \{S, M, T, W, Th, Fr, Sat\} \Rightarrow n(S) = 7$

Let E be the event of getting 53 Sundays on non-leap year.

Then, $n(E) = 1 \Rightarrow \therefore P(E) = \frac{n(E)}{n(S)} = \frac{1}{7}$.

d) Let E be the event of getting 52 Sundays exactly on non-leap year.

Then, $E = \{T, W, Th, Fr, Sat, S\} \Rightarrow n(E) = 6 \Rightarrow P(E) = \frac{6}{7}$.

30. Given, $P(A) = 0.5, P(B) = 0.4, P(A \cup B) = 0.6$

We have,

i) $P(\bar{A}) = 1 - P(A) = 0.5$

ii) $P(A \cap \bar{B}) = P(A - B) = P(A) - P(A \cap B) = 0.5 - 0.3 = 0.2$

iii)$P(\bar{A} \cap B) = P(B \cap \bar{A}) = P(B - A) = P(B) - P(A \cap B) = 0.4 - 0.3 = 0.1$

iv) $P(\bar{B}) = 1 - P(B) = 0.6$

31. As, A, B are disjoint sets, then $A \cap B = \emptyset$.

$\Rightarrow P(A \cap B) = 0 \Rightarrow P(A \cup B) = P(A) + P(B) - p(A \cap B) = \frac{2}{7} + \frac{3}{5} - 0 = \frac{10+21}{35} \frac{=31}{35}$.

32. $P(\bar{A}) - P(\bar{B}) = (1 - 0.65) - (1 - 0.75) = 0.1$

33. $P(A) = 0.4, P(A \cup B) = 0.7$ and A ∩ B = $\emptyset \Rightarrow P(A \cap B) = 0$

$\Rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow P(B) = P(A \cup B) + P(A \cap B) - P(A)$

$= 0.7 + 0 - 0.4 = 0.3.$

34. A bag contains $[4(R) + 3(W) + 5(b)]12$ balls

Let S be the event of drawing three balls randomly. Then, $n(S) = {}^{12}C_3$.

a) Let E_1 be the event of drawing 1 red, 1 white, 1 blue balls $\Rightarrow n(E_1) =$ ${}^4C_1 \times {}^3C_1 \times {}^5C_1$.

Now, $P(E_1) = \frac{4\times3\times5}{{}^{12}C_3} = \frac{30}{110} = \frac{3}{11}$

b) Let E_2 be the event of getting ' 2 ' white and 1 blue balls.

Then $n(E_2) = {}^3C_2 \cdot {}^5C_1 = {}^3C_1 \cdot {}^5C_1$.

Now, $P(E_2) = \frac{n(E_2)}{n(S)} = \frac{3\times5}{\frac{12.11.10}{1.2.3}} = \frac{3}{44}$.

c) Let E_3 be the event of getting 2 red &1 white ball. Then, $n(E_3) = {}^4C_2 \times {}^3C_1$.

Now, $P(E_3) = \frac{n(E_3)}{n(S)} = \frac{{}^4C_2 \times {}^3C_1}{{}^{12}C_3} \Rightarrow \frac{\frac{4.3}{1.2}\times3}{\frac{12.11.10}{3.21}} = \frac{9}{110}$.

d) Let E_4 be the event of selecting none of the balls is white. Then, $n(E_4) = 9C_3$.

Now, $P(E_4) = \frac{9C_3}{12C_3} = \frac{\frac{9\cdot8\cdot7}{3\cdot2\cdot1}}{\frac{12\cdot11\cdot10}{3\cdot2\cdot1}} = \frac{9\cdot8\cdot7}{12\cdot11\cdot10} = \frac{21}{55}$.

WORKSHEET – 2
Level – I

1) A 2) B 3) D 4) B 5) D 6) A 7) A 8) D

1. Given $p(A) = \frac{3}{8}, P(B) = \frac{1}{3}, P(A \cap B) = \frac{1}{4}$

we have $P(A \cup B) = P(A) + P(B) - \big((A \cap B) = \frac{3}{8} + \frac{1}{3} - \frac{1}{4}$

$P(A \cup B) = \frac{11}{24}$

2. Let S be the event of selecting one ticket randomly out of 30 tickets num bered from 1 to 30.

   $\therefore n(s) = {}^{30}C_1 = 30.$

   Let E_1 be the event of getting a ticket numbered multiple of 5.

   Then, $E_1 = \{5,10,15,20,25,30\} \Rightarrow n(E_1) = 6.$

   Let E_2 be the event of getting a ticket numbered multiple of 7.

   $E_2 = \{7,14,21,28\} \Rightarrow n(E_2) = 4 \Rightarrow P(E_1 \cap E_2) = 0$

   we have $p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2) = \frac{6}{30} + \frac{4}{30} - 0 = \frac{10}{30} = \frac{1}{3}$

3. Given A, B, C are mutually exclusive and exhaustive events

   $\Rightarrow A \cap B = \phi \Rightarrow p(A \cap B) = 0 \Rightarrow B \cap C = \phi \Rightarrow p(B \cap C) = 0$

   $C \cap A = \phi \Rightarrow p(C \cap A) = 0 \Rightarrow A \cap B \cap C = \phi \Rightarrow p(A \cap B \cap C) = 0$

   $p(A \cup B \cup C) = p(A) + p(B) + p(C) \Rightarrow 1 = 11 \cdot \frac{p(A)}{6} \Rightarrow p(A) = \frac{6}{11}$

4. Let S be the event of rolling two dice simultaneously $\Rightarrow n(S) = 6^2 = 36.$

   Let E_1 be the event of getting even numbers on both the faces.

   Then, $E_1 = \{(2,2),(2,4),(2,6),(4,2),(4,4),(4,6),(6,2),(6,4),(6,6)\}$

   $\Rightarrow n(E_1) = 9$

   Let E_2 be the event of getting the sum on the faces be 10 .

   Then, $E_2 = \{(4,6),(5,5),(6,4)\} \Rightarrow n(E_2) = 3.$

   Now, $E_1 \cap E_2 = \{(4,6)(6,4)\} \Rightarrow n(E_1 \cap E_2) = 2.$

   Now $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) = \frac{9}{36} + \frac{3}{36} - \frac{2}{36} = \frac{10}{36}$

   $\Rightarrow P(E_1 \cup E_2) = \frac{5}{18}$

5. Let S be the event of rolling two fair dice.

   Then, $n(S) = 6^2 = 36.$

   Let E be the event of getting the sum of digits on their faces to be greater than (or) equal to 10 .

   Then, $E = \{(4,6)(5,5)(5,6)(6,4)(6,5)(6,6)\} \Rightarrow n(E) = 6.$

   Then, $P(E) = \frac{n(E)}{n(S)} = \frac{6}{36} \Rightarrow P(E) = \frac{1}{6}.$

6. $P(A) = 0.4, P(B) = 0.7, A \cap B = \phi \Rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B)$

   $0.7 = 0.4 + P(B) - 0 \Rightarrow P(B) = 0.7 - 0.4 \Rightarrow P(B) = 0.3$

7. A bag contains $[4(g) + 6(b) + 7(w)] = 17$ balls

   Let S be the event of drawing a ball randomly $\Rightarrow n(s) = {}^{17}C_1 = 17.$

   Let E_1 be the event of getting a black ball $\Rightarrow n(E_1) = {}^{6}C_1.$

Let E_2 be the event of getting a white ball $\Rightarrow n(E_2) = 7$.

$E_1 \cap E_2 = \phi \Rightarrow n(E_1 \cap E_2) = 0$.

$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) = \frac{6+7-0}{17} \Rightarrow P(E_1 \cup E_2) = \frac{13}{17}$.

8. $P(B) = \frac{4}{5}, P(A \cap B) = \frac{3}{10}$

Now $P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{3}{10}}{\frac{4}{5}} = \frac{3}{10} \times \frac{5}{4} \Rightarrow P(A/B) = \frac{3}{8} = \frac{3}{2k}$

$\Rightarrow 8 = 2k \Rightarrow k = 4$

Level – II

1) D	2) D	3) A	4) A	5) A	6)C	7)12
8) D	9) B	10)C	11)C	12)B	13)D	14)C
15)A	16)B	17)B	18)C	19)B	20)D	21)C
22)C	23)A,C	24) C	25)C	26)C	27)A	
28) a-q, b-r,c-s, d-r		29) D	30) B,C	31)C	32)A	33)A
34)A,D	35)a-s,b-q, c-r, d-p					

1. ASSISTANT word contains $(A^1s - 2,\ S^1s - 3, T^1 - 2, I^1s - 1, N^1s - 1)$9 letters.

STATISTICS word contains $[A^1s - 1, S^1s - 3,\ T^1s - 3, I^1s - 2, C^1s - 1]$10 letters.

Probability of getting same letters $= P(A) + P(S) + P(T) + P(I)$

$= \frac{2}{9} \times \frac{1}{10} + \frac{3}{9} \times \frac{3}{10} + \frac{2}{9} \times \frac{3}{10} + \frac{1}{9} \times \frac{2}{10} = \frac{2+9+6+2}{90} = \frac{19}{90}$.

2. Let S be the event of selecting a number randomly from 1 to 90.

Then, $n(S) = {}^{90}C_1 = 90$.

Let E_1 be the event of selecting numbers that are divisible by 6.

Then, $E_1 = \{6,12,18, \ldots..90\} \Rightarrow n(E_1) = 15$

Let E_2 be the event of numbers that are divisible by 8 . Then, $E_2 =$ $\{8,16,24, \ldots..,88\}$.

$\Rightarrow n(E_2) = 11$ and $E_1 \cap E_2 = \{24,48,72\} \Rightarrow n(E_1 \cap E_2) = 3$

$\therefore P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) = \frac{15}{90} + \frac{11}{90} - \frac{3}{90}$

$\Rightarrow P(E_1 \cup E_2) = \frac{23}{90}$.

3. Three groups of childrens

$(3G + 1B) + (2G + 2B) + (1G + 3B) = 12$

Let E be the event of selecting one child from each group containing 1 girl & 2 boys

Then, $E = \{GBB, BGB, BBG\}$

$P(E) = \frac{3}{2} \cdot \frac{2}{4} \cdot \frac{3}{4} + \frac{1}{4} \cdot \frac{2}{4} \cdot \frac{3}{4} + \frac{1}{4} \cdot \frac{2}{4} \cdot \frac{1}{4} = \frac{18+6+2}{64} = \frac{26}{64} \Rightarrow P(E) = \frac{13}{32}$.

4. $P(E) = \frac{40}{100} = 0.4\ P(J) = 25\% = \frac{25}{100} = 0.25$

   $P(E \cap J) = 15\% = \frac{15}{100} = 0.15$

   The probability that the person selected read Eanadu but not jyothi is $P(E - J) = P(E) - P(E \cap J) = \frac{40}{100} - \frac{15}{100} = \frac{25}{100} = \frac{1}{4} = \frac{40}{100} - \frac{15}{100} = \frac{25}{100} = \frac{1}{4}$

5. Let A be the event that the student passed in maths $\Rightarrow P(A) = \frac{70}{125}$

   Let B be the event that the student passed in statistics $\Rightarrow P(B) = \frac{55}{125}$; and

   $P(A \cap B) = \frac{30}{125}$.

   The probability that a student selected at random has passed in only one subject

   $P(A - B) + P(B - A) = P(A) - P(A \cap B) + P(B) - P(A \cap B)$

   $= P(A) + P(B) - 2P(A \cap B) = \frac{70}{125} + \frac{55}{125} - 2\left(\frac{30}{125}\right) = \frac{65}{125} = \frac{13}{25}$.

6. $P(A) = 0.4, P(B) = 0.5, P(C) = 0.6$

   $P(A \cap B) = 0.2, P(B \cap C) = 0.3, P(C \cap A) = 0.25$

   $P(A \cap B \cap C) = 0.1 \Rightarrow P(A \cup B \cup C) = 0.85$

7. $P(A) = \frac{2}{5}, P(B \cap A) = \frac{3}{10} \Rightarrow P(B/A) = \frac{P(B \cap A)}{P(A)} = \frac{3}{10} \times \frac{5}{2} = \frac{3}{4}. = \frac{k}{l}$

   k = 3,1 = 4 product $k \times 1 = 3 \times 4 = 12$

8. A,B are mutually exclusive events $\Rightarrow A \cap B = \phi$

   $P(A/\bar{B}) = \frac{P(A \cap \bar{B})}{P(\bar{B})} = \frac{P(A - B)}{P(\bar{B})} = \frac{P(A) - P(A \cap B)}{1 - P(B)} \Rightarrow P(A/\bar{B}) = \frac{P(A)}{1 - P(B)}$

9. Given digits are 1,2,3,4 ……..9

   Let A be the event that the sum of the numbers is an even number and B be the event that the two numbers are odd

   i, e = A = $\left.\begin{matrix}(1,3),(1,5)(1,7),(1,9),(3,5),(3,7),(3,9),(5,7)\\(5,9),(7,9)(2,4)(2,6)(2,8),(4,6),(4,8)(6,8)\end{matrix}\right\} \Rightarrow n(A) = 16$

   $A \cap B = \{(1,3)(1,5)(1,7)(1,9)(3,5)(3,7)(3,9)(5,7)(5,9),(7,9)\}$

   $P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{10}{16} = \frac{5}{8} \Rightarrow P(B/A) = \frac{5}{8}$

10. Let E_1 be the event that one of the selected mangoes is good and E_2 be the event that the other mango is also good.

    $P(E_1) = 1 - \frac{{}^4C_2}{{}^{10}C_2} = 1 - \frac{2}{15} = \frac{13}{15} \Rightarrow P(E_1 \cap E_2) = \frac{{}^6C_2}{{}^{10}C_2} = \frac{6.5}{10.9} = \frac{1}{3}$

    Now $P(E_2/E_1) = \frac{P(E_2 \cap E_1)}{P(E_1)} = \left(\frac{1}{3}\right) \times \frac{15}{13} \Rightarrow P(E_2/E_1) = \frac{5}{13}$

11. Let A be the event that selected ticket has a number whose sum of the digits = 7 Then, $A = \{07,16,25,34,43,52,61,70\}$

Let B be the event that selected ticket has number whose product of digits is

Then, $B = \{00,01,02,03,04,05,06,07,08,09,10,20,30,40,50,60,70,80,90\}$.

Now, $A \cap B = \{07,70\} \Rightarrow n(A \cap B) = 2 \Rightarrow P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{2}{19} = \frac{a}{b}$

a = 2, b = 9 now, $\frac{a+b}{7} = \frac{2+19}{7} = \frac{21}{7} = 3$.

12. Let A, B, C be the events such that

$P(A) = \frac{1+3p}{3}, P(B) = \frac{1-p}{4}, \mathrm{P}(C) = \frac{1-2p}{2}$

But, we know that

$0 \le P(A) \le 1 \Rightarrow 0 \le \frac{1+3p}{3} \le 1 \Rightarrow -\frac{1}{3} \le p \le \frac{2}{3} \rightarrow p \in \left[-\frac{1}{3}, \frac{2}{3}\right]$

$0 \le P(B) \le 1 \Rightarrow 0 \le \frac{1-p}{4} \le 1 \Rightarrow -3 \le p \le 1 \Rightarrow p \in [-3,1]$

$0 \le P(C) \le 1 \Rightarrow 0 \le \frac{1-2p}{2} \le 1 \Rightarrow -\frac{1}{2} \le p \le \frac{1}{2} \Rightarrow p \in \left[-\frac{1}{2}, \frac{1}{2}\right]$

Since A,B,C are mutually exclusive events

$A \cap B = B \cap C = C \cap A = A \cap B \cap C = \phi$

Now $0 \le P(A \cup B \cup C) \le 1 \Rightarrow 0 \le P(A) + P(B) + P(C) \le 1$

$0 \le \frac{1+3p}{3} + \frac{1-p}{4} + \frac{1-2p}{2} \le 1 \Rightarrow 0 \le 13 - 3p \le 12 \Rightarrow \frac{1}{3} \le p \le \frac{13}{3}$

$p \in \left[\frac{1}{3}, \frac{13}{3}\right] \ldots\ldots (4) \Rightarrow$ From 1,2,3,4 we get $\Rightarrow p \in \left[\frac{1}{3}, \frac{1}{2}\right]$

13. Give A,B,C are participating in a competition.

such that $P(A) = 2.P(B); 2P(C) = P(B)$

If $P(C) = x$ then $P(B) = 2x \Rightarrow P(A) = 2 \cdot P(B) = 2 \cdot (2x) = 4x$

A,B,C are mutually exclusive

$P(A \cap B) = P(B \cap C) = P(C \cap A) = P(A \cap B \cap C) = 0$ and $P(A \cup B \cup C) = 1$

$P(A) + P(B) + P(C) = 1$

$4x + 2x + x = 1$

$\Rightarrow 7x = 1 \Rightarrow x = \frac{1}{7} \Rightarrow P(A) = 4x = \frac{4}{7} \Rightarrow P(\bar{A}) = 1 - P(A) = 1 - \frac{4}{7} = \frac{3}{7}$.

14. Given $P(E_1) = \frac{1}{4}, P(E_2/E_1) = \frac{1}{2}, P(E_1/E_2) = \frac{1}{4}$

Now,

$$= \frac{1}{4} + \frac{P(E_1 - E_2)}{1 - P(E_2)} = \frac{1}{4} + \frac{P(E_1) - P(E_1 \cap E_2)}{1 - P(E_2)} = \frac{1}{4} + \frac{\left(\frac{1}{4}\right) - P(E_2)\left(P(E_1/E_2)\right)}{1 - P(E_2)}$$

$$= \frac{1}{4} + \frac{\frac{1}{4}\left(1 - P(E_2)\right)}{\left(1 - P(E_2)\right)} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

15. Three horses A, B, C runs in a race, $\Rightarrow A \cap B = B \cap C = C \cap A = A \cap B \cap C = \phi$

Given that, $P(A) = 2 \cdot P(B); P(B) = 3 \cdot P(C)$

Let, $P(C) = x \Rightarrow P(B) = 3x \Rightarrow P(A) = 2P(B) = 2.(3x)6x$

Now,

$P(A \cup B \cup C) = 1 \Rightarrow P(A) + P(B) + P(C) = 1 \Rightarrow 6x + 3x + x = 1$

$10x = 1 \Rightarrow x = \frac{1}{10} \Rightarrow P(C) = \frac{1}{10} \Rightarrow P(B) = 3x = \frac{3}{10} \Rightarrow P(A) = 6x = \frac{6}{10}$

$P(A), P(B), P(C)$ winning chances are $\frac{6}{10}, \frac{3}{10}, \frac{1}{10}$

16. Let A, B, C be the events that the problem being solved by three students respectively $\Rightarrow A, B, C$ are independent

Now, we have

$P(A \cup B \cup C) = 1 - P(\overline{A \cup B \cup C}) = 1 - P(A^c)P(B^c)P(C^c)$

But $P(A) = \frac{1}{3} \Rightarrow P(A^c) = 1 - P(A) = 1 - \frac{1}{3} = \frac{2}{3}$

$P(B) = \frac{1}{4} \Rightarrow P(B^c) = 1 - P(B) = 1 - \frac{1}{4} = \frac{3}{4}$

$P(C) = \frac{1}{5} \Rightarrow P(C^c) = 1 - P(C) = 1 - \frac{1}{5} = \frac{4}{5}$

i,e $P(A \cup B \cup C) = 1 - P(A^c) \cdot (B^c) \cdot (C^c) = 1 - \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5} = 1 - \frac{2}{5}$

$P(A \cup B \cup C) = \frac{3}{5} = \frac{36}{60}$.

17. Given, $P(A) = 0.5$

$P(A \cap B) \leq 0.3$. Given, A, B are independent. Then, $P(A \cap B) = P(A) \cdot P(B)$.

$\Rightarrow P(A) \cdot P(B) \leq 0.3 \Rightarrow 0 \leq 0.5 \cdot P(P(B)) \leq 0.3$

$\Rightarrow 0 \leq \frac{5}{10}.P(B) \leq \frac{3}{10} \Rightarrow \frac{0}{5} \leq P(B) \leq \frac{3}{5} \Rightarrow 0 \leq p(B) \leq 0.6.$

18. Let $S = \{MM, MF, FM, FF\}$

Let E_1 be the event that atleast one of them is a male

$E_1 = \{MM, MF, FM\}$

Let E_2 be the event that both are male

$E_2 = \{M, M\} \Rightarrow E_1 \cap E_2 = \{M, M\}$; Now, $P(E_2/E_1) = \frac{P(E_2 \cap E_1)}{P(E_1)} = \frac{1}{3} = \frac{a}{b}$

a = 1, b = 3, now $a^b \times b^a = 1^3 \times 3^1 = 3$.

19. Given that A, B, C, D, E are contesting in an election .

    Since, out of ' 3 ' one [C] is selected unanimously.

    Now, from remainig $4[A, B, D, E]$, we have to select ' 2 ' persons.

    $\therefore$ Probability of selecting ' D ' $= \frac{1}{{}^4C_2} = \frac{1}{6}$.

20. Given two dices are rolled

    Let E_1 be the event that the sum of the numbers on them is ' 6 '

    $E_1 = \{(1,5), (2,4)(3,3), (4,2), (5,1)\}$

    Let E_2 be the event of getting number '4' on any one of the dice.

    $E_2 = \{(2,4)(4,2)\} \Rightarrow E_1 \cap E_2 = \{(2,4), (4,2)\}$

    Now, $P(E_2/E_1) = \frac{P(E_2 \cap E_1)}{P(E_1)} = \frac{2}{5}$

21. Given $P(\bar{A}) = 0.2 \Rightarrow P(A) = 1 - P(\bar{A}) = 1 - 0.2 = 0.8$

    $P(\bar{B}) = 0.3 \Rightarrow P(B) = 1 - P(\bar{B}) = 1 - 0.3 = 0.7$

    $P(\bar{A} \cup \bar{B}) = P\overline{(A \cap B)} = 1 - P(A \cap B) = 1 - P(A) \cdot P(B) = 1 - 0.56 = 0.44$

22. $P(E_1)$ = probability of getting a head $= \frac{1}{2}$

    $P(E_2)$ = probability of getting an odd number $= \frac{3}{6} = \frac{1}{2}$

    Now ,

    The probability of getting a head on the coin and odd number on the die is

    $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

23. A bag containing $7[4 - \text{Re}\, d + 3 -$ black] balls

    Let E_1 be the event of drawing a black ball. $\Rightarrow P(E_1) = \frac{3}{7}$

    Let E_2 be the event of drawing a black ball

    [where the first ball is replaced again] $\Rightarrow P(E_2) = \frac{3}{7}$

    Now, $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2) = \left(\frac{3}{7}\right) \cdot \left(\frac{3}{7}\right) \Rightarrow P(E_1 \cap E_2) = \frac{9}{49}$.

24. $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2/E_1) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8} \cdot \Rightarrow P(E_2) = \frac{P(E_1 \cap E_2)}{P(E_1/E_2)} = \frac{1/8}{1/2} = \frac{1}{4}$.

25. $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) = \frac{1}{4} + \frac{1}{2} - \frac{1}{8} = \frac{5}{8}$.

26. $P(\bar{E}_1/\bar{E}_2) = \frac{P(\bar{E}_1 \cap \bar{E}_2)}{P(\bar{E}_2)} = \frac{1 - P(E_1 \cup E_2)}{1 - P(E_2)} = \frac{1 - (5/8)}{1 - (1/2)} = \frac{3}{4}$.

27. As per Statement- I, the first drawn ball is red. It can be done in ${}^8C_1$ ways.

    We can select the second ball either red or green in $7_{C_1}, 10_{C_1}$ ways respectively.

    But our condition is that second ball drawn is green when it is known that first ball drawn is red. Let this event be E_1.

Then, $P(E_1) = \frac{{}^{8}C_1 \cdot {}^{10}C_1}{8_{C_1} \cdot {}^{7}C_1 + 8_{C_1} \cdot {}^{10}C_{C_1}} = \frac{10}{17}$.

Similarly, as per Statement-II, $P(E_2) = \frac{8}{17}$.

28. a) Give $P(A) = \frac{1}{3}, P(B) = \frac{1}{4}$

(i) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A) \cdot P(B)$

$= \frac{1}{3} + \frac{1}{4} - \frac{1}{(3)(4)} = P(A \cup B) = \frac{1}{2}$.

[$\therefore A, B$ are independent]

b) $P(A/A \cup B) = \frac{P(A \cap (A \cup B))}{P(A \cup B)} = \frac{P((A \cap A) \cup (A \cap B))}{P(A \cup B)} = \frac{P(A)}{P(A \cup B)} = \frac{2}{3}$.

c) $P(B/A^c \cap B^c) = P(B/(A \cup B)^c) = \frac{P(B \cap (A \cup B)^c)}{P(A \cup B)^c} = \frac{0}{P(A \cup B)^c} = 0$.

d) $P(A^c/B) = \frac{P(A^c \cap B)}{P(B)} = P(A^c) = 1 - P(A) = 1 - \frac{1}{3} = \frac{2}{3}$.

(A is independent, A^1 is also independent)

29. As, A,B are mutually exclusive and exhaustive events of a random experiment, $P(A) + P(B) = 1 \Rightarrow P(B) = \frac{1}{4}$[$\because P(A) = 2P(B)$]; Therefore, k = 4.

30. $P(E_1) = 1 - P(\overline{E_1}) = \frac{3}{4}; P(E_2) = 1 - P(\overline{E_2}) = \frac{1}{3}$

$P(E_1 \cap E_2) = P(E_1) + P(E_2) - P(E_1 \cup E_2) = \frac{1}{4} = P(E_1)P(E_2)$.

Therefore, E_1, E_2 are independent events.

Similarly we can prove that, $\overline{E_1}, \overline{E_2}$ are independent events.

Given three horses A, B and C run a race and $P(A) = 2 \cdot P(B); P(B) = 3 \cdot P(C)$

A,B,C are mutually exclusive and exhaustive events.

$\therefore A \cap B \cap C = \phi$ and $A \cup B \cup C = S \Rightarrow P(A \cap B \cap C) = 0 \& P(A \cup B \cup C) = 1$

$\Rightarrow P(A) + P(B) + P(C) = 1 \Rightarrow 6x + 3x + x = 1$[$\because P(C) = x$] $\Rightarrow x = \frac{1}{10}$

31. Probability for A to win the race $= 6x = \frac{6}{10} = \frac{3}{5}$.

32. Probability for B to win the race $= 3x = \frac{3}{10}$.

33. Probability for C to win the race $= x = \frac{1}{10}$.

34. $P(E_2) = \frac{4}{5}, P(E_1 \cap E_2) = \frac{3}{10}$

Now $P(E_1/E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{\left(\frac{3}{10}\right)}{\left(\frac{4}{5}\right)} = \frac{3}{10} \times \frac{5}{4} = \frac{3}{8}$.

35. $a \to s; b \to q; c \to r; d \to p$

a) Let E be the event of getting sum of the digits on the dice is six or less. and

S be the sample space. Then, $n(S) = 216; n(E) = 20 \Rightarrow P(E) = \frac{20}{216} = \frac{5}{54}$

b) $P(\overline{A \cup B}) = 1 - P(A \cup B) = 0.5$

c) Let 'A' be the event of getting a Spade card and ' B ' be the event of getting a King card.

$n(A \cup B) = n(A) + n(B) - n(A \cap B) = 13 + 4 - 1 = 16.$

$n(S) = 52. \ \therefore P(E) = \frac{n(E)}{n(S)} = \frac{16}{52} = \frac{4}{13}.$

d) As, A and B are mutually exclusive events and

$A \cup B = S, P(A) + P(B) = 1 \Rightarrow P(B) = \frac{3}{4} \& P(A) = \frac{1}{4}.$

WORKSHEET – 3

Level – I

1) B 2) C 3) A 4) B 5) C 6) A

1. $P(A) = P(A_1) \cdot P(A/A_1) + P(A_2) \cdot P(A/A_2) = \frac{1}{3} \times \frac{3}{7} + \frac{2}{3} \times \frac{4}{7} = \frac{11}{21}.$
2. $P(E) = P(A) \cdot P(E/A) + P(B) \cdot P(E/B) + P(C) \cdot P(E/C) \Rightarrow \frac{1}{8} + \frac{1}{16} + \frac{3}{128} = \frac{27}{128}.$
3. $P(A) = P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2)$

   $\frac{7}{2} = \frac{1}{n+1} \times 1 + \frac{n}{n+1} \times \frac{1}{2} \Rightarrow \frac{7}{2} = \frac{2+n}{2(n+1)} \Rightarrow 12 + 6n = 7n + 7 \Rightarrow n = 5.$
4. $P(E) = P(F) \cdot P(E/F) + P(\bar{F}) \cdot P(E/\bar{F}) = 0.10 \times 0 + 0.90 \times 0.99 = 0.891.$
5. $P(E_1/A) = \frac{P(E_1) \cdot P(A/E_1)}{\Sigma P(E_i) \cdot P(A/E_i)} = \frac{\frac{1}{6} \times \frac{1}{100}}{\frac{1}{6} \times \frac{1}{100} + \frac{1}{3} \times \frac{3}{100} + \frac{1}{2} \times \frac{15}{100}} = \frac{1}{52}.$
6. $P(A_2/E) = \frac{P(A_2) \cdot P(E/A_2)}{P(A_1) \cdot P(E/A_1) + P(A_2) \cdot P(E/A_2)} = \frac{\frac{1}{10} \times \frac{1}{4}}{\frac{9}{10} \times 1 + \frac{1}{10} \times \frac{1}{4}} = \frac{1}{37}.$

Level – II

1) B	2) D	3) C	4) D	5) B	6) B	7) B
8) A	9) D	10)B	11)C	12) C	13) A	14) C
15) B	16) C	17) D	18)10	19) C	20) A,B,D	21) A
22) D	23) a-q, b-r,c-s,d-p		24) C	25. 18	26. a-q; b-p; c-r; d-q	
27. A,B,C	28. A,C	29. a-q,r; b-r; c-p; d-p,q,r,s				

JEE MAIN & ADV

1. Given

	WHITE	BLACK
BAG – A	4	3
BAG – B	5	2

E = Event of selecting a ball from each bag.

E_1, E_2 = selection of two balls of different colours

$P(E_1) = \frac{4}{7} \cdot \frac{2}{7}$, $P(E_2) = \frac{3}{7} \cdot \frac{5}{7}$

∴ Probability $P(E) = P(E_1) + P(E_2) = \frac{6}{49} + \frac{15}{49} = P(E) = \frac{23}{49}$

2.

	25 ps coins	1 rupee coins
Compartment-I	3	2
Compartment-II	2	3

$P(E_1)$ = P(Selecting compartment I) = $\frac{1}{2}$

$P(E_2)$ = P(Selecting compartment II) = $\frac{1}{2}$

E be the event of selecting 1 rupee coin.

$P(E) = P(E_1) \cdot \mathrm{P}(E/E_1) + P(E_2) \cdot \mathrm{P}(E/\mathrm{E}_2) = \frac{1}{2} \times \frac{2}{5} + \frac{1}{2} \times \frac{3}{5} = \frac{2}{10} + \frac{3}{10} = \frac{1}{2}$

3.

	RED	**BLACK**
BAG-A	5	3=8
BAG-B	4	5=9

Probability of selecting a bag is

$$P(A) = \frac{1}{2}\ ,\ P(B) = \frac{1}{2}$$

E = Event of drawing a two balls of different colours

$P(E/A) = \frac{{}^5C_1\ {}^3C_1}{{}^8C_2} = \frac{15 \times 2}{8 \times 7} = \frac{15}{28}$

$P(E/B) = \frac{4c_1 \cdot 5}{9c_2} = \frac{20}{\frac{9.8}{1.2}} = \frac{5}{9}$

$P(E) = P(A) \cdot P(E/A) + P(B) \cdot P(E/B) = \frac{1}{2} \cdot \frac{15}{28} + \frac{1}{2} \cdot \frac{5}{9} = \frac{275}{504}$.

4. Given words are "RANGE" "PAGE" common letters are A,G,E.

Let E be the event of selecting a common letter is

$P(E) = \frac{1}{5} \cdot \frac{1}{4} + \frac{1}{5} \cdot \frac{1}{4} + \frac{1}{5} \cdot \frac{1}{4} = \frac{1}{20} + \frac{1}{20} + \frac{1}{20} \Rightarrow P(E) = \frac{3}{20}$.

5.

Given	**White**	**Black**
Bag A	4	3 = 7
Bag B	5	2 = 7

Probability of selecting a bag

$P(A) = P(B) = \frac{1}{2}$

E = Event of drawing two balls of same colour

E_1, E_2 are events of selecting two balls of same colour

$P(E_1) = \frac{{}^4C_1}{{}^7C_1} \cdot \frac{{}^5C_1}{{}^7C_1}$; $P(E_2) = \frac{{}^3C_1}{{}^7C_1} \cdot \frac{{}^2C_1}{{}^7C_1}$

$P(E) = P(E_1) + P(E_2) = \frac{4}{7} \cdot \frac{5}{7} + \frac{3}{7} \cdot \frac{2}{7} = \frac{26}{49}$.

6. Let B_1, B_2, B_3 are ' 3 ' boxes

Probability of selection of boxes

$P(B_1) = P(B_2) = P(B_3) = \frac{1}{3}$

Let ' E ' be the event of getting white ball

Now

$P(E/B_1) = \frac{1}{6}$, $P(E/B_2) = \frac{2}{4}$, $P(E/B_3) = \frac{4}{12}$

Now $P(B_2/E) = \frac{\frac{1}{3} \times \frac{1}{2}}{\frac{1}{3}.\frac{1}{6} + \frac{1}{3}.\frac{1}{2} + \frac{1}{3}.\frac{4}{12}} = \frac{\frac{1}{2}}{\frac{1}{6} + \frac{1}{2} + \frac{1}{3}} = \frac{1}{2}$

7.

BK	Red (k-ball)	white $(\mathbf{k}+1)$ balls
K = 1 B_1	1	2
K = 2 B_2	2	3
K = 3 B_3	3	4

Given the probability of selection of bags

$P(B_1) = \frac{1}{2}, P(B_2) = \frac{1}{3}, P(B_3) = \frac{1}{6}$

Let ' E ' be the event of selecting a Red ball

$P(E/B_1) = \frac{1}{3}, P(E/B_2) = \frac{2}{5}, P(E/B_3) = \frac{3}{7}$

Now

$$P(B_2/E) = \frac{\frac{1}{3} \cdot \frac{2}{5}}{\frac{1}{2} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{2}{5} + \frac{1}{6} \cdot \frac{3}{7}} \Rightarrow \frac{\frac{1}{3} \cdot \frac{2}{5}}{\frac{1}{3}\left[\frac{1}{2} + \frac{2}{5} + \frac{3}{14}\right]} \Rightarrow P(B_2/E) = \frac{14}{39}$$

8. In a chess board, there are $(8 \times 8)64$ squares [32-white and 32 -black] Three square can be choosen as

(i) 2- white -1 black = $\frac{{}^{32}C_2 \cdot {}^{32}C_1}{{}^{64}C_3}$

(ii) 1 - white -2 black $= \frac{{}^{32}C_1 \cdot {}^{32}C_2}{{}^{64}C_3}$

Now ,

The probability $= \frac{{}^{32}C_2 \cdot {}^{32}C_1 + {}^{32}C_1 \cdot {}^{32}C_2}{{}^{64}C_3} = \frac{2 \cdot ({}^{32}C_2 \cdot {}^{32}C_1)}{{}^{64}C_3}$

$= 2.\left(\frac{32 \times 31}{2.1}\right) \times \frac{32}{1} \times \frac{3.2.1}{64.63.62} = \frac{16}{21}$

9. $P(A) = \frac{40}{100}, \mathrm{P}(B) = \frac{30}{100}, \mathrm{P}(A \cap B) = \frac{20}{100}$

$P[(A - B) \cup (B - A)] = P(A - B) + P(B - A)$

$P(A) + P(B) - 2 \cdot \mathrm{P}(A \cap B)$

$= \frac{40}{100} + \frac{30}{100} - 2 \cdot \left(\frac{20}{100}\right) = \frac{40}{100} + \frac{30}{100} - \frac{40}{100} = \frac{30}{100} = \frac{3}{10}$

10.

	Black	Red
Box-A	2	3
Box-B	3	4

One box is selected randomly

$P(A) = 2P(B)$

Let $P(B) = x \Rightarrow P(A) = 2x$

E = Event of selecting a Red ball

$P(E/\mathrm{A}) = \frac{3}{5},\ P(E/B) = \frac{4}{7}$

Probability of selecting a red ball from box B is

$P(B/E) = \frac{x \cdot \frac{4}{7}}{2x \cdot \frac{3}{5} + x \cdot \frac{4}{7}} = \frac{\frac{4x}{7}}{x\left[\frac{6}{5} + \frac{4}{7}\right]} = \frac{\frac{4}{7}}{\frac{42+20}{5.7}} = \frac{4.5}{62} = \frac{10}{31} \Rightarrow P(B/E) = \frac{10}{31}.$

11. Let B_1, B_2, B_3 be the events of containing 2,3,4 white balls respectively

$$P(B_1) = P(B_2) = P(B_3) = \frac{1}{3}[\text{ Selection of bags }]$$

Let ' E ' be the event of selecting two white balls

$$P(E/B_1) = \frac{{}^2C_2}{{}^4C_2} = \frac{1}{6} \Rightarrow P(E/B_2) = \frac{{}^3C_2}{{}^4C_2} = \frac{3.2}{4.3} = \frac{1}{2} \Rightarrow P(E/B_3) = \frac{{}^4C_2}{{}^4C_2} = 1$$

The probability that all the balls in the bag are white is

$$P(B_3/E) = \frac{1 \cdot \frac{1}{3}}{\frac{1}{3} \cdot \frac{1}{6} + \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot 1} = \frac{1}{\frac{1}{6} + \frac{1}{2} + 1} = \frac{1}{\frac{2}{6} + 1} = \frac{3}{5}.$$

12. W_1 be the event of drawing a white ball at any draw

w_2 be the event of drawing a black ball

$$P(W_1) = \frac{a}{a+b} = p;\ P(w_2) = \frac{b}{a+b} = q$$

$P(A$ wins the game $) = p + pq^2 + pq^4 +$

$$= \frac{p}{1-q^2} = \frac{\frac{a}{a+b}}{1-\left(\frac{b}{a+b}\right)^2} = \frac{a+b}{a+2b}$$

From given condition $\Rightarrow \frac{a+b}{a+2b} = 3 \cdot \frac{b}{a+2b}$

$\left[\because P(B \text{ wins the game }) = 1 - P(A \text{ wins the game }) = 1 - \frac{a+b}{a+2b} = \frac{b}{a+2b}\right]$

$\Rightarrow a + b = 3b \Rightarrow a = 2b \Rightarrow a : b = 2 : 1$

Now $\Rightarrow a + b = 2 + 1 = 3$

13. A bag contains 5 balls.

Two balls are drawn & found to be red. From this, we can say that a bag contains atleast ' 2 ' red balls.

Let R_k be the event that the bag contains k - red, balls, k = 2,3,4,5 Possibilities of different colour are

Bag $B_1 = 2\ 3$

Bag $B_2 = 3\ 2$

Bag $B_3 = 4\ 1$

Bag $B_4 = 5\ 0$

probability of selecting a bag.

$$P(B_1) = P(B_2) = P(B_3) = \frac{1}{4} = P(B_4)$$

Let ' A ' be the event that the two balls are red. Probability of selecting the balls are red.

$$P(\ \mathrm{B}_4/\mathrm{A}) = \frac{P(B_4) \cdot P(A/B_4)}{P(B_1) \cdot P(A/B_1) + P(B_2) \cdot P(A/B_2) + P(B_3) \cdot P(A/B_3) + P(B_4) \cdot P(A/\mathrm{B}_4)}$$

$$P(A/B_1) = \frac{{}^2C_2}{{}^5C_2} = \frac{1}{10} \Rightarrow P(A/B_2) = \frac{{}^3C_2}{{}^5C_2} = \frac{3}{10} \Rightarrow P(A/\mathrm{B}_3) = \frac{{}^4C_2}{{}^5C_2} = \frac{6}{10}$$

$$P(A/B_4) = \frac{{}^5C_2}{{}^5C_2} = \frac{{}^5C_2}{{}^5C_2} = \frac{10}{10} = 1$$

Now ,

$$P(B_4/A) = \frac{1}{\frac{1}{10}+\frac{3}{10}+\frac{6}{10}+\frac{10}{10}} = \frac{10}{20} = \frac{1}{2}$$

14.

Given	Urn A	Urn B
Black	8	6

White	5	7
Total	13	13

Probability of selecting Urns A and B is $P(A) = P(B) = \frac{1}{2}$

Let E = Event of selecting a white ball

$P(E/A) = \frac{5}{13}$, $P(E/B) = \frac{7}{13}$

Now ,

Probability of selecting a white ball is

$P(E) = P(A) \cdot P(E/A) + P(B) \cdot P(E/B) = \frac{1}{2} \cdot \frac{5}{13} + \frac{1}{2} \cdot \frac{7}{13}$

$= \frac{1}{2 \cdot 13}[7 + 5] = \frac{1}{2 \cdot (13)}(12) \Rightarrow P(E) = \frac{6}{13}$.

15. Let ' A ' be the event of drawing a black ball

A_1 = Event of choosing the first box

A_2 = Event of choosing the second box

$$\therefore P\left(\frac{A_2}{A}\right) = \frac{P(A_2) \cdot P\left(\frac{A}{A_2}\right)}{P(A_1) \cdot P\left(\frac{A}{A_1}\right) + P(A_2) \cdot P\left(\frac{A}{A_2}\right)} = \frac{\frac{4}{7} \cdot \frac{1}{2}}{\frac{3}{7} \cdot \frac{1}{2} + \frac{4}{7} \cdot \frac{1}{2}} = \frac{4}{3+4} = \frac{4}{7}$$

16.

Given	White	Black
B_1	4	2= 6
B_2	3	5= 8

Let ' E ' be the event of drawing a white ball from each bag

$P(E) = \frac{4}{6} \cdot \frac{3}{8} \Rightarrow P(E) = \frac{1}{4}$.

17. Given $\Rightarrow P\left(\frac{B}{A}\right) = 0.8$

$P(A \cap B) = 0.8 \times P(A)$

But $P(A \cup B) = 0.6 \Rightarrow P(A) = \frac{1}{2} = P(B)$ and $p(A \cap B) = \frac{2}{5}$

Now $P[(A \cap \bar{B}) \cup (\bar{A} \cap B)] = P(A \cap \bar{B}) + P(\bar{A} \cap B)$

$= P(A) - P(A \cap B) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{1}{2} - 2 \times \left(\frac{2}{5}\right) = \frac{1}{5}$

18. Let E_1 be an event when coin with two heads is selected &E_2 be the event when a fair coin is selected. Let ' A ' be the event when the toss results in heads. Then

$$P(E_1) = \frac{n}{2n+1}, P(E_2) = \frac{n+1}{2n+1} \Rightarrow P\left(\frac{A}{E_1}\right) = 1, P\left(\frac{A}{E_2}\right) = \frac{1}{2}, \text{ then}$$

$$\Rightarrow P(A) = 1 \cdot \frac{n}{2n+1} + \frac{1}{2} \cdot \frac{n+1}{2n+1} \Rightarrow \frac{31}{42} = \frac{2n+n+1}{2(2n+1)}$$

$\Rightarrow 31(2n+1) = 21(3n+1) \Rightarrow 62n+31 = 63n+21 \Rightarrow \therefore n = 10$

19. $P = \frac{13}{52} = \frac{1}{4}, q = \frac{3}{4}$

Probability of A winning $= \frac{p}{1-q^2} = \frac{16}{37}$

Probability of B winning $= \frac{pq}{1-q^2} = \frac{12}{37}$

Probability of C winning $= \frac{pq^2}{1-q^3} = \frac{9}{37}$

20. E_1 is the event of selecting n-urns

E_2 is the event of selecting $(n+1)$ - urns

A is the event of drawing two black balls.

Given $P\left(\frac{E_2}{A}\right) = \frac{1}{16}$

By Baye's theorem, we have

$$P\left(\frac{E_2}{A}\right) = \frac{P(E_2)\cdot P\left(\frac{A}{E_2}\right)}{P(E_1)\cdot P\left(\frac{A}{E_1}\right) + P(E_2)\cdot P\left(\frac{A}{E_2}\right)} \Rightarrow P(E_2) = \frac{1}{n+1}, P(E_1) = \frac{n}{n+1}$$

$$P\left(\frac{A}{E_1}\right) = \frac{{}^6C_2}{{}^{10}C_2}, P\left(\frac{A}{E_2}\right) = \frac{{}^5C_2}{{}^{10}C_2}$$

Now

$$\frac{1}{16} = \frac{\left(\frac{1}{n+1}\right)\cdot\frac{{}^5C_2}{{}^{10}C_2}}{\frac{n}{n+1}\cdot\frac{{}^6C_2}{{}^{10}C_2} + \frac{1}{n+1}\cdot\frac{{}^5C_2}{{}^{10}C_2}} = \frac{\left(\frac{1}{n+1}\right)\cdot\frac{1}{{}^{10}C_2}\cdot({}^5C_2)}{\frac{1}{(n+1)\,{}^{10}C_2}[n\cdot{}^6C_2 + {}^5C_2]} = \frac{{}^5C_2}{n\cdot{}^6C_2 + {}^5C_2}$$

$$\frac{1}{16} = \frac{4}{6n+4} = \frac{2}{3n+2} \Rightarrow 3n+2 = 32 \Rightarrow 3n = 30 \Rightarrow n = 10$$

21. Let A_1 be the event that 'P' rides A,

A_2 be the event that Q rides A.

E be the event that A wins the race.

Now ,

$$P(A_1) = \frac{2}{3},\ P(A_2) = \frac{1}{3} \Rightarrow P\left(\frac{E}{A_1}\right) = \frac{1}{5} \Rightarrow P\left(\frac{E}{A_2}\right) = 3\cdot\left(\frac{1}{5}\right) = \frac{3}{5}$$

$$P(E) = P(A_1)\cdot P\left(\frac{E}{A_1}\right) + P(A_2)\cdot P\left(\frac{E}{A_2}\right) = \frac{2}{3}\cdot\frac{1}{5} + \frac{1}{3}\cdot\frac{3}{5} = \frac{2+3}{15} = \frac{5}{15} = \frac{1}{3}$$

The odds favour to E is

$$P(E):P(\bar{E}) = \frac{1}{3}:\frac{2}{3} = 1:2$$

22. The odd againts to favour to E is $P(E):P(E) = \frac{2}{3}:\frac{1}{3} = 2:1$

23. (1) Let ' A ' be the event of selecting the first b ag

' B ' be the event of selecting the second bag

' C ' 'be the event of selecting the third bag

' E ' ; be the event of drawing a red ball.

$P(A) = \frac{1}{3} = P(B) = P(C) \Rightarrow P\left(\frac{E}{A}\right) = \frac{3}{7}, P\left(\frac{E}{B}\right) = \frac{4}{9}, P\left(\frac{E}{C}\right) = \frac{5}{7}$

$P(E) = P(A) \cdot P\left(\frac{E}{A}\right) + P(B) \cdot P\left(\frac{E}{B}\right) + P(C) \cdot \mathrm{P}(C) P\left(\frac{E}{C}\right)$

$= \frac{1}{3} \cdot \frac{3}{7} + \frac{1}{3} \cdot \frac{4}{9} + \frac{1}{3} \cdot \frac{5}{7} = \frac{27 + 28 + 45}{189} = \frac{100}{189}$

(b) Let ' A ' be the event of selecting bag one

' B ' be the event of selecting the second bag

$P(A) = P(B) = \frac{1}{2}$

E be the event of selecting number ' 4 '

$P\left(\frac{E}{A}\right) = \frac{1}{4}, P\left(\frac{E}{B}\right) = \frac{1}{6}$

Now, $P(E) = P(A) \cdot P\left(\frac{E}{A}\right) + P(B) \cdot P\left(\frac{E}{B}\right) = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{4}\left[\frac{1}{2} + \frac{1}{3}\right] = \frac{1}{4}\left[\frac{5}{6}\right]$

$= \frac{5}{24}$

c) Let A, be the event of selecting a win having heads both sides A_2 be its complementary

E be the event of getting head when the selected coin is tossed.

$P(A_1) = \frac{2}{16}, P(E_2) = \frac{14}{16} \Rightarrow P\left(\frac{E}{A_1}\right) = 1, P\left(\frac{E}{A_2}\right) = \frac{1}{2}$

$P(E) = P(A_1) \cdot P\left(\frac{E}{A_1}\right) + P(A_2) \cdot P\left(\frac{E}{A_2}\right) = \frac{2}{16} \cdot (1) + \frac{14}{16} \cdot \frac{1}{2} = \frac{1}{16}[2 + 7]$

$P(E) = \frac{9}{16}$

d) Let E_1 be the event of selecting bag A

E_2 be the event of selecting bag B.

$P(E_1) = \frac{2}{6} = \frac{1}{3} \Rightarrow P(E_2) = 1 - P(E_1) = \frac{2}{3}$

Let ' E ' be the event of getting a black ball from the selecting bag $P\left(\frac{E}{E_1}\right) = \frac{8}{13}$, $P\left(\frac{E}{E_2}\right) = \frac{6}{13}$

Now $P(E) = P(E_1) \cdot P\left(\frac{E}{E_1}\right) + P(E_2) \cdot P\left(\frac{E}{E_2}\right) = \frac{1}{3} \cdot \frac{8}{13} + \frac{2}{3} \cdot \frac{6}{13}$

$P(E) = \frac{8 + 12}{39} \Rightarrow P(E) = \frac{20}{39}$

24. REASONING TYPE:

Statement-I

Let A_1 be the event of selecting a pair of consecutive letters from the word "LONDON"

A_2 be the event of selecting a pair of consecutive letters from the word CLIFTON.

E be the event of selecting a pair of letters 'ON'

$$P(A_1 \cap E) = \frac{2}{5}, P(A_2 \cap E) = \frac{1}{6} \Rightarrow P\left(\frac{A_1}{E}\right) = \frac{P(A_1 \cap E)}{P(A_1 \cap E) + P(A_2 \cap E)} = \frac{\frac{2}{5}}{\frac{2}{5} + \frac{1}{6}} = \frac{12}{17}$$

Statement-II

Let $A_1, A_2, A_3, A_4 A_5$ denote the events that the bag contains 2,3,4,5,6 red balls respectively and ' C ' be the event that balls are drawn from the bag

Since A_1, A_2, A_3, A_4, A_5 are equally likely

$$P(A_1) = P(A_2) = P(A_3) = P(A_4) = P(A_5) = \frac{1}{5}$$

$$P\left(\frac{E}{A_1}\right) = \frac{^2C_2}{^6C_2} = \frac{1}{15} \Rightarrow P\left(\frac{E}{A_2}\right) = \frac{^3C_2}{^6C_2} = \frac{3}{15} = \frac{1}{5} \Rightarrow P\left(\frac{E}{A_3}\right) = \frac{^4C_2}{^6C_2} = \frac{6}{15} = \frac{2}{5}$$

$$P\left(\frac{E}{A_4}\right) = \frac{^5C_2}{^6C_2} = \frac{10}{15} = \frac{2}{3} \Rightarrow P\left(\frac{E}{A_5}\right) = \frac{^6C_2}{^6C_2} = 1$$

Now,

$$P\left(\frac{A_4}{E}\right) = \frac{P(A_4) \cdot P\left(\frac{E}{A_4}\right)}{P(A_1) \cdot P\left(\frac{E}{A_1}\right) + P(A_2) \cdot P\left(\frac{E}{A_2}\right) + \cdots \ldots + P(A_5) \cdot P\left(\frac{E}{A_5}\right)}$$

$$= \frac{\frac{1}{5} \cdot \frac{2}{3}}{\frac{1}{5}\left[\frac{1}{15} + \frac{1}{15} + \frac{2}{5} + \frac{2}{3} + 1\right]} = \frac{\frac{2}{15}}{\frac{1}{5} \times \frac{1 + 15 + 10 + 15}{15}} = \frac{10}{41}.$$

25. Let ' A ' be the event that one of the selected mangoes is good. B be the event that the selected Mango is also good

$$P(A) = 1 - \frac{^4C_2}{^{10}C_2} = 1 - \frac{2}{15} = \frac{13}{15} \Rightarrow P(A \cap B) = \frac{^6C_2}{^{10}C_2} = \frac{1}{3}$$

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{3}}{\frac{13}{15}} = \frac{1}{3} \times \frac{15}{13} = \frac{5}{13} = \frac{a}{b}$$

$$a + b = 5 + 13 = 15$$

26. Let E_1 denote the event that the bag contains i black and $(12 - i)$ white balls $(i = 0,1,2, \ldots,12)$ and A denote the event that the four balls drawn are all black. Then

$P(E_1) = \frac{1}{13}(i = 0,1,2,\ldots,12) \Rightarrow P\left(\frac{A}{E_1}\right) = 0$ for $i = 0,1,2,3 \Rightarrow P\left(\frac{A}{E_1}\right) = \frac{{}^{i}C_4}{{}^{12}C_4}$ for $i \geq 4$

a) $P(A) = \sum_{i=0}^{12} P(E_1)P\left(\frac{A}{E_1}\right) = \frac{1}{13} \times \frac{1}{{}^{12}C_4}[{}^{4}C_4 + {}^{5}C_4 + \cdots .. + {}^{12}C_4] = \frac{{}^{13}C_5}{13 \times {}^{12}C_4} = \frac{1}{5}$.

b) Clearly, $P\left(\frac{A}{E_{10}}\right) = \frac{{}^{10}C_4}{{}^{12}C_4} = \frac{14}{13}$

c) By Bayer's theorem, $P\left(\frac{E_{10}}{A}\right) = \frac{P(E_{10})\left(\frac{A}{E_{10}}\right)}{P(A)} = \frac{\frac{1}{13} \times \frac{14}{13}}{\frac{1}{5}} = \frac{70}{429}$

d) Let B denote the probability of drawing 2 white and 2 black balls. Then

$P\left(\frac{B}{E_1}\right) = 0$ if $i = 0,1$ or $11,12 \Rightarrow P\left(\frac{B}{E_1}\right) = \frac{{}^{i}C_2 \times {}^{12-i}C_2}{{}^{12}C_4}$ for $i = 2,3,\ldots,10$

$\therefore P(B) = \sum_{i=0}^{12} P(E_1)P\left(\frac{B}{E_1}\right) = \frac{1}{13} \times \frac{1}{{}^{12}C_4}[2\{{}^{2}C_2 \times {}^{10}C_2 \times {}^{3}C_2 + \cdots . + {}^{10}C_2 \times {}^{2}C_2]$

$= \frac{1}{13} \times \frac{1}{{}^{12}C_4}[2\{{}^{2}C_2\,{}^{10}C_2 + {}^{3}C_2^{9}C_2 + \cdots . + {}^{5}C_2 \times {}^{7}C_2\} + {}^{6}C_2 \times {}^{6}C_2] = \frac{1}{13} \times \frac{1}{495}(1287) = \frac{1}{5}$.

27. Let ' H ' be the event that married amn watches the show and ' W ' be the probability that married woman watches the show.

$\therefore P(H) = 0.4, P(W) = 0.5, P(H/W) = 0.7$

1. $P(H \cap W) = P(W)P(H/W) = 0.5 \times 0.7 = 0.35$.
2. $P(W/H) = \frac{P(H \cap W)}{P(H)} = \frac{0.35}{0.4} = \frac{7}{8}$.
3. $P(H \cup W) = P(H) + P(W) - P(H \cap W) = 0.4 + 0.5 - 0.35 = 0.55$.

28. Let p_1, p_2 be the chances of happening of the first and second events, respectively. Then according to the given conditions. We have $p_1 = p_2 2$ and $\frac{1-p_1}{p_1} = \left(\frac{1-p_2}{p_2}\right)^3$

Hence,

$\frac{1-p_2 2}{p_2 2} = \left(\frac{1-p_2}{p_2}\right)^3$ or $p_2(1 + p_2) = (1 - p_2)^2$ or

$3p_2 = 1$ or

$p_2 = \frac{1}{3}$ and so $p_1 = \frac{1}{9}$

29. a) $\frac{{}^{r}C_2}{{}^{r+b}C_2} = \frac{1}{2} 2r(r-1) = (r-b)(r+b-1) = 2r(r-1) = (r+b)(r+b-1)$

or $2r^2 - 2r = r^2 + (2b-1)r + b^2 - 1$

or $r^2 - 1(1 + 2b)r + 1 - b^2 = 0$

or $b^2 + 2br + r - r^2 - 1 = 0$

or $b = \frac{-2r \pm \sqrt{4r^2 - 4(r - r^2 - 1)}}{2} = -r \pm \sqrt{2r^2 - r + 1}$

Since b is integer, possible values of r are 3 and 8 .

b) ${}^4C_2\left(\frac{r}{r+10}\right)^2\left(\frac{10}{r+10}\right)^2 = \frac{3}{8}$

c) $\left(\frac{r}{r+10}\right)^2\left(\frac{10}{r+10}\right)^2 = \frac{1}{16} \Rightarrow r = 10$

d) Probability of getting exactly n red balls in $2n$ draws is always equal to probability of getting exactly n black ballds in $2n$ draws for any value of r and b, hence the ratio r/b can be 10,3,8,2.

11. Functions

WORKSHEET – 1

Level – I

1) A	2) D	3) C	4) A	5) D	6) B
7) B	8) D	9)D			

Level – II

1) B	2)B	3)B	4)D	5)A	6)B	7)C
8) B	9)A	10)A	11)A	12)D	13)A	14)B
15)B	16)B	17)A	18)B	19)B	20)D	21)A
22)D	23)A,B	24)D	25) *a-q,b-s, c- p, d -r*		26) C	27) D
28) B	29)B,C	30)C	31)C	32)B	33)5	34)A,B

35) *a-q,b-r, c-s,d - p*

1. $f(x)=\frac{1-x^2}{1+x^2} \Rightarrow f(\tan\theta)=\frac{1-\tan^2\theta}{1+\tan^2\theta}=\cos 2\theta$
2. $f(x)=2x+1$ if $x<-3$

   $f(-4)=2(-4)+1=-8+1=-7$
3. $f(1)=1, f(2)=3$

   $1=a(1)+b; 3=a(2)+b$

   $a+b=1$

   $(1)\ ; 2a+b=3$

   solving (1) & (2) we get a $=2$; b $=-1$

   $(a,b)=(2,-1)$

   $4\ f(x)=\frac{\alpha x}{x+1}, x\neq 1$

   $f(f(x))=f\left(\frac{\alpha x}{x+1}\right)=\frac{\alpha\left(\frac{\alpha x}{x+1}\right)}{\frac{\alpha x}{x+1}+1}=\frac{\alpha^2 x}{\alpha x+x+1}=x$(Given)

   $\Rightarrow \alpha^2=(\alpha+1)x+1 \forall x\neq -1 \Rightarrow \alpha+1=0,\ 1-\alpha^2=0 \Rightarrow \alpha=-1.$
5. Conceptual
6. $f(-x)=\frac{-x}{e^{-x}-1}-\frac{x}{2}+1=\frac{-xe^x}{1-e^x}+1-\frac{x}{2}=\frac{-xe^x+x-x}{1-e^x}+1-\frac{x}{2}$

   $=x+\frac{x}{e^x-1}+1-\frac{x}{2}=\frac{x}{e^x-1}+\frac{x}{2}+1=f(x)$
7. Here $\cos(\cos x)$ has period π, as even also $\cos(\sin x)$ has been period π as even

   Thus period of $f(x)=\frac{1}{2}\{$ L.C.M of $\pi,\pi\}=\frac{\pi}{2}$
8. $\frac{f(-3)+f(2)+f(5)}{f(1)}=\frac{(9+1)+(3)+(23)}{1}=36.$
9. $fof(x)=f[f(x)]=f\left[(a-x^n)^{\frac{1}{n}}\right]=x$
10. Conceptual

11. $f(x)$ is even then $\frac{1}{2}[f(x)+f(-x)]=\frac{2f(x)}{2}=f(x)$ even

    $f(x)$ is odd then $\frac{1}{2}[f(x)+f(-x)]=0$ even.

12. $f(0)=\text{Tan}\,0=0, f\left(\frac{\pi}{6}\right)=\text{Tan}\left(\frac{\pi}{12}\right)=\frac{\sqrt{3}-1}{\sqrt{3}+1}$

    $\frac{1}{5}\times f(5)=\frac{1}{5}\times\frac{5}{25-2}=\frac{1}{5}\times\frac{5}{23}=\frac{1}{23}$

    $\therefore\sqrt{0+\frac{\sqrt{3}-1}{\sqrt{3}+1}+\frac{1}{23}-\frac{1}{23}}=\sqrt{\frac{\sqrt{3}-1}{\sqrt{3}+1}\times\frac{\sqrt{3}-1}{\sqrt{3}-1}}=\sqrt{\frac{(\sqrt{3}-1)^2}{3-1}}=\frac{\sqrt{3}-1}{\sqrt{2}}$

13. $f(x)=\sin^2 x, g(x)=\sqrt{x}$ then

    $f(g(x))=f(\sqrt{x})=(\sin\sqrt{x})^2, g(f(x))=g(\sin^2 x)=\sqrt{\sin^2 x}=|\sin x|$

14. $f(f(x))=f(x)=x=f(1-x)=1-(1-x)=x$

15. $f(-x)=\ln\left(-x+\sqrt{(-x)^2+1}\right)$

    here it is not obvious whether " f " is an odd or even function. In such case we check

    $f(x)+f(-x)=0\Rightarrow$ f is an odd function

    $f(x)-f(-x)=0\Rightarrow$ f is an even function

    Now, $f(x)+f(-x)=\ln\left(x+\sqrt{x^2+1}\right)+\ln\left(-x+\sqrt{x^2+1}\right)=\ln\left(\left(\sqrt{x^2}+1+x\right)\left(\sqrt{x^2-1}-x\right)\right)$

    $=\ln(x^2+1-x^2)=0$

    $\therefore$ f is an odd function.

16. Period of $f(x)=\frac{\pi}{\sqrt{[K]}}\Rightarrow\pi=\frac{\pi}{\sqrt{[K]}}\Rightarrow[K]=1\Rightarrow 1\le k<2.$

17. when $-3<x\le -1\ f(x)=[x]\Rightarrow f(x)<0$

    when $-1<x<1, f(x)=|x|$ &$1\le x<3, f(x)=|[x]|\Rightarrow f(x)\ge 0$

    $\Rightarrow\{x: f(x)\ge 0\}=(-1,3)$

18. $\left(\frac{2f}{3g}+\frac{3g}{2f}\right)(0)=\frac{2f(0)}{3g(0)}+\frac{3g(0)}{2f(0)}=\frac{2(6)}{3(3)}+\frac{3(3)}{2(6)}=\frac{25}{12}$

19. $f(x)=3+4x$

    $f(f(x))=f(3+4x)=3+4(3+4x)=3+12+16x=15+16x=4^2(x+1)-1$

    $f(f(f(x)))=f(15+16x)=3+4(15+16x)=3+60+64x=63+64x$

    $=4^3(x+1)-1$

    $f(f\ldots\ldots\text{ntimes})(x)=4^n(x+1)-1$

20. $f(x)=\frac{4^x}{4^x+2}, f(1-x)=\frac{4^{1-x}}{4^{1-x}+2}$

$$f(x)+f(1-x)=\frac{4^x}{4^x+2}+\frac{\frac{4}{4^x}}{\frac{4+24^x}{4^x}}=\frac{4^x}{4^x+2}+\frac{4}{2(2+4^x)}=\frac{4^x+2}{4^x+2}=1\Rightarrow f(1-x)$$

$$=1-f(x)$$

21. $f(x)=2kf(x)$.

22. Period of $\cos x$ is 2π, period of $\{x\}$ is 1

$\therefore$ L.C.M $\{2\pi,1\}$ doesn't exist.

23. $f(n+2)=f(n)+(n+1)!+(n+2)!=f(n)+(n+1)![1+n+2]$

$f(n+2)=f(n)+(n+1)![n+3]=f(n)+[f(n+1)-f(n)](n+3)$

$=f(n)[-n-3+1]+(n+3)f(n+1)$

$f(n+2)=f(n)[-n-2]+(n+3)f(n+1)$

$\therefore Q(n)=-n-2\Rightarrow Q(x)=-x-2$

$P(n)=n+3\Rightarrow P(x)=x+3$.

24. Here domain & co-domains of $f(x)\&g(x)$ are not equal so, $f(x)\&g(x)$ are not equal.

25. $\cos[\pi^2]x+\cos[-\pi^2]x=\cos 9x+\cos(-10x)=\cos 9x+\cos 10x$

$f\left(\frac{\pi}{2}\right)=\cos\frac{9\pi}{2}+\cos 5\pi=0-1=-1$

$f\left(\frac{\pi}{4}\right)=\cos\frac{9\pi}{4}+\cos\frac{5\pi}{4}=\cos\frac{\pi}{4}=\frac{1}{\sqrt{2}}$

$f(\pi)=\cos 9\pi+\cos 10\pi=-1+1=0$

$f(0)=\cos 0+\cos 0=2$

26. number of constant functions $=n(B)=n$

27. The number of functions $(n(B))^{n(A)}=20^{10}$

28. Number of functions $(n(B))^{n(A)}=4^3$

29. $f(\cos x)=\sqrt{1-\cos^2 x}=|\sin x|=\max\{\sin x,-\sin x\}$

30. $$f(x)=\begin{Bmatrix} x+1 & \le 1 \\ 2x+1 & 1<x\le 2\end{Bmatrix}, g(x)=\begin{Bmatrix} x^2 & 1\le x<2 \\ x+2 & 2\le x<3\end{Bmatrix}$$

$$f(g(x))=\begin{Bmatrix} g(x)+1, & g(x)\le 1 \\ 2g(x)+1, & 1<g(x)\le 2\end{Bmatrix}\Rightarrow f(g(x))=\begin{Bmatrix} x^2+1, x^2\le 1, -1\le x<2 \\ x+2+1, x+2\le 1, 2\le x\le 3 \\ 2x^2+1, 1<x^2\le 2, -1\le x<2 \\ 2(x+2)+1, 1<x+2\le 2, 2\le x\le 3\end{Bmatrix}$$

$$f(g(x))=\begin{Bmatrix} x^2+1 & -1\le x\le 1 \\ 2x^2+1 & 1<x\le\sqrt{2}\end{Bmatrix}$$

Hence domain of $f(g(x))$ is $[-1,\sqrt{2}]$

31. for $-1 \le x \le 1$, we have $x^2 \in [0,1] \Rightarrow x^2 + 1 \in [1,2]$

    for $1 < x \le \sqrt{2}$, we have $x^2 \in (1,2] \Rightarrow 2x^2 + 1 \in (3,5]$

    hence range is $[1,2] \cup (3,5]$

32. For $f(g(x)) = 2 \Rightarrow x^2 + 1 = 2 \& 2x^2 + 1 = 2$

    $\Rightarrow x = \pm 1$ or $x = \pm\frac{1}{\sqrt{2}} \Rightarrow x = \pm 1$ only.

    Hence 2 roots.

33. Given $f(x+y) = f(x) + f(y) \forall x, y \in R, f(1) = \frac{1}{2}$

    $\Rightarrow f(2) = f(1+1) = f(1) + f(1) = 2f(1)$

    $\Rightarrow f(r) = rf(1) \Rightarrow \sum_{r=1}^{4} f(r) = +\sum_{r=1}^{4} rf(1) = f(1)\left[\frac{4^2(4+1)}{2}\right] = 10 \times \frac{1}{2} = 5.$

34. $f(x) = 1 + \sin x; 0 \le x \le 3\frac{\pi}{4} = 1 - \cos x; \frac{3\pi}{4} \le x \le \frac{3\pi}{4} = 1; = \frac{3\pi}{2} \le x \le 2\pi,$

    $g(x) = 1 - x; x \le 0$

    $= 1\ ; 0 \le x \le 2$

    $= x - 1; x \ge 2$

    $f(0) = 1 \Rightarrow g(f(1)) = 1, f(1) = 1 + \sin 1 \& f(g(0)) = 1 + \sin 1$

35. $a \rightarrow$ when $-1 \le x \le 0, f(x) \in [0,1)$ and $f(x) = x + 1$

    $\therefore f(f(x)) = (x+1)^2 - 1 = x^2 + 2x$

    $b \rightarrow \frac{[\cos 2x + 1][\sec^2 x + 2\tan x]}{2} = \frac{1 + \tan^2 x + 2\tan x}{1 + \tan^2 x} = 1 + \frac{2\tan x}{1 + \tan^2 x}.$

    i.e $f\left(\frac{2\tan x}{1+\tan^2 x}\right) = 1 + \frac{2\tan x}{1+\tan^2 x} \Rightarrow f(x) = 1 + x$

    $c \rightarrow$ Put $x = y = 0 \rightarrow f(1) = (1+1)^2 = 2^2$

    Put $x = 0, y = 1 \rightarrow f(2) = 3^2$

    $d \rightarrow f(x) = 2x + 3 \rightarrow f^{-1}(x) = \frac{x-3}{2}$

WORKSHEET – 2
Level – I

1) A	2) A	3) C	4) B	5) B	6) C
7) A	8) D	9) A			

1. Substitute values
2. $16 - x^2 \ge 0 \Rightarrow x^2 - 16 \le 0 \Rightarrow (x-4)(x+4) \le 0 \Rightarrow x \in [-4,4]$
3. $x^2 - 25 > 0 \Rightarrow (x-5)(x+5) > 0 \Rightarrow x \in (-\infty, -5) \cup (5, \infty)$
4. $x^2 - 7x + 10 \ne 0 \Rightarrow x^2 - 5x - 2x + 10 \ne 0 \Rightarrow (x-5)(x-2) \ne 0 \Rightarrow x \ne 2,5$
5. $2 - 3x \ne 0 \Rightarrow x \ne \frac{2}{3} \Rightarrow x \in R - \left\{\frac{2}{3}\right\}$
6. $x - 2 > 0 \Rightarrow x > 2 \Rightarrow x \in (2, \infty)$

7. $3-x>0 \& 3-x\neq 1 \Rightarrow x<3, x\neq 2 \Rightarrow x\in(-\infty,3)-\{2\}$
8. $\frac{x-2}{x-3}\geq 0, \& \, x-3\neq 0 \Rightarrow x\in(-\infty,2]\cup(3,\infty)$
9. A
10. A
11. B
12. C

Level – II

1) C	2) B	3) A	4) A	5) B	6) C	7) C
8) C	9) B	10) A	11) C	12) B	13) C	14) D
15) C	16) A	17) C	18) A	19) B	20) C	21) A, C
22) A	23) D	25) B	26) a-r,b-s,c-q,d-p		27) A	28) A,B,C
29) A	30) D	31) A	32) B	33) B		

34) $a-s, b-t, c-p, d-p$

1. $f(x)=\log(x-[x])\in R \Leftrightarrow x-[x]>0 \Leftrightarrow x>[x]$

   $\Leftrightarrow x$ is a non- Integer.

   $\therefore$ Domain of f is $R-Z$.

2. $f(x)=\sqrt{\log_{10}\left(\frac{3-x}{x}\right)}\in R \Leftrightarrow \log_{10}\left(\frac{3-x}{x}\right)\geq 0 \& \frac{3-x}{x}>0$

   $\Leftrightarrow \frac{3-x}{x}\geq 10^0=1 \& 3-x>0, x\neq 0, x>0 \Leftrightarrow 3-x\geq x \& 0<x<3$

   $\Leftrightarrow x\leq \frac{3}{2} \& 0<x<3$

   $\Leftrightarrow x\in\left(-\infty,\frac{3}{2}\right]\cap(0,3)=\left(0,\frac{3}{2}\right]$

   $\therefore$ Domain of f is $\left(0,\frac{3}{2}\right]$

3. $x^2-5x+4\neq 0$

   $x^2-4x-x+4\neq 0$

   $(x-4)(x-1)\neq 0$

   $x\neq 1,4 \Rightarrow x\in R-\{1,4\}$

4. $|x|-x>0$

   $|x|>x$

   $x\in(-\infty,0)$

5. $(x-3)(6-x)>0$

   $(x-3)(x-6)<0$

$x \in (3,6)$.

6. $-1 \le \cos x \le 1 \Rightarrow -1 \le 1 - 2\cos x \le 3$

7. Range $= [1-3, 1+3] = [-2,4]$. Then, range of $f(x) = (-\infty, -1] \cup \left[\frac{1}{3}, \infty\right)$

8. $f(x) = y \Rightarrow \frac{2+x}{2-x} = y \Rightarrow x = \frac{2(y-1)}{(y+1)}$, clearly x is not defined if $y = -1$.

9. $y = f(x) = \frac{x}{1+x^2} \Rightarrow x^2 y - x + y = 0 \Rightarrow x = \frac{1 \pm \sqrt{1-4y^2}}{2y}$ is a real number ($\because$ Domain is R)

   If $1 - 4y^2 \ge 0 \Leftrightarrow (1-2y)(1+2y) \ge 0 \Rightarrow y \in \left[\frac{-1}{2}, \frac{1}{2}\right]$

   Range of $f = \left[\frac{-1}{2}, \frac{1}{2}\right]$.

10. $5x \in R$

11. $x - 2 \ge 0, 4 - x > 0, x \ne 3$

12. Since domain of $\sin^{-1} x$ is $[-1,1]$

$\therefore f(x) = \sin^{-1}\left\{\log_2\left(\frac{x^2}{2}\right)\right\}$ exists if

$-1 \le \log_2\left(\frac{x^2}{2}\right) \le 1 \Rightarrow 2^{-1} \le \frac{x^2}{2} \le 2^1 \Rightarrow 1 \le x^2 \le 4 \Rightarrow 1 \le |x| \le 2 \Rightarrow x \in [-2,-1] \cup [1,2]$

13. we have $f(x) = \frac{e^{-x}}{1+[x]}$

$1 + [x] \ne 0 \Rightarrow x \in [-1,0)$

$f(x) = y \Rightarrow \frac{e^{-x}}{1+[x]} = y$

$(1 + [x])y > 0$

If $1 + [x] > 0, y > 0$ i.e $x \in [0, \infty) \Rightarrow y > 0$

If $1 + [x] < 0, y < 0$ i.e $x \in (-\infty, -1) \Rightarrow y < 0 \ \ \therefore y \in R - \{0\}$

Range of $f(x) = R - \{0\}$

14. $f(x)$ is defined $\Rightarrow -1 \le x - 3 \le 1 \& 9 - x^2 > 0 \Rightarrow 2 \le x \le 4 \& -3 < x < 3$ domain $[2,3)$.

15. If $2 \le x < 12$ then range of $f = \{10\}$

    If $x \ge 12$ then the range of $f = [10, \infty)$

    If $x \le 2$ then the range of $f = [10, \infty)$.

16. $|[|x| - 1]| - 5 > 0 \Rightarrow |[|x|] - 1| > 5 \Rightarrow |[|x|] - 1| > 5$ or $[|x| - 1] < -5$

    $\Rightarrow |x| - 1 > 6$ or $|x| - 1 < -5 \Rightarrow |x| \ge 7$ (or $)|x| < -4 \Rightarrow |x| \ge 7 (\therefore |x| - /4)$

    $\therefore x \in (-\infty, -7] \cup [7, \infty)$

17. We have $\frac{\pi^2}{16} - x^2 \geq 0 \Rightarrow \frac{\pi^2}{16} - x^2 \leq \frac{\pi^2}{16}$

$$0 \leq \frac{\pi^2}{16} - x^2 \leq \frac{\pi^2}{16} \Rightarrow 0 \leq \sqrt{\frac{\pi^2}{16} - x^2} \leq \frac{\pi}{4} \Rightarrow 0 \leq \sin\sqrt{\frac{\pi^2}{16} - x^2} \leq \frac{1}{\sqrt{2}}$$

$$0 \leq 3\sin\left(\sqrt{\frac{\pi^2}{16} - x^2}\right) \leq \frac{3}{\sqrt{2}}$$

Hence Range of f is $\left[0, \frac{3}{\sqrt{2}}\right]$

18. $\frac{\sqrt{4-x^2}}{1-x} > 0; 4 - x^2 > 0 \Rightarrow x \in (-2,1)$ and range is $[-1,1]$

19. for $f(x)$ to be defined $|x^2 - 10x + 9| < 4x$

$\Rightarrow x^2 - 10x + 9 < 4x$ and $x^2 - 10x + 9 > -4x$

$\Rightarrow x^2 - 14x + 9 < 0$ and $x^2 - 6x + 9 > 0$

$\Rightarrow x \in (7 - \sqrt{40}, 7 + \sqrt{40})$ and $x \in R - \{-3\}$

20. least value of $x^2 + x + 1 = \frac{3}{4}$ at $x = \frac{1}{2}$

$f(-1) = 2$, $f(1) = 0$ incase of decreasing function $-a + b = 2$

$a + b = 0 \Rightarrow b = 1$

$a = -1 \Rightarrow f(x) = -x + 1$

$\therefore$ least value of $\sqrt{x^2 + x + 1} \leq$ is $\frac{\sqrt{3}}{2}$, also $\sqrt{x^2 + x + 1} \leq 1$ Thus

$$\frac{\sqrt{3}}{2} \leq \sqrt{x^2 + x + 1} \leq 1 \Rightarrow \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) \leq \sin^{-1}\left(\sqrt{x^2 + x + 1}\right) \leq \sin^{-1}(1)$$

$$\Rightarrow \frac{\pi}{3} \leq \sin^{-1}\left(\sqrt{x^2 + x + 1}\right) \leq \frac{\pi}{2}.$$

21. 1 & 3 22. 1 23. 4

24. $\sqrt{\frac{3-x}{4-x}}$ is defined $\Rightarrow \frac{3-x}{4-x} \geq 0, 4 - x \neq 0$

$\Rightarrow (3 - x)(4 - x) \geq 0,\ x \neq 4 \Rightarrow (x - 3)(x - 4) \geq 0,\ x \neq 4 \Rightarrow x \in (-\infty, 3] \cup (4, \infty)$.

25. $\frac{2-x}{x-7} \geq 0 \& x \neq 7 \Rightarrow (x - 2)(x - 7) \leq 0, x \neq 7 \Rightarrow 2 \leq x < 7$.

26. a-r,b-s,c-q,d-p

$a \Rightarrow f(x) = \log_3(5 + 4x - x^2) = \log_3(9 - (x - 2)^2)$

Now $-\infty < 9 - (x - 2)^2 \leq 9$ but $f(x)$ to get defined

$0 < 9 - (x - 2)^2 \leq 9 \Rightarrow -\infty < \log_3(9 - (x - 2)^2) \leq \log_3^9 \Rightarrow$ range $= (-\infty, 2]$

$b \Rightarrow f(x) = \log_3(x^2 - 4x + 5) = \log_3((x - 2)^2 - 9)$

but here $0 < (x - 2)^2 - 9 < \infty, -\infty < f(x) < \infty$

$c \Rightarrow \log_3(x^2 - 4x + 5) = \log_3((x - 2)^2 + 1)$ but $(x - 2)^2 + 1 \in [1, \infty)$

Range $[0,\infty)$

$d \Rightarrow \log_3(4x-5-x^2)$ but $-x^2+4x-5$ is always negative.

$\Delta < 0$, leading co-efficient < 0.

27 $[x^2+1]$ is an Integer $\Rightarrow \sin n\pi = 0 \forall x \in R \Rightarrow f(x) = 0 \forall x \in R$.

28. [1,2,3] Domain of $f(x)$ is IR $= (-\infty,\infty)$

$|x| \geq 0, |1+x| \geq 1$

$|x| + |1+x| \geq 1$

Range $[1,\infty)$

29. $x^2+x-6 \neq 0 \Rightarrow (x+3)(x-3) \neq 0, x \neq -3, x \neq 2$

$R - \{-3,2\}$

30. $(x-3)(x-5) \neq 0 \Rightarrow x \neq 3,5$

$R - \{3,5\}$

31. $y = \frac{x^2-x+1}{x^2+x+1} \Rightarrow (y-1)x^2 + (y+1)x + (y-1) = 0$

$x \in R \Rightarrow \Delta \geq 0 \Rightarrow (y+1)^2 - 4(y-1)^2 \geq 0 \Rightarrow -3y^2+10y-3 \geq 0$

$\Rightarrow 3y^2 - 10y + 3 \leq 0 \Rightarrow (3y-1)(y-3) \leq 0 \Rightarrow \frac{1}{3} \leq y \leq 3$

32. Domain $[-1,1]$

$-1 \leq x \leq 1 \Rightarrow \frac{-\pi}{4} \leq \tan^{-1} x \leq \frac{\pi}{4} \Rightarrow \frac{\pi}{4} \leq \frac{\pi}{2} + \tan^{-1} x \leq \frac{3\pi}{4}$

$\Rightarrow \frac{\pi}{4} \leq f(x) \leq \frac{3\pi}{4}$.

33.

$f(x) = \cos^{-1}\left(\frac{2-|x|}{4}\right) + \frac{1}{\log_e(3-x)}$

$-1 \leq \frac{2-|x|}{4} \leq 1 \;\&\; 3-x > 0, 3-x \neq 1$

by solving the above In equality we get $x \in [-6,2) \cup (2,3]$

34. $a-s, b-t, c-p, d-p$

a) $\sin^{-1}(\log_2 x) \geq 0$

$\log_2 x \geq 0 \Rightarrow x \geq 1$

$-1 \leq \log_2 x \leq 1 \Rightarrow \frac{1}{2} \leq x \leq 2$

from (1) & (2)

$\Rightarrow x \in [1,2]$

b) $\cos 0 \sin x \geq 0 \Rightarrow \frac{-\pi}{2} \leq \sin x \leq \frac{\pi}{2}$ which is always true $\therefore x \in R$

c) $-1 \leq \frac{1+x^2}{2x} \leq 1$

$\Rightarrow \frac{1+x^2}{2x} \geq -1$ & $\frac{1+x^2}{2x} \leq 1$

$\Rightarrow \frac{x^2+1+2x}{2x} \geq 0$ & $\frac{1+x^2-2x}{2x} \leq 0$

$\Rightarrow \frac{(x+1)^2}{2x} \geq 0$ & $\frac{(x-1)^2}{2x} \leq 0$

which holds good only if $x = -1$ or 1

$\therefore$ domain $= \{-1,1\}$

d) if $f(x) = g(x) + h(x)$

Domain of $f(x) = R \cap \{-1,1\} = \{-1,1\}$.

35. $x + 2 \neq 0, x \neq -2 \Rightarrow R - \{-2\}$

WORKSHEET – 3

Level – I

1) A	2) D	3) A	4) D	5) A	6) C	7) C
8) A	9) C	10) B				

Level – II

1) D	2) D	3) B	4) B	5) C	6) B	7) C
8) C	9) B	10) D	11)A	12)C	13)7	14)D
15)B	16)B	17)A	18)A	19)D	20)C	21)D
22)D	23)A,B	24)A	25)	26)A	27)D	28)1
29)A,B	30) $a \rightarrow s$; $b \rightarrow p$; $c \rightarrow q$; $d \rightarrow r$					

1. $f(1) = f(-1) = 1$ $\therefore f$ is not one-one

   so, $|x| \geq 0 \forall x \in R$ $\therefore f(x)$ is not onto.

2. $f(1) = f(2) = 0 \Rightarrow f$ is not one-one

   So, -2 has no pre-image $\Rightarrow f$ is not onto

3. $f(z) = RC(z) \in R \Rightarrow f$ is onto

4. Number of constant function's $n(B) = 12$

5. $f^{-1}(17) \Rightarrow x^2 + 1 = 17 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$

   $\therefore f^{-1}(17) = \{4, -4\}$

   $x^2 + 1 = -3 \Rightarrow x^2 = -4 \Rightarrow x = \sqrt{-4} \notin R$ $f^{-1}(-3) = \emptyset$.

6. $f(x) = e^{5x+13} = y \Rightarrow 5x + 13 = \log y \Rightarrow x = \frac{\log y - 13}{5} \Rightarrow f^{-1}(x) = \frac{-13+\log x}{5}$

7. " 0 " if $m > n$ number of one-one mappings $= 0$.

8. $x|x| = \begin{cases} -x^2 & \text{if } x < 0 \\ x^2 & \text{if } x \geq 0 \end{cases}$ $\therefore f$ is one-one & onto.

9. $x > 0$

   $x < 0$

$\Rightarrow f(x_1) = f(x_2)$

$\Rightarrow f(x_1) = f(x_2)$

$\Rightarrow \frac{x_1}{1+x_1} = \frac{x_2}{1+x_2}$

$\Rightarrow \frac{x_1}{1+x_1} = \frac{x_2}{1+x_2}$

and also $\frac{x}{1+|x|} \in (-1,1) \forall x \in R$

$\Rightarrow x_1 + x_1x_2 = x_2 + x_1x_2 \Rightarrow x_1 - x_1x_2 = x_2 - x_1x_2$ f is not onto

$\Rightarrow x_1 = x_2 \Rightarrow x_1 = x_2$

$\therefore f$ is one-one $\therefore f$ is one-one

10. $f(2) = f\left(\frac{1}{2}\right), f(1)$ has no-pre-image

11. Clearly, f is a bijection and hence invertible.

Let $f(x) = y$. Then, $y < 1$, Now, $f(x)y \Rightarrow \frac{e^x - e^{-x}}{e^x + e^{-x}} = y$

$\Rightarrow \frac{e^{2x} - 1}{e^{2x} + 1} = y \Rightarrow \frac{2e^{2x}}{-2} = \frac{y+1}{y-1} \Rightarrow e^{2x} = \frac{1+y}{1-y}$

$\Rightarrow 2x = \log\left(\frac{1+y}{1-y}\right) \Rightarrow x = \frac{1}{2}\log\left(\frac{1+y}{1-y}\right) = f^{-1}(y) \Rightarrow f^{-1}(x) = \frac{1}{2}\log\left(\frac{1+x}{1-x}\right)$

12. Take $f^{-1}(x) = y \Rightarrow x = f(y)$

$f^{-1}(-100) = x \Rightarrow f(x) = -100 \Rightarrow \frac{-(x-1)}{2} = -100 \Rightarrow x = 201$

13. Number of surjections $= 2^4 - 2 = 14 = m$. Then $\frac{m}{2} = 7$.

14. $f(x) = x - [x] = \{x\}$

We know, $0 \leq \{x\} < 1 f$ is .not onto

$f^{-1}(x) = x \Rightarrow f^{-1}(2010) = 2010$ is not one-one

15. Clearly, $g: R \to R$ given by $g(x) = 2x$ is a bijection. Also it is strictly increasing function on " R ". Therefore $f(x) = 2x + |\cos x|$ is also strictly increasing function on " R ". Hence, it is one-one & onto function.

16. $f(x) = \frac{1}{1-x}; f(x) = y \Rightarrow x = f^{-1}(y)$

$\Rightarrow \frac{1}{1-x} = y \Rightarrow \frac{1}{y} = 1 - x \Rightarrow 1 - \frac{1}{y} = x \Rightarrow \frac{y-1}{y} = x = f^{-1}(y) \therefore f^{-1}(x) = \frac{x-1}{x}$

17. $y = \frac{x^2+2x+1}{x^2+2x+7}$

$\Rightarrow x^2y + 2xy + 7y = x^2 + 2x + 1 \Rightarrow x^2(y-1) + 2x(y-1) + (7y-1) = 0$

$x = \frac{-2(y-1) \pm \sqrt{(2y-2)^2 - 4(y-1)(7y-1)}}{2(y-1)}$.

To define " x " $(2y-2)^2 - 4(y-1)(7y-1) \geq 0, y - 1 \neq 0 \Rightarrow y \neq 1$

$4y^2 + 4 - 8y - 4(7y^2 - 8y + 1) \geq 0$

$\Rightarrow -24y^2 + 24y \geq 0 \Rightarrow 24y^2 - 24y \leq 0 \Rightarrow y^2 - y \leq 0 \Rightarrow y(y-1) \leq 0$

$y \in [0,1]$ and $y \neq 1 \Rightarrow y \in [0,1) \Rightarrow 0 \leq y < 1.$

18. Use graph

19. $y = 2^{x^2-x} \Rightarrow \log_2 y = x^2 - x$

$\Rightarrow x^2 - x - \log_2 y = 0$

$\Rightarrow x = \dfrac{1 \pm \sqrt{1 + 4\log_2 y}}{2} = f^{-1}(x) = \dfrac{1 + \sqrt{1 + 4\log_2 x}}{2}$

20. $f(x) = 4x^3 - 3x$

Let $x = \cos\theta \Rightarrow y = f(\cos\theta) = 4\cos^3\theta - 3\cos\theta = \cos 3\theta$

$\Rightarrow 3\theta = \cos^{-1}(y) \Rightarrow \theta = \frac{1}{3}\cos^{-1} y \Rightarrow \cos^{-1} x = \frac{1}{3}\cos^{-1} y.$

$\Rightarrow x = \cos\left(\dfrac{1}{3}\cos^{-1} y\right) = f^{-1}(y) \Rightarrow f^{-1}(x) = \cos\left(\dfrac{1}{3}\cos^{-1} x\right)$

21. We have $f(x) = x + \sqrt{x^2} = x + |x| = \begin{Bmatrix} 2x & x \geq 0 \\ 0 & x < 0 \end{Bmatrix}$

Clearly, f is many-one into function

22. Clearly, from the graph, f is many-one into function

23. Number of one-one function from set A to B where, $n(A) = m, \mathrm{n}(B) = n$ is

$\begin{cases} {}^nP_m & \text{if} \quad n \geq m \\ 0 & \text{if} \quad n < m \end{cases} \Rightarrow$ one-one function's $= 0$

Number of onto functions $= 2^{n(A)} - 2 = 2^4 - 2 = 14.$

Number of bijections from B to A $= 0$ as $n(A) \neq n(B)$.

24. Conceptual.

25. $(a \to p, q, s; b \to q, r; c \to p, r; d \to s)$

$(-6, -23)$ satisfies $y = 4x + 1$ and $y = 3x - 5$

$(1,5)$ satisfies $y = 4x - 1, y = x + 4$

$\left(\frac{9}{2}, \frac{17}{2}\right)$ satisfies $y = x + 4, 3x - 5$

$(-1, -3)$ satisfies $y = 4x + 1, y = 5x + 2$

26. If $f(x) = x, f(-x) = -f(x)$

$f(x+1) = x + 1 = f(x) + 1 \Rightarrow f\left(\frac{1}{x}\right) = \frac{1}{x} = \frac{f(x)}{x^2}.$

27. $f^{-1}(x) = x$

$k(f^{-1}of^{-1})(x) = f^{-1}(x) = x \Rightarrow (\therefore f^{-1}of^{-1} \ldots\ldots f^{-1})(1) = 1.$

28. $f^{-1}(x) = x \Rightarrow f^{-1}(2010) = 2010.$

29. Conceptual

30. $a \to s; b \to p; c \to q; d \to r$

a) f is neither injection nor surjection

b) $f(x) = x^2, f: [0, \infty) \to R$ is a function

Let $x_1 x_2 \in [0, \infty)$ and $f(x_1) = f(x_2) = x_1^2 = x_2^2 \Rightarrow x_1 = x_2$

Different elements in $[0, \infty)$ have different ' f ' images in R

$\therefore f: [0, \infty) \to R$ is one-one but not onto

c) if $f(x) = x^2$ then, $f: R \to [0, \infty)$ is a function $f(-2) = 4 = f(2) \Rightarrow f$ is not oneone

Let $y \in [0, \infty)$ then $y > 0$ let $x = \sqrt{y} \in R$

Now, $f(x) = x^2 = (\sqrt{y})^2 = y \Rightarrow f: R \to [0, \infty)$ is onto.

d) If $f(x) = x^2$, then $f: [0, \infty) \to [0, \infty)$ is a function

Let $x_1, x_2 \in [0, \infty)$ and $f(x_1) = f(x_2) \Rightarrow x_1^2 = x_2^2 \Rightarrow x_1 = x_2$

$\therefore f: [0, \infty) \to [0, \infty)$ is one-one

Let $y \in [0, \infty)$, then $y > 0$ let $x = \sqrt{y} \in R$.

Now, $f(x) = x^2 = (\sqrt{y})^2 = y \Rightarrow f: [0, \infty)$ is onto. $\Rightarrow f$ is bijection.

31. $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\tan^{-1} x - 1 \le x \le 1$

$= \pi - 2\tan^{-1} x > 1$

$= -\pi + 2\tan^{-1} x < -1$

$g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = 2\tan^{-1} x; x \ge 0$

$= -2\tan^{-1} x; x < 0$

But given that $f(x)$ and $g(x)$ are identical functions

$\Rightarrow$ domain of $f(x)$ = domain of $g(x)$

32. $(a \to p, r, t; b \to s; c \to q)$

a) Let $y = f(x) = \sin(\tan^{-1} x) \Rightarrow x = f^{-1}(y) \Rightarrow \sin^{-1} y = \tan^{-1} x \Rightarrow x = \tan(\sin^{-1} y)$

$\Rightarrow f^{-1}(y) = \tan(\sin^{-1} y) \Rightarrow f^{-1}(x) = \tan(\sin^{-1} x) = \tan\left(\frac{\pi}{2} - \cos^{-1} x\right)$

$= \cot(\cos^{-1} x) = \cot\left(\cot^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)\right) = \frac{x}{\sqrt{1-x^2}}$

(b) $y = f(x) = 1 - 2^{-x} \Rightarrow y = 1 - 2^{-x} \Rightarrow 2^{-x} = 1 - y$

$\Rightarrow -x = \log_2^{(1-y)} \Rightarrow x = -\log_2^{(1-y)} \Rightarrow f^{-1}(y) = -\log_2^{(1-y)} \Rightarrow f^{-1}(x) = -\log_2^{(1-x)}$

(c) Let $y = f(x) = \frac{x}{2^{x-1}} \Rightarrow x = f^{-1}(y)$ and

$$\log_2{}^{y} = \frac{x}{x-1} \Rightarrow x\log_2{}^{y} - \log_2{}^{y} = x \Rightarrow x = \frac{\log_2{}^{y}}{\log_2{}^{y-1}} \Rightarrow f^{-1}(y) = \frac{\log_2{}^{y}}{\log_2{}^{y-1}} \Rightarrow f^{-1}(x)$$

$$= \frac{\log_2{}^{x}}{\log_2{}^{x-1}}$$

33. $f(x) = |\cos x|$ which will be in $\left[\frac{-\pi}{2}, 0\right]$

34. $f(x) = |x| \Rightarrow B = [0, \infty)$

$[x] + [-x] = 1; x \notin Z = 0, x \in Z \Rightarrow B = \{-1,0\}$

12. Limits

WORKSHEET – 1

Level – I

1)B	2)A	3)D	4)C	5)B	6)A	7)D
8)B	9)B	10)D				

Level – II

1) D	2) A	3) A	4) A	5) B	6)D	7)D
8) D	9)A	10)D	11)C	12)8	13)C	14)D
15)D	16)	17)D	18)B	19)B	20)B	21)B
22)C	23)C	24)0	25)B,D	26) a-p,b-s,c-q,d-r		27)A
28)D	29)C	30)A	31)A	32)A,B,C	33) a-q; b-q; c-r; d-p	
34)D	35) a-q; b-p; c-p; d-s:					

1. $\lim_{x\to -1}[x]$ doesn't exist.
2. $\lim_{x\to \frac{\pi}{2}} \sin^2 x + \cos^2 x = 1$
3. $\lim_{x\to 2} \frac{7x^2-11x-6}{3x^2-x-10} = \lim_{x\to 2} \frac{(7x+3)(x-2)}{(3x+5)(x-2)} = \frac{17}{11}.$
4. $\lim_{x\to 2^-} \sqrt{2-x} = \lim_{\substack{h\to 0 \\ h>0}} \sqrt{2-(2-h)} = 0.$
5. $\lim_{x\to 2^-} f(x) = 2 = (2)^2 - 1 = 3 = \alpha;\ \lim_{x\to 2^+} f(x) = 2(2) + 3 = 7 = \beta$

   Now, required quadratic equation is

   $x^2 - (\alpha+\beta)x + \alpha\beta = 0 \Rightarrow x^2 - 10x + 21.$
6. $\lim_{x\to 1^-} f(x) = \frac{1}{2}; \lim_{x\to 1^+} f(x) = 2(1)^2 - 2(1) + \frac{3}{2} = \frac{3}{2}.$

   As $\lim_{x\to 1^-} f(x) \neq \lim_{x\to 1^+} f(x), \lim_{x\to 1} f(x)$ doesn't exist.
7. $\lim_{x\to 1} \frac{4x^4+3x^2-1}{x^3+7} = \frac{4+3-1}{1+7} = \frac{3}{4}.$
8. $\lim_{x\to 0^-} \frac{|\sin x|}{x} = -1$ and $\lim_{x\to 0^+} \frac{|\sin x|}{x} = 1$

   $\therefore \lim_{x\to 0} \frac{|\sin x|}{x}$ doesn't exist.
9. $\lim_{x\to \frac{-1^+}{2}} \sqrt{1+2x} = \lim_{x\to 0} \sqrt{1+2\left(-\frac{1}{2}+h\right)} = \sqrt{h} = 0.$
10. $\lim_{x\to 1} \frac{2x^2+5x-7}{3x^2-x-2} = \lim_{x\to 1} \frac{(2x+7)(x-1)}{(3x+2)(x-1)} = \frac{9}{5}.$
11. $f(x) = |\operatorname{sgn} x|$

    $\lim_{x\to 0^-} f(x) = 1; \lim_{x\to 0^+} f(x) = 1 \ \therefore \lim_{x\to 0} f(x) = 1$
12. $\lim_{x\to 4^+} \{[x] + x\} = 4 + 4 = 8.$
13. (3) $\lim_{x\to 2^+} \frac{[x]^3}{3} - [x/3]^3 = (2)^3/3 = 8/3.$

14. (4) $\lim_{x\to1^-}(1-x+[x-1]+|1-x|)$

$$=\lim_{\substack{h\to0\\h>0}}(1-(1-h)+[1-h-1]+|1-(1-h)|)$$
$$=\lim_{h\to0}h+[-h]+|h|=-1$$

$\lim_{x\to1^+}1-x+[x-1]+|1-x|=\lim_{h\to0}-h+[h]+h=0$

$\therefore \lim_{x\to1}f(x)$ doesn't exist.

15. $\lim_{x\to1^-}\frac{x^2-1}{|x-1|}=\lim_{x\to1}-(x+1)=-2$

$$\lim_{x\to1^+}\frac{x^2-1}{|x-1|}=\lim_{x\to1}x+1=2\ \therefore\lim_{x\to1}\frac{x^2-1}{|x-1|}\text{ doesn't exist.}$$

16. $\lim_{x\to0^-}\frac{3|x|+x}{7|x|-5x}=\frac{-3+1}{-7-5}=\frac{-2}{-12}=\frac{1}{6}\ \lim_{x\to0^+}\frac{3|x|+x}{7|x|-5x}=\frac{3+1}{7-5}=\frac{4}{2}=2\ \therefore\lim_{x\to0}\frac{3|x|+x}{7|x|-5x}$ doesn't exist.

17. $\lim_{x\to0^+}\frac{|x|}{x}+x^2+3=1+0+3=4.$

18. $\sin x+\cos x=\sqrt{2}\left(\sin\left(x+\frac{\pi}{4}\right)\right)$

$\therefore \text{Lim}_{x\to\frac{5\pi}{4}}[\sin x+\cos x]=\text{Lim}_{x\to\frac{5\pi}{4}}\left[\sqrt{2}\left(\sin\left(x+\frac{\pi}{4}\right)\right)\right]=\left[\sqrt{2}\sin\frac{3\pi}{2}\right]=[-\sqrt{2}]=-2$

19.

$$\text{Lim}_{x\to0}\frac{x\sqrt{y^2-(y-x)^2}}{\left(\sqrt{8xy-4x^2}+\sqrt{(8xy)}\right)^3}=\text{Lim}_{x\to0}\frac{x^{3/2}\sqrt{2y-x}}{x^{3/2}[2\sqrt{2y-x}+2\sqrt{2y}]^3}$$
$$=\frac{\sqrt{2y}}{(2\sqrt{2y}+2\sqrt{2y})^3}=\frac{\sqrt{2y}}{34\cdot24\cdot\sqrt{24}}=\frac{1}{128y}$$

20. $\text{Lt}_{x\to0+}\frac{2^{1/x}-1}{2^{1/x}+1}=\text{Lt}_{x\to0+}\frac{1-2^{-1/x}}{1+2^{-1/x}}=\frac{1-0}{1+0}=1$

21. $g(x)=0\forall x\Rightarrow\text{Lim}_{x\to0^+}\left(\log_{|\sin x|}x\right)^{g(x)}=\text{Lim}_{x\to0^+}\left(\log_{|\sin x|}x\right)^0=1$

22. $x\to2-, x<2\Rightarrow[x]=1.\ \text{Lt}_{x\to2-}\{x+(x-[x])^2\}=2+(2-1)^2=3$

23. $f(2+)-f(2-)=[3(2)+4]-[4(2)^2-9]=10-7=3$

24. $\text{Lt}_{x\to0}x=0, \sin\frac{1}{x^2}$ is a bounded function $\Rightarrow \underset{x\to0}{\text{Lt}}\,x\sin\frac{1}{x^2}=0$

25. L.H.L = R.H.L $\Rightarrow \text{Lim}_{x\to1^-}f(x)=\text{Lim}_{x\to1^+}f(x)\Rightarrow\text{Lim}_{h\to0}f(1-h)=\text{Lim}_{h\to0}f(1+h)$

$\Rightarrow \text{Lim}_{h\to0}\left(1+\frac{2(1-h)}{\lambda}\right)=\text{Lim}_{h\to0}\lambda(1+h)\ \Rightarrow1+\frac{2}{\lambda}=\lambda\Rightarrow\lambda^2-\lambda-2=0\Rightarrow\lambda=-1,2$

26. $[a-p, b-s, c-q, d-r]$:

a) L.H.L $=\text{Lim}_{x\to0^-}|x|=\text{Lim}_{x\to0}(-x)=0$

R.H.L $=\text{Lim}_{x\to0^+}|x|=\text{Lim}_{x\to0}(x)=0\ \therefore\text{Lim}_{x\to0}|x|=0$

b) L.H.L $=\text{Lim}_{x\to2^-}[x]=\text{Lim}_{x\to2}1=1$; R.H.L $=\text{Lim}_{x\to2^+}[x]=\text{Lim}_{x\to2}2=2$

$\therefore \text{Lim}_{x\to 2}[x]$ doesn't exist

c) L.H.L $= \text{Lim}_{x\to 2010.1^-}[x] = \text{Lim}_{h\to 0}[2010.1 - h] = [2010.1] = 2010$

R.H.L $= \text{Lim}_{x\to 2010.1^+}[x] = \text{Lim}_{h\to 0}[2010.1 + h] = [2010.1] = 2010$

d) 2011

27. L.H.L $= \text{Lim}_{x\to 0^-} \frac{|x|}{x} = \text{Lim}_{x\to 0^-} \frac{-x}{x} = -1$

R.H.L $= \text{Lim}_{x\to 0^+} \frac{|x|}{x} = \text{Lim}_{x\to 0^+} \frac{x}{x} = 1$ Since L.H.L ≠ R.H.L, $\lim_{x\to 0} f(x)$ doesn't exist.

28. : L.H.L $= \text{Lim}_{x\to 1^-} f(x) = \text{Lim}_{x\to 1}(1 + x^2) = 1 + 1 = 2$

R.H.L $= \text{Lim}_{x\to 1^+} \text{f(x)} = \text{Lim}_{\text{x}\to 1}(2 - \text{x}) = 2 - 1 = 1$ $\therefore \text{Lim}_{\text{x}\to 1}\,\text{f(x)}$ doesn't exist.

29. $\text{Lim}_{x\to 0} f(x)$ exists $\Rightarrow \text{Lim}_{x\to 0^-} f(x) = \text{Lim}_{x\to 0^+} f(x)$

$\Rightarrow \text{Lim}_{x\to 0}\, x + k = \text{Lim}_{x\to 0} \cos x \Rightarrow 0 + k = \cos 0 \Rightarrow k = 1$

30. : : L.H.L of the function at 2 is not defined in real number system. Where as R.H.L is 2

31. R.H.L $= \text{Lim}_{x\to 4^+} f(x) = \text{Lim}_{h\to 0} f(4 + h) = \text{Lim}_{h\to 0} \frac{|4+h-4|}{4+h-4}$

$= \text{Lim}_{h\to 0} \frac{|h|}{h} = \text{Lim}_{h\to 0} 1 = 1$(As $x \to 4^+ h \to 0^+ |h| = h$)

32. $\text{Lim}_{x\to 1^-} f(x) = \text{Lim}_{h\to 0} f(1 - h) = \text{Lim}_{h\to 0}(1 - h)^2 = 1^2 = 1$

$\text{Lim}_{x\to 1^+} f(x) = \text{Lim}_{h\to 0} f(1 + h) = \text{Lim}_{h\to 0}(1 + h) = 1$

$\text{Lim}_{x\to 4^-} f(x) = \text{Lim}_{h\to 0} f(4 - h) = \text{Lim}_{h\to 0}(4 - h) = 4$

$\text{Lim}_{x\to 4^+} f(x) = \text{Lim}_{h\to 0} f(4 + h) = \text{Lim}_{h\to 0}[4 - (4 + h)] = 0$

33. a-q; b-q; c-r; d-p

a) $\lim_{x\to 1^-} g(x) = (1)^2 = 1.$

b) $\lim_{x\to 1^+} g(x) = 1.$

c) $\lim_{x\to 4^-} g(x) = (4) = 4.$

d) $\lim_{x\to 4^+} g(x) = 4 - 4 = 0.$

34. $\lim_{x\to 0^-} \frac{\sin[x]}{[x]} = \lim_{\substack{h\to 0 \\ h>0}} \frac{\sin[-h]}{[-h]} = \frac{\sin 1}{1}$. $\lim_{x\to 0^+} \frac{\sin[x]}{[x]} = \lim_{\substack{h\to 0 \\ h>0}} \frac{\sin[h]}{[h]} = \frac{\sin 0}{0}$ (Doesn't Exist).

$\therefore \lim_{x\to 0} \frac{\sin[x]}{[x]}$ doesn't exist.

35. a-q; b-p; c-p; d-s:

a) $\text{Lim}_{x\to 0^-} f(x) = \text{Lim}_{x\to 0^-} \frac{\sin(1+[x])}{[x]} = \text{Lim}_{x\to 0^-} \frac{\sin(1+[x])}{1+[x]} \times \frac{1+[x]}{[x]} = 1 \times \frac{0}{(-1)} = 0$

b) $\text{Lim}_{x\to a^+} f(x) = \text{Lim}_{x\to a^+}(x + 2) = a + 2$; $\text{Lim}_{x\to a^-} f(x) = \lim_{x\to a^-} 2x + 3 = 2a + 3$

$\text{Lim}_{\text{x}\to \text{a}}\,\text{f(x)}$ exists $\Rightarrow \text{a} + 2 = 2\text{a} + 3 \Rightarrow \text{a} = -1$ $\therefore$ Required limit $= \text{a} + 2 = -1 + 2 = 1$

d)L.H.L $= \text{Lim}_{x\to -1^-} \frac{1}{\sqrt{|x|-\{-x\}}} = \text{Lim}_{x\to -1^-} \frac{1}{\sqrt{-x-(x+2)}} = \text{Lim}_{x\to -1^-} \frac{1}{\sqrt{-2x-2}} = \infty$

R.H.L $= \text{Lim}_{x\to -1^+}\frac{1}{\sqrt{|x|-\{x\}}} = \text{Lim}_{x\to -1^+}\frac{1}{\sqrt{-x-(x+1)}} = \text{Lim}_{x\to -1^+}\frac{1}{\sqrt{-2x-1}} = 1$

Hence the limit doesn't exist.

WORKSHEET – 2

Level – I

1)A	2)C	3)A	4)D	5)A	6)A	7)C
8)B	9)C	10)B				

Level – II

1)A	2) C	3) A	4)B	5)C	6)A	7)A
8)B	9)A	10)A	11)D	12)A	13)D	14)B
15)B	16)A	17)A	18)C	19)A	20)B	21)C
22)A,C	23)A	24)A	25)B	26)A		
27) $a\to q;b\to p;c\to s;d\to r$			28)0	29) B,C,D	30) B	31) D
32) D	33) B	34) $a\to q;b\to p;c\to q;d\to s$				

1. Using $\lim_{x\to a}\frac{x^p-a^p}{x^q-a^q} = \frac{p}{q}a^{p-q}$
2. Using L-Hospital
3. Using L-Hospital $= \lim_{x\to 2}\frac{14x-11}{6x-1} = \frac{17}{11}$
4. Given limit $\lim_{x\to 0}\frac{(2x-3)(\sqrt{x}-1)}{(\sqrt{x}-1)(\sqrt{x}+1)(2x+3)} = \frac{(-1)}{2(5)} = \frac{-1}{10}$
5. $\lim_{x\to 1^-} f(x) \neq \lim_{x\to 1^+} f(x)$
6. $\lim_{x\to\frac{1}{2}}\left(\frac{8x-3}{2x-1}-\frac{4x^2+1}{4x^2-1}\right) = \lim_{x\to\frac{1}{2}}\left(\frac{2(2x-1)(3x+2)}{(2x-1)(2x+1)}\right) = \frac{7}{2}.$
7. $\lim_{n\to\infty}\frac{2^n-1}{2^n+1} = \lim_{n\to\infty}\frac{1-\frac{1}{2^n}}{1+\frac{1}{2^n}} = 1.$
8. $\lim_{n\to\infty}\frac{n(n+1)}{2n^2} = \frac{1}{2}$
9. $\lim_{x\to\infty}\frac{|3x^2+1|}{2x^2+1} = \lim_{x\to\infty}\frac{3+\frac{1}{x^2}}{2+\frac{1}{x^2}} = \frac{3}{2}.$
10. $\lim_{n\to\infty}\frac{2^{3n}}{3^{3n}} = \lim_{n\to\infty}\left(\frac{2}{3}\right)^{3n} = 0.$
11. $\lim_{x\to 1}\frac{\sum_{k=1}^{200}(x^k-200)}{x-1} = \lim_{x\to 1}\frac{(x-1)+(x^2-1)+\cdots.(x^{200}-1)}{x-1} = 1+2+..+200 = 20100.$
12. $\text{n}\frac{\text{Lim}}{x\to\infty}\frac{8|+3x}{3|x|-2x} = \text{Lim}_{x\to\infty}\frac{8x+3x}{3x-2x} = 11$
13. $\text{lt}_{x\to a}\frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3x+x}-2\sqrt{x}} = \text{lt}_{x\to a}\left(\frac{\frac{1}{\sqrt{a+2x}}-\frac{\sqrt{3}}{2\sqrt{x}}}{\frac{1}{2\sqrt{3a+x}}-\frac{1}{\sqrt{x}}}\right) = -\left(\frac{1}{2\sqrt{3}\cdot\sqrt{a}}\right)\left(\frac{4\sqrt{a}}{-3}\right) = \frac{2}{3\sqrt{3}}.$
14. $\lim_{x\to 4}\frac{3-\sqrt{5+x}}{x-4} = \lim_{x\to 4}\frac{1}{3+\sqrt{5+x}} = \frac{1}{6}.$

15. Hint: $\text{lit}_{x\to\infty}\frac{x^2\left(3+\frac{5}{x}+\frac{2}{x^2}\right)}{x^2\left(2-\frac{3}{x}+\frac{1}{x^2}\right)}=\frac{3}{2}$.

16. $\lim_{x\to\infty}\frac{(2+x)^{40}(4+x)^5}{(2-x)^{45}}=\lim_{\frac{1}{x}\to 0}\frac{\left(\frac{2}{x}+1\right)^{40}\left(\frac{4}{x}+1\right)^5}{\left(\frac{2}{x}-1\right)^{45}}=-1.$

17. Given limit (if $r<1$ then sum of terms in G.P is

$$a\left(\frac{1-r^n}{1-r}\right)=\lim_{n\to\infty}\left(\frac{1+\frac{1}{2}+\frac{1}{4}+\cdots+\infty}{1+\frac{1}{3}+\frac{1}{9}+\cdots+\infty}\right)=\frac{4}{3}$$

18. Taking q^n common.

19. $\lim_{x\to 1}\frac{x-1-\ln x}{\ln x(x-1)}=\lim_{x\to 1}\frac{1-\frac{1}{x}}{\frac{1}{x}(x-1)+\ln x}(L-\text{Hopital rule})$

$=\lim_{x\to 1}\frac{\frac{1}{x^2}}{\frac{1}{x}(1)+(x-1)\left(\frac{-1}{x^2}\right)+\frac{1}{x}}(L-\text{Hopital rule})=\frac{1}{2}.$

20. $\lim_{n\to\infty}\frac{\frac{n(n+1)(2n+1)}{6}}{2n^3+3n^2+4n+1}=\lim_{n\to\infty}\frac{\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)}{6\left(2+\frac{3}{n}+\frac{4}{n^2}+\frac{1}{n^3}\right)}=\frac{1}{6}.$

21. $\lim_{n\to\infty}an-\frac{\left(1+n^2\right)}{1+n}=b\Rightarrow\lim_{n\to\infty}\frac{n^2(a-1)+(an-1)}{n+1}=b.$

As, b is finite, $a-1=0\&a=b\Rightarrow a=b=1.$

22. $\lim_{x\to 3}\frac{(x^2-9)(x-3+x+3)}{(x^2-9)}=6.$

23. Hint: $\text{lt}_{x\to 0}\frac{5+8x}{\sqrt{9+5x+4x^2}+3}=\frac{5}{6}$.

24. $\lim_{x\to\infty}\frac{[2x+3]}{x}=2.$

25. $\lim_{x\to\infty}\frac{[3x+5]}{x}=3.$

26. $\lim_{x\to\infty}\frac{[4x+5]}{2x}=2.$

27. $a\to q; b\to p; c\to s; d\to r.$

a) $\lim_{x\to\infty}\frac{(x+1)(2x+3)}{(x+2)(3x+4)}=\frac{(1+0)(2+0)}{(1+0)(3+0)}=\frac{2}{3}.$

b) $\lim_{x\to\infty}\frac{2x+7}{x^2+5x+4}=0\cdot\frac{2+0}{(1+0+0)}=0.$

c) $\lim_{x\to\infty}\frac{2x^2+1}{x^2+3}=\frac{2+0}{(1+0)}=2.$

d) $\lim_{x\to\infty}\frac{3x^2+5x+2}{2x^2-3x+1}=\frac{3+0+0}{2-0+0}=\frac{3}{2}.$

28. $\lim_{x\to 0}\frac{\sqrt{1+x^3}-\sqrt{1-x^3}}{x^2}=\lim_{x\to 0}\frac{2x}{\sqrt{1+x^3}+\sqrt{1-x^3}}=0.$

29. $\text{Lim}_{x\to 0}\frac{\left(1+a^3\right)e^{\frac{1}{x}}+8}{e^{\frac{1}{x}}+(1-b^3)}=2\Rightarrow\frac{0+8}{0+1-b^3}=2\Rightarrow -b^3=3\Rightarrow b=(-3)^{\frac{1}{3}}.$

30. $\lim_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}}{x}=\lim_{x\to 0}\frac{2}{\sqrt{1+x}+\sqrt{1-x}}=1.$

31. $\lim_{x\to 0}\frac{\sqrt{2+x^2}-\sqrt{2-x^2}}{x^2}=\lim_{x\to 0}\frac{2}{\sqrt{2+x^2}+\sqrt{2-x^2}}=\frac{1}{\sqrt{2}}.$

32. $\lim_{x\to 3}\frac{\sqrt{x-1}-\sqrt{2}}{x^2-9}=\lim_{x\to 3}\frac{1}{(x+3)(\sqrt{x+1}+\sqrt{2})}=\frac{1}{12\sqrt{2}}.$

33. Given $\lim_{x\to\infty}\left(\frac{x^2+x+1}{x+1}-ax-b\right)=4\Rightarrow\lim_{x\to\infty}\frac{x^2+x+1-ax^2-ax-bx-b}{(x+1)}=4$

$\Rightarrow\lim_{x\to\infty}\frac{(1-a)x^2+(1-a-b)x+(1-b)}{(x+1)}=4\Rightarrow 1-a=0$ and $1-a-b=4\Rightarrow b=-4, a=1.$

34. $a\to q; b\to p; c\to q; d\to s.$

(A): Here a > 0, if a ≤ 0, then limit = ∞

$$\therefore\lim_{x\to\infty}\frac{\left(\sqrt{x^2-x+1}-ax-b\right)\left(\sqrt{x^2-x+1}+ax+b\right)}{\left(\sqrt{x^2-x+1}+ax+b\right)}\Rightarrow\lim_{x\to\infty}\frac{(x^2-x+1)-(ax+b)^2}{\left(\sqrt{x^2-x+1}+ax+b\right)}$$

$$\Rightarrow\lim_{x\to\infty}\frac{(1-a^2)x^2-(1+2ab)x+(1-b^2)}{\left(\sqrt{x^2-x+1}+ax+b\right)}$$

This is possible only when $1-a^2=0$ and $1+2ab=0$

$\therefore a=\pm 1\Rightarrow a=1\ (\because a>0)\ \ldots(1)\ \Rightarrow b=-1/2\Rightarrow(a,2b)=(1,-1).$

(B) : Divide numerator and denominator by $e^{1/x}$, then

$$\lim_{x\to\infty}\frac{(1+a^3)e^{1/x}+8}{e^{1/x}+(1-b^3)}=2\Rightarrow\frac{0+8}{0+1-b^3}=2\Rightarrow 1-b^3=4\ \therefore b^3=-3\to b=-3^{1/3}$$

Then $a\in R\Rightarrow(a,b^3)=(a,-3).$

(C) : $\lim_{x\to\infty}\left(\sqrt{(x^4-x^2+1)}-ax^3-b\right)=0$ Put $x=\frac{1}{t}$,

$$\therefore\lim_{x\to 0}\left(\sqrt{\left(\frac{1}{t^4}-\frac{1}{t^1}+1\right)}-\frac{a}{t^2}-b\right)=0\Rightarrow\lim_{t\to 0}\frac{\sqrt{(1-t^2+t^4)}-a-bt^2}{t^2}=0$$

Since RHS is finite, numerator must be equal to 0 at t → 0. ∴ 1 − a = 0 ∴ a = 1

from equation (1), $\lim_{t\to 0}\frac{\sqrt{(1-t^2+t^4)}-1-bt^2}{t^2}=0$

$$\lim_{t\to 0}(-1+t^2)\left(\frac{(1-t^2+t^4)^{1/2}-(1)^{1/2}}{(1-t^2+t^4)-1}\right)=b\Rightarrow(-1)\left(\frac{1}{2}\right)=b\Rightarrow a=1, b=-\frac{1}{2}$$

$$\Rightarrow(a,-2b)=(1,2)$$

(D): $\lim_{x\to -a}\frac{a^7-(-x)^7}{a-(-x)}=7\Rightarrow 7a^6=7\Rightarrow a^6=1\ \Rightarrow a=-1.$

WORKSHEET – 3
Level – I

1) B	2) B	3) B	4) A	5) D	6) D
7) C	8) B	9) D	10) B		

Level – II

1) B	2) C	3) D	4) A	5) C	6) A
7) D	8) D	9) C	10) D	11)A	12)D
13)A	14)B	15)D	16)B	17)B	18)A
19)D	20)B	21)B	22)A,B,C	23)A	24)A
25)C	26)C	27) a-r,b-s,c-p,d-q		28)A	29)A,B
30) a-q, b-p, c-q, d-p		31)A,B	32)D	33)B	34)A

1. $\lim_{x\to 0}\frac{e^{\sin x}\left(e^{x-\sin x}-1\right)}{2(x-\sin x)}=\frac{1}{2}$

2. Given limit is $\lim_{x\to 0}\frac{(1-\cos x)\left(1+\cos x+\cos^2 x\right)}{x\sin 2x}=\frac{3}{4}$

3. $\lim_{x\to 1}\frac{(1-x)}{\cot(\pi x/2)}=\frac{2}{\pi}\lim_{x\to 1}\frac{1}{-\operatorname{cosec}^2(\pi x/2)}=2/\pi$

4. $\mathrm{Lt}_{x\to 0}\frac{3\sin x-\sin 3x}{x^3}=\mathrm{Lt}_{x\to 0}\frac{4\sin^3 x}{x^3}=4$

5. $\mathrm{Lt}_{x\to 0}\frac{1-\cos mx}{1-\cos nx}=\underset{x\to 0}{Lt}\frac{m\cdot\sin mx}{n\cdot\sin nx}=\frac{m^2}{n^2}$

6. $\mathrm{Lt}_{x\to \pi/2}\frac{\cos x}{\frac{\pi}{2}-x}=\underset{x\to \pi/2}{\mathrm{Lt}}\frac{-\sin x}{-1}=\sin\frac{\pi}{2}=1$

7. $\mathrm{Lt}_{x\to 0}\frac{\tan x-\sin x}{x^3}=\mathrm{Lt}_{x\to 0}\frac{\tan x}{x}\left(\frac{1-\cos x}{x^2}\right)=1\cdot\frac{1}{2}=\frac{1}{2}$

8. $\mathrm{Lt}_{x\to 0}\frac{\cos 7x-\cos 9x}{\cos 3x-\cos 7x}=\frac{81-49}{49-9}=\frac{32}{40}=\frac{4}{5}$

9. $\mathrm{Lt}_{x\to 0}\frac{1-\cos^3 x}{x\cdot\sin x\cdot\cos x}=\mathrm{Lt}_{x\to 0}\frac{(1-\cos x)}{x^2}\frac{\left(1+\cos x+\cos^2 x\right)}{(\sin x/x)\cos x}=\frac{1}{2}\cdot\frac{1+1+1}{1\cdot 1}=\frac{3}{2}$

10. $\mathrm{Lt}_{x\to 0}\frac{1-\cos 2x^0}{x^2}=\mathrm{Lt}_{x\to 0}\frac{1-\cos\left(\frac{\pi x}{90}\right)}{\left(\frac{\pi x}{90}\right)^2}\left(\frac{\pi}{90}\right)^2=\frac{1}{2}\cdot\left(\frac{\pi}{90}\right)^2=2\left(\frac{\pi}{180}\right)^2$

11. $\mathrm{Lt}_{x\to 0}\frac{\sin 7x-\sin 5x+\sin 3x-\sin x}{\sin 6x-\sin 4x+\sin 2x}=\frac{7-5+3-1}{6-4+2}=\frac{4}{4}=1$

12. $\underset{x\to \pi/2}{\mathrm{Lt}}(\sec 5x\cdot\cos 7x)=\underset{x\to \pi/2}{\mathrm{Lt}}\frac{\cos 7x}{\cos 5x}=\underset{x\to \pi/2}{\mathrm{Lt}}\frac{-7\sin 7x}{-5\sin 5x}=\frac{-7(-1)}{-5(1)}=-\frac{7}{5}$

13. $\mathrm{Lt}_{x\to \pi/2}\frac{1-\sin^3 x}{\cos^2 x}=\mathrm{Lt}_{x\to \pi/2}\frac{(1-\sin x)\left(1+\sin x+\sin^2 x\right)}{(1-\sin x)(1+\sin x)}=\frac{1+1+1}{1+1}=\frac{3}{2}$

14. $\mathrm{Lt}_{x\to a}\frac{\operatorname{cosec}(x-a)}{x-a}=\mathrm{Lt}_{x\to a}\frac{1}{(s-a)\sin(x-a)}=\underset{x\to a}{\mathrm{Lt}}\frac{1}{(x-a)^2}=\infty$

15. [4] $\underset{x\to a}{\mathrm{Lt}}\frac{\tan x-\tan a}{\sin a-\sin x}=\underset{x\to a}{\mathrm{Lt}}\frac{\sec^2 x}{-\cos x}=-\sec^3 a$

16. $\mathrm{Lt}_{y\to x}\frac{\sin^2 y-\sin^2 x}{y-x}=\underset{yt\to x}{\mathrm{Li}}\frac{2\sin y\cos y}{1}=2\sin x\cos x=\sin 2x$

17. $\mathrm{Lt}_{x\to \pi/6}\frac{3\sin\pi x-\sin 3\pi x}{6x-\pi}=\mathrm{Lt}_{x\to \pi/6}\frac{3\cos x+\sqrt{3}\sin x}{6}=\frac{\frac{3\sqrt{3}}{2}+\frac{\sqrt{3}}{2}}{6}=\frac{1}{\sqrt{3}}$

18. $\mathrm{Lt}_{x\to 0}\frac{\sqrt{1+\tan x}-\sqrt{1-\tan x}}{\sin x}=\mathrm{Lt}_{x\to 0}\frac{\frac{\sec^2 x}{2\sqrt{1+\tan x}}+\frac{\sec^2 x}{2\sqrt{1-\tan x}}}{\cos x}\frac{\frac{\sec^2 0}{2\sqrt{1+\tan 0}}+\frac{\sec^2 0}{2\sqrt{1-\tan 0}}}{\cos 0}=\frac{1}{2}+\frac{1}{2}=1$

19. $\mathrm{Lt}_{x\to 0}\frac{\sin^{-1}x-\tan^{-1}x}{x^3}=\underset{x\to 0}{\mathrm{Lt}}\frac{\frac{1}{\sqrt{1-x^2}}-\frac{1}{1+x^2}}{3x^2}=\mathrm{Lt}_{x\to 0}\frac{-\frac{1}{2(1-x^2)^{3/2}}(-2x)+\frac{1}{(1+x^2)^2}\cdot 2x}{6x}=\frac{\frac{1}{2}+1}{3}=\frac{3/2}{3}=\frac{1}{2}$

20. $\mathrm{Lt}_{x\to -1}\frac{\sqrt{\pi}-\sqrt{\cos^{-1}x}}{\sqrt{x+1}}=\mathrm{Lt}_{x\to -1}\frac{\frac{1}{2\sqrt{\cos^{-1}x}}\cdot\frac{1}{\sqrt{1-x^2}}}{\frac{1}{2\sqrt{x+1}}}=\mathrm{Lt}_{x\to -1}\frac{1}{\sqrt{\cos^{-1}x}\sqrt{1-x}}=\frac{1}{\sqrt{\pi}\cdot\sqrt{2}}=\frac{1}{\sqrt{2\pi}}$

21. Put $x=\sin\theta$

$$\mathrm{Lt}_{x\to 0}\frac{x\left(1-\sqrt{1-x^2}\right)}{\sqrt{1-x^2}(\sin^{-1}x)}=\underset{\theta\to 0}{\mathrm{Lt}}\frac{\sin\theta(1-\cos\theta)}{\cos\theta\cdot\theta^3}=\mathrm{Lt}_{\theta\to 0}\frac{\tan\theta}{\theta}\cdot\frac{1-\cos\theta}{\theta^2}=1\cdot\frac{1}{2}=\frac{1}{2}$$

22. $L=\mathrm{Lt}_{x\to 0}\frac{\frac{\sin x^n}{x^n}\cdot x^n}{\frac{(\sin x)^m}{x^n}\cdot x^m}=\mathrm{Lt}_{x\to 0}\,x^{n-m}$

If n = m, then L = (a very small value near to zero) exactly zero = 1

If n > m, then L = (a very small value near to zero) positive integer = 0

If n < m, then L = (a very small value near to zero) negative integer = ∞

23.

24. $ax^2+bx+c=a(x-\alpha)(x-\beta)$

$$L=\mathrm{Lt}_{x\to\alpha}\frac{1-\cos[a(x-\alpha)(x-\beta)]}{(x-\alpha)^2\cdot(x-\beta)^2}\cdot(x-\beta)^2=\frac{a^2}{2}(\alpha-\beta)^2$$

25. ; $L=\underset{x\to 0}{\mathrm{Lt}}\left[\frac{1-\cos(1-\cos x)}{(1-\cos x)^2}\right]\left[\frac{1-\cos x}{x^2}\right]^2=\frac{1}{2}\left(\frac{1}{2}\right)^2=\frac{1}{8}$

26. $[a-r,b-s,c-p,d-q];(a)L=\underset{x\to 0}{\mathrm{Lt}}\frac{\frac{\sin 5x}{x}-\frac{\sin 3x}{x}}{\frac{\sin 4x}{x}-\frac{\sin 2x}{x}}=\frac{5-3}{4-2}=\frac{2}{2}=1$

(b) $L=\frac{6\times 5}{3\times 5}=2$

(c) $\mathrm{L}=\frac{9^2-7^2}{7^2-3^2}=\frac{32}{40}=\frac{4}{5}$

(d) $\mathrm{L}=\frac{9-25}{2}=\frac{-16}{2}=-8$

28. $x\to\frac{\pi}{2}\frac{\cot x}{2}-x=1$

29. $L=\mathrm{Lt}_{x\to 0}\frac{2\sin x-\cos x+a\sin x}{x^3}=\underset{x\to 0}{\mathrm{Lt}}\frac{\sin x}{x}\frac{2\cos x+a}{x^2}$

Now denominator tends to 0 when x → 0, then numerator also must tends to zero for which $\mathrm{Lt}_{x\to 0}(2\cos x+a)=0\Rightarrow a=-2$

Now, $L=\mathrm{Lt}_{x\to 0}\frac{2\cos x-2}{x^2}=-2\underset{x\to 0}{\mathrm{Lt}}\frac{2\sin^2\frac{x}{2}}{x^2}=-1$

30. a-q, b-p, c-q, d-p ;

a) $L=\mathrm{Lt}_{x\to 0}\frac{1-1+\tan x}{\sin x(1+\sqrt{1-\tan x})}$

$\text{Lt}_{x\to 0}\dfrac{1}{\cos x(1+\sqrt{1-\tan x})}=\dfrac{1}{1+1}=\dfrac{1}{2}$

b) $L=\text{Lt}_{x\to 0}\left(\dfrac{\cos x}{2\sqrt{1+\sin x}}+\dfrac{\cos x}{2\sqrt{1-\sin x}}\right);\dfrac{1}{2}+\dfrac{1}{2}=1$

c) $\text{Lt}_{x\to 0}\dfrac{\sec\left(\frac{\pi}{2}-x\right)-\tan\left(\frac{\pi}{2}-x\right)}{\frac{\pi}{2}-\left(\frac{\pi}{2}-x\right)}=\text{Lt}_{x\to 0}\dfrac{\text{cosec}\,x-\cot x}{x}=\text{Lt}_{x\to 0}\dfrac{1-\cos x}{x\sin x}=\dfrac{1}{2}$

d) $L=\underset{x\to\frac{\pi}{2}}{\text{Lt}}\dfrac{\frac{\pi}{2}-x}{\cot x}=\underset{x\to\frac{\pi}{2}}{\text{Lt}}\dfrac{\frac{\pi}{2}-x}{\tan\left(\frac{\pi}{2}-x\right)}=1$

31. $\lim_{x\to 0+}f(x)=\lim_{h\to 0}f(0+h)=\lim_{h\to 0}\dfrac{\sin^{-1}(1-h)}{(1-h)}\lim_{h\to 0}\dfrac{\cos^{-1}(1-h)}{\sqrt{2}h}=\infty$

$\lim_{x\to 0-}f(x)=\lim_{h\to 0}f(0-h)=\lim_{h\to 0}\dfrac{\sin^{-1}(1+h-1)\cdot\cos^{-1}(1+h-1)}{\sqrt{2}(-h+1)\cdot(1+h-1)}=1\cdot\dfrac{\pi/2}{\sqrt{2}}$

$=\dfrac{\pi}{2\sqrt{2}}$

32. Since $x-1\le[x]\le x\forall x\in R$, so $\frac{1}{x^3}-1\le\left[\frac{1}{x^3}\right]\le\frac{1}{x^3}$

$\Rightarrow x^8\left(\frac{1}{x^3}-1\right)\le x^8\left[\frac{1}{x^3}\right]\le\frac{x^8}{x^3}$ for all x

But $\text{Lim}_{x\to 0}\,x^5=0=\text{Lim}_{x\to 0}(x^5-x^8)$; So, $\text{Lim}_{x\to 0}\,x^8\left[\frac{1}{x^3}\right]=0$

33. Since $\sqrt{x^4+4x^2+7}\ge 1$, so $0\le\frac{|x|}{\sqrt{x^4+4x^2+7}}\le|x|$

$\Rightarrow 0\le\text{Lim}_{x\to 0}\dfrac{|x|}{\sqrt{x^4+4x^2+7}}\le\text{Lim}_{x\to 0}|x|=0\Rightarrow\text{Lim}_{x\to 0}\dfrac{|x|}{\sqrt{x^4+4x^2+7}}=0$

34. Since $-1\le\sin\left(\frac{1}{3\sqrt{x}}\right)\le 1$, so $-x^4\le x^4\sin\left(\frac{1}{3\sqrt{x}}\right)\le x^4$

But $\text{Lim}_{x\to 0}\,x^4=0$, hence $0\le\text{Lim}_{x\to 0}\,x^4\sin\left(\frac{1}{3\sqrt{x}}\right)\le 0;\therefore\text{Lim}_{x\to 0}\,x^4\sin\left(\frac{1}{3\sqrt{x}}\right)=0$

WORKSHEET - 4

Level – I

1) B	2) B	3) B	4) C	5) C	6) A	7) C
8) D	9) A	10) A				

Level – II

1) D	2) D	3) C	4) B	5) D	6) B	7) B
8) A	9) B	10) C	11)D	12)A	13)D	14)D
15)D	16)D	17)B	18)B	19)		

1. $\text{Lt}_{x\to 0}\dfrac{x\cdot 2^x-x}{1-\cos x}=\text{Lt}_{x\to 0}\dfrac{x(2^x-1)/x^2}{(1-\cos x)/x^2}=\dfrac{\log 2}{1/2}=\log 4$

2. $\text{Lt}_{x\to 0}\frac{5^x+3^x-2^x-1}{x} = \underset{x\to 0}{\text{Lt}}(5^x\log 5 + 3^x\log 3 - 2^x\log 2) = \log 5 + \log 3 - \log 2 = \log\left(\frac{15}{2}\right)$

3. $\text{Lt}_{x\to 0}\left(\frac{a^x+b^x}{2}\right)^{1/x} = \sqrt{ab}$

4. $\text{Lt}_{x\to\infty}\left[\frac{a^{1/x}+b^{1/x}+c^{1/x}}{3}\right]^x = \sqrt[3]{abc}$

5. $\text{Lt}_{x\to\infty}\left[\frac{a_1^{1/x}+a_2^{1/x}+\cdots+a_n^{1/x}}{n}\right]^{nx} = \text{Lt}_{x\to 0}\left[\frac{a_1^x+a_2^x+\cdots+a_n^x}{n}\right]^{n/x} = \left[(a_1a_2\ldots a_n)^{1/n}\right]^n = a_1a_2\ldots a_n$

6. $\text{Lt}_{x\to 0}\left[\frac{1^x+2^x+\cdots+n^x}{n}\right]^{1/x} = \sqrt[n]{1.2\ldots n} = (n!)^{1/n}$

7. $\text{Lt}_{x\to 0}(1+x)^{2/x} = e^2$

8. $\text{Lt}_{x\to 0}\left(\frac{x+1}{x+2}\right)^{x+5} = \exp\left[\text{Lt}_{x\to\infty}\left(\frac{x+1}{x+2}-1\right)(x+5)\right] = \exp\left[\text{Lt}_{x\to\infty}\frac{-1}{x+2}\times(x+5)\right] = e^{-1}.$

9. $\underset{x\to 1}{\text{Lt}}(\log_5 5x)^{\log_x 5} = \underset{e^x\to 1}{\text{Lt}}(\log_5 5x - 1)\cdot\log_x 5 = \underset{e^x\to 1}{\text{Lt}}\log_5 x\cdot\log_x 5 = e^1 = e$

10. $\underset{x\to\pi/2}{\text{Lt}}(\sin x)^{\tan x} = \exp[\underset{x\to\pi/2}{\text{Lt}}(\sin x - 1)\tan x] = \exp\left[\underset{x\to\pi/2}{\text{Lt}}\frac{\sin x-1}{\cos x}\right] = \exp\left[\underset{x\to\pi/2}{\text{Lt}}\frac{\cos x}{-\sin x}\right] =$
$e^0 = 1$

11. $\text{Lt}_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{\sin x}{x-\sin x}} = \underset{e^x\to 0}{\text{Lt}}\frac{\sin x-x}{x}\cdot\frac{\sin x}{x-\sin x} = \underset{e^x\to 0}{\text{Lt}}\frac{-\sin x}{x} = e^{-1} = \frac{1}{e}$

12. $\text{Lt}_{x\to 0}\left(\frac{1+\tan x}{1+\sin x}\right)^{\text{cosec}\,x} = \text{Lt}_{e^x\to 0}\frac{\tan x-\sin x}{1+\sin x}\text{cosec}\,x$
$= \text{Lt}_{e^x\to 0}\frac{\tan x-\sin x}{(1+\sin x)\sin x} = \text{Lt}_{e^x\to 0}\frac{\tan x(1-\cos x)}{\sin x(1+\sin x)} = e^0 = 1$

13. $\text{L} = \exp\left[\underset{x\to\infty}{\text{Lt}}\frac{-1}{x+2}(x+5)\right] = e^{-1} = 1/e$

14. $\text{L} = e^{\text{Lt}\to 0\frac{\sin x-x}{x}\times\frac{\sin x}{x-\sin x}} = e^{\text{Lt}\to 0\frac{-\sin x}{x}} = e^{-1} = 1/e$

15. $L = \underset{x\to 0}{\text{Lt}}\left[\frac{e^{x^2}-1}{x^2}+\frac{1-\cos x}{x^2}\right] = 1+\frac{1}{2} = \frac{3}{2}$

16. [4] $\underset{x\to\infty}{Lt}\left(\frac{x^2+5x+3}{x^2+x+2}\right)^x\left(1^\infty form\right) = \exp\left\{\underset{x\to\infty}{Lt}\left(\frac{x^2+5x+3}{x^2+x+2}-1\right)x\right\} = \exp\left\{\underset{x\to\infty}{Lt}\frac{4x^2-x}{x^2+x+2}\right\} = e^4.$

17. $\log_{\sqrt{5}}\left(\frac{1}{4}+\frac{1}{8}+\cdots\ldots+\infty\right) = \log_{5^{1/2}}\left(\frac{1}{2}\right) = -\log_5(4)$

$\lim_{n\to\infty}\left(\frac{1}{5}\right)^{\log_{\sqrt{5}}\left(\frac{1}{4}+\frac{1}{8}+\cdots..\right)} = 5^{\log_5(4)} = 4$

18. $\lim_{x\to 2}\frac{2^x+2^{3-x}-6}{\sqrt{2^{-x}}-2^{1-x}} = \lim_{x\to 2}\frac{(2^x)^2-6.2^x+2^3}{\sqrt{2^x}-2}$

[Multiplying N^r and D^r by 2^x] $= \lim_{x\to 2}(2^x-2)\left(\sqrt{2^x}+2\right) = (2^2-2)(2+2) = 8$

19. Use L-Hospital

20. Given limit $= e^{\lim_{x\to\infty}(x+b)\left[\frac{x+a}{x+b}-1\right]}$

21. [4] $\overset{lt}{x \to \pi}\left(1-4\tan x\right)^{\cot x} = \overset{lt}{e}x \to \pi^{\cot x(1-4\tan x-1)} = e^{-4}$

22. [2] $\overset{lt}{x \to 1}^{(2-x)^{\tan\left(\frac{\pi x}{2}\right)}} = e^{\overset{lt}{x \to 1}^{\tan\left(\frac{\pi x}{2}\right)(2-x-1)}} = e^{\overset{lt}{x \to 1}^{\frac{(1-x)}{\cot\frac{\pi x}{2}}}}$ and using L hospital rule.

23. [1,2,4]; $L = e^{\underset{x \to 1}{Lt}\left(\log_2 2x-1\right)\log_x 5} = e^{\log_2 5}$

25. [3]; $L = \left[\underset{x \to \infty}{Lt}\left(1+\frac{3}{x}\right)^{x/3}\right]^3 = \left[\underset{\frac{3}{x} \to 0}{Lt}\left(1+\frac{3}{x}\right)^{x/3}\right]^3 = e^3$

26. [4]; $\underset{x \to 0}{Lt}(1+\tan x)^{\frac{1}{\tan x}} = e$

27. [2] ; $L = \left[\underset{x \to 0}{Lt}(1+x)^{1/x}\right]^2 = e^2$

28. a-s,b-q,c-p,d-r ; (a) $Lt_{x\to 0}\frac{2^x-1}{\sqrt{1+x}-1} = \underset{x\to 0}{Lt}\frac{2^x-1}{\sqrt{1+x}-1}\times\frac{\sqrt{1+x}+1}{\sqrt{1+x}+1}$

$= Lt_{x\to 0}\frac{2^x-1}{x}\cdot Lt_{x\to 0}(\sqrt{1+x}+1) = (\log 2)2 = \log 4$

(b) $\underset{x\to 0}{Lt}\frac{e^{\tan x}-1}{\sin x} = \underset{x\to 0}{Lt}\frac{e^{\tan x}-1}{\tan x\cos x} = \underset{x\to 0}{Lt}\frac{e^{\tan x}-1}{\tan x}\times\underset{x\to 0}{Lt}\frac{1}{\cos x} = 1\times 1 = 1$

(c) $L = \underset{x \to 0}{Lt}\frac{x\left(10^x-1\right)}{1-\cos x} = \underset{x \to 0}{Lt}\frac{10^x-1}{x}.\underset{x \to 0}{Lt}\frac{x^2}{1-\cos x}$ = 2 log 10 = log 100

(d) $L = Lt_{x\to 0}\frac{5^x-1}{x}\cdot\frac{2^x-1}{x} = \log 5\cdot\log 2$

29. $L = \underset{x \to 0}{Lt}[\log(1-3x)]^{1/x} = \left[\underset{x \to 0}{Lt}\log(1-3x)^{\frac{1}{-3x}}\right]^{-3} = \log\underset{x \to 0}{Lt}\left[\log(1-3x)^{\frac{1}{-3x}}\right]^{-3} = \log_e e^{-3} = -3$

30. $L = Lt_{x\to 0}\frac{e^{\tan x}-1+1-e^x}{\tan x-x}$

$= Lt_{x\to 0}\left\{\frac{\tan x\left(\frac{e^{\tan x}-1}{\tan x}\right)-\left(\frac{e^x-1}{x}\right)x}{\tan x-x}\right\} = Lt\frac{\tan x-x}{\underset{x\to 0}{\tan x-x}} = 1$

31. a-s,b-t,c-q,d-p ;

(a) $L = e^{Lt\to\infty\frac{(4x+1)x}{x^2+x+2}}$

(b) $L = e^{Lt\left(\frac{x}{x\to\infty}-1\right)2x} = e^{Lt\left(\frac{-4x}{x+2}\right)} = e^{-4}$

(c) $L = e^{Lt\left(\frac{a^x+b^x}{2}-1\right)\frac{1}{x}} = e^{Lt\left[\frac{a^x+b^x-2}{2}\right]\frac{1}{x}} = e^{\frac{1}{2}xt\left[0\left[\frac{a^x-1}{x}+\frac{b^x-1}{x}\right]\right.} = e^{\frac{1}{2}} \cdot [(\log a + \log b)] = e^{\log\sqrt{ab}} = \sqrt{ab}$

(d) L - exp. $\left[\underset{x\to\infty}{Lt}\frac{a-b}{x+b}\times(x+b)\right] = e^{a-b}$

32. $L = -Lt_{x\to 0}\frac{(1-\cos x)(1+\cos x)(\cos x-e^x)}{(1+\cos x)x^n}$

$$= Lt_{x\to 0}\frac{\left(\frac{\sin x}{x}\right)^2\left[\frac{1-\cos x}{x}+\frac{e^x-1}{x}\right]}{x^{n-3}}\cdot\frac{1}{1+\cos x}$$

since $-[\cos x - e^x] = 1 - \cos x + e^x - 1$

L is finite non-zero then n = 3.

For n = 1,2 ⇒ L = 0 and for n = 4 ⇒ $L = \infty$

33. Standard

34. Given limit is ⇒ $e^{\lim_{x\to\infty}\left[1+\frac{\lambda}{x}+\frac{\mu}{x^2}-1\right]\cdot 2x} = e^2 \Rightarrow e^{\lim_{x\to\infty}\left(\lambda+\frac{\mu}{x}\right)\cdot 2} = e^2 \Rightarrow e^{2\lambda} = e^2 \Rightarrow \lambda = 1, \mu \in R$

35. Given limit $= \lim_{x\to 0} n\cdot\left[\left(\frac{1}{n}\right)^{\frac{1}{\sin^2 x}}+\left(\frac{2}{n}\right)^{\frac{1}{\sin^2 x}}+\cdots+\left(\frac{n-1}{n}\right)^{\frac{1}{\sin^2 x}}+1\right] = n(0+0+\cdots+0+1) = n$

13. Continuity

WORKSHEET – 1

Level – I

1) A	2) B	3) C	4) B	5) A	6) C	7) C

Level – II

1) B	2) A	3) C	4) B	5) B	6) A	7) C
8) A	9) D	10) B	11) B	12) A	13) C	14) C
15) C	16) D	17) D	18) A	19) A	20) A,B,C	21) A
22) a-q; b-p; c-s; d-r		23) A	24) A	25) B	26) A,B	27) B
28) C	29) C	30) A,C	31) a-q; b-r; c-q; d-s		32) C	

1. $|x|$ is continuous everywhere.
2. L.H.L $= \lim_{x\to\pi^-} \tan x = +\infty$ and R.H.L $= \lim_{\pi^+} \tan x = -\infty$. $x \to \frac{\pi^-}{2}$ $x \to \frac{\pi^+}{2}$

   $\therefore f(x)$ is discontinuous at $x = \frac{\pi}{2}$.
3. $f(c) = \frac{f(a)+f(b)}{2}$ (intermediate value theorem).
4. e^x is continuous on ' R '.
5. $\lim_{x\to2} f(x) = \lim_{x\to2} |x| = 2 = f(2)$

   $\therefore f(x)$ is continuous at x = 2.
6. $f(x) = \cos^{-1}(x)$ is continuous on $[0, \pi]$.
7. $f(x) = \frac{1}{x}$ is discontinuity at $x = 0$.

Level – II

1. $\lim_{x\to0} f(x) = \lim_{x\to0} \frac{x\sin 5x}{\tan 2x \tan 7x} = \lim_{x\to0} \frac{x^2 \sin 5x}{x\tan 2x \tan 7x} = \frac{\lim_{5x\to0} \sin\frac{5x}{5x}\times5}{\lim_{2x\to0}\frac{\tan 2x}{2x}\times2\lim_{7x\to0}\frac{\tan 7x}{7x}} = \frac{5}{2\times7} = \frac{5}{14}$.

   But, $f(0) = \frac{5}{9}$. As $\lim_{x\to0} f(x) \neq f(0), f(x)$ is discontinuous at x = 0.
2. $\lim_{x\to0} f(x) = \lim_{x\to0} (1+x)^{\frac{5}{x}} = \left[\lim_{x\to0} (1+x)^{\frac{1}{x}}\right]^5 = e^5 = f(0)$. $\therefore f(x)$ is continuous at $x = 0$.
3. $\lim_{x\to\frac{\pi}{2}} f(x) = \lim_{x\to\frac{\pi}{2}} \frac{k\cos x}{\pi-2x} = f\left(\frac{\pi}{2}\right) \Rightarrow \lim_{x\to\frac{\pi}{2}} \frac{k\cos x}{\pi-2x} = 3 \Rightarrow \lim_{x\to\frac{\pi}{2}} \frac{k\sin\left(\frac{\pi}{2}-x\right)}{2\left(\frac{\pi}{2}-x\right)} = 3 \Rightarrow \frac{k}{2} = 3 \Rightarrow k = 6$.

4. $\text{Lim}_{x\to 2^-} f(x) = \text{Lim}_{x\to 2^-}[x] = 1$ and $\text{Lim}_{x\to 2^+} f(x) = \text{Lim}_{x\to 2^+}[x] = 2$

   $\therefore \lim_{x\to 2^-} f(x) = 1 \neq \lim_{x\to 2^+} f(x) = 2 = f(2)$ $\therefore f(x)$ is right continuous.

5. Given, $f(x) = \frac{\sin x}{x}$ is continuous at $x = 0$.

   $\therefore \lim_{x\to 0} f(x) = f(0) \Rightarrow \lim_{x\to 0} \frac{\sin x}{x} = f(0) \Rightarrow f(0) = 1.$

6. Given, $f(x) = \frac{x^2-10x+25}{x^2-7x+10}$ is continuous at $x = 5$.

   then, $\lim_{x\to 5} f(x) = f(5) \Rightarrow \lim_{x\to 5} \frac{x^2-10x+25}{x^2-7x+10} = f(5)$

   $\Rightarrow \lim_{x\to 5} \frac{(x-5)(x-5)}{(x-2)(x-5)} = f(5) \Rightarrow f(5) = 0.$

7. Given, $f(x) = \frac{\sqrt{1+x}-1}{x}$ is continuous at $x = 0$.

   $\Rightarrow \lim_{x\to 0} f(x) = f(0) \Rightarrow \lim_{x\to 0} \frac{\sqrt{1+x}-1}{x} = f(0) \Rightarrow \lim_{x\to 0} \frac{(1+x)-1}{x(\sqrt{1+x}+1)} = f(0)$ $\therefore f(0) = \frac{1}{2}.$

8. Given $f(x) = \frac{\cos x-\sin x}{\cos(2x)}$ is continuous on $\left[0, \frac{\pi}{2}\right]$

   $f(x)$ is continuous at $x = \frac{\pi}{4}$.

   $\Rightarrow \lim_{x\to \frac{\pi}{4}} f(x) = f\left(\frac{\pi}{4}\right) \Rightarrow \lim_{x\to \frac{\pi}{4}} \frac{\cos x - \sin x}{\cos(2x)} = f\left(\frac{\pi}{4}\right)$

   $\Rightarrow \lim_{x\to \frac{\pi}{4}} \frac{\cos x-\sin x}{\cos^2 x-\sin^2 x} = f\left(\frac{\pi}{4}\right) \Rightarrow \frac{1}{\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}} = f\left(\frac{\pi}{4}\right) \Rightarrow f\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}.$

9. Case (1): Continuity at $x = 0$

   $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} x^2 = 0.$

   and $f(0) = 0$

   $\therefore$ f is continuous at $x = 0$.

   Case (2): Continuity at x = 1.

   $\lim_{x\to 1^-} f(x) = (1)^2 = 1$ and $\lim_{x\to 1^+} f(x) = (1) = 1$

   $\therefore \lim_{x\to 1} f(x) = 1 = f(1)$. So, f is continuous at x = 1.

   Case (3): Continuity at x = 2

   $f(x) = 2; \lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} x = 2 = f(2).$

   $\therefore f$ is continuous at $x = 2$

   Hence, $f(x)$ continuous on [0,2].

10. $f(x)$ is continuous at $x = 0$.

    $\Rightarrow \lim_{x\to 0^+} f(x) = f(0) \Rightarrow \lim_{x\to 0^+} e^{ax+b} = a^2\cos^2(0) + b^2\sin^2(0) \Rightarrow e^b = a^2$

    $\Rightarrow b = \log(a)^2 \Rightarrow b = 2\log|a|$ and $a \neq 0$.

11. [2] Given, $f(x)$ is continuous at $x = 0$.

    $\lim_{x\to 0^-} f(x) = f(0) = \lim_{x\to 0^+} f(x)$

$\Rightarrow \lim_{x\to 0^-}(4-3x) = \lim_{x\to 0^+}(2a+x) \Rightarrow 4-3(0) = 2a+0 \Rightarrow 4 = 2a \Rightarrow a = 2.$

12. Let us take ' f ' is continuous at $x = 1$.

Then, $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x) = f(1)$

$\Rightarrow 3(1)^2 - 4(a) + 2b = 10 \Rightarrow 4a - 2b = -7 \rightarrow (1)$

Take f is continuous at $x = -1$

$\Rightarrow \lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x) = f(-1) \Rightarrow -a - b = 3 + 4a + 2b \Rightarrow 5a + 3b = -3 \rightarrow (2)$

from (1) & (2)

$a = \frac{-27}{22} \& b = \frac{23}{22}.$

13. $f(x)$ is continuous on R

$f(x)$ is continuous at $x = 0,1,3$

(i) $f(x)$ is continuous at 0 .

$\therefore \lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x) = f(0)$

$\Rightarrow \lim_{x\to 0^-} \sin x = \lim_{x\to 0^+}(x^2 + a) = 0 \Rightarrow a = 0$

(ii) $f(x)$ is continuous at $x = 3$

$\therefore \lim_{x\to 3^-} f(x) = f(3) = \lim_{x\to 3^+} f(x) \Rightarrow 3b + 3 = -3 \Rightarrow b = -2$

$\therefore a = 0, b = -2.$

14. $f(x)$ is continuous at $x = 0$

$\Rightarrow \lim_{x\to 0^-} f(x) = f(0) = \lim_{x\to 0^+} f(x)$

$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} \frac{\sin(a+1)x + \sin x}{x} = (a+1) + 1 = a + 2$$

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \frac{(x+bx^2)^{\frac{1}{2}} - x^{\frac{1}{2}}}{bx^{\frac{3}{2}}} = \lim_{x\to 0^+} \frac{(x+bx^2) - x}{bx^{\frac{3}{2}}\left[(x+bx^2)^{\frac{1}{2}} + x^{\frac{1}{2}}\right]}$$

$$= \lim_{x\to 0^+} \frac{bx^2}{bx^{\frac{3}{2}}\left[(x+bx^2)^{\frac{1}{2}} + x^{\frac{1}{2}}\right]} = \lim_{x\to 0^+} \frac{x^2}{x^2\left[(1+bx)^{\frac{1}{2}} + 1\right]} (b \neq 0)$$

$$= \lim_{x\to 0^+} \frac{x^2}{\left[(1+bx)^{\frac{1}{2}} + 1\right]} = \frac{1}{2}$$

From (1), (2) and (3), $a + 2 = c = \frac{1}{2}$ and $b \neq 0 \Rightarrow a = \frac{-3}{2}$ and $c = \frac{1}{2}$.

15. Given, $f(x)$ is continuous at $x = 0$.

i.e $f(0) = \lim_{x\to 0} f(x)$

$$\Rightarrow = \lim_{x\to 0} \frac{36^x - 9^x - 4^x + 1}{2\sin^2\left(\frac{x}{4}\right)} \Rightarrow k = \lim_{x\to 0} \frac{\frac{9^x - 1}{x} \times \frac{4^x - 1}{x}}{\frac{2\sin^2\left(\frac{x}{4}\right)}{\left(\frac{x}{4}\right)^2} \times \frac{1}{16}}$$

$\Rightarrow k = \dfrac{\log 9 \times \log 4}{2 \times \frac{1}{16}}$

$\Rightarrow k = 8\log 3^2 \times \log 2^2 \Rightarrow k = (8 \times 2 \times 2)\log 3 \times \log 2 \Rightarrow k = 32\log 2\log 3.$

16. $f(x)$ is continuous at $x = 0 \Rightarrow \lim_{x\to 0} f(x) = f(0)$

$\Rightarrow \lim\limits_{x\to 0} \dfrac{(4^x - 1)^3}{\dfrac{\sin\left(\frac{x}{p}\right)\log\left(1 + \frac{x^2}{3}\right)}{x} x^2} = 12(\log 4)^3$

$\Rightarrow \lim_{x\to 0} \dfrac{\left(\frac{4^x - 1}{x}\right)^3}{\dfrac{\sin\left(\frac{x}{p}\right)}{p\left(\frac{x}{p}\right)} \times \dfrac{\log\left(1 + \frac{x^3}{3}\right)}{3\left(\frac{x^2}{3}\right)}} = 12(\log 4)^3$

$\Rightarrow \lim_{x\to 0} \dfrac{(\ln 4)^3}{\frac{1}{p} \times \frac{1}{3}} = 12(\log 4)^3$

$\Rightarrow 3p(\log 4)^3 = 12(\log 4)^3 \Rightarrow 3p = 12 \Rightarrow p = 4.$

17. $\lim\limits_{x\to \frac{\pi^-}{2}} \dfrac{\sin(\cos x) - \cos x}{(\pi - 2x)^3} = \lim_{h\to 0} \dfrac{\sin\left(\cos\left(\frac{\pi}{2} - h\right)\right) - \cos\left(\frac{\pi}{2} - h\right)}{\left(\pi - 2\left(\frac{\pi}{2} - h\right)\right)^3}$

$= \lim_{h\to 0} \dfrac{\sin(\sinh) - \sinh}{(2h)^3} = \lim_{h\to 0} \dfrac{\sin(\sinh) - \sinh}{8h^3}$

$= \lim_{h\to 0} \dfrac{\left(\sinh - \frac{\sin^3 h}{3!} + \frac{\sin^5 h}{5!} \cdots..\right) - \sinh}{8h^3}$

$= \lim_{h\to 0} \dfrac{\sin^3 h}{h^3} \dfrac{\left(\frac{-1}{3!} + \frac{\sin^2 h}{5!} - \cdots.....\right)}{8} = \dfrac{(1)\left(\frac{-1}{3!}\right)}{8} = \dfrac{-1}{6\times 8} = \dfrac{-1}{48}.$

18. $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} x \dfrac{e^{\frac{1}{x}} - e^{\frac{-1}{x}}}{e^{\frac{1}{x}} + e^{\frac{-1}{x}}} = \lim_{x\to 0^+} x \dfrac{1 - e^{\frac{-2}{x}}}{1 + e^{\frac{-2}{x}}} = 0.1 = 0$

$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} x \dfrac{e^{\frac{1}{x}} - e^{\frac{-1}{x}}}{e^{\frac{1}{x}} + e^{\frac{-1}{x}}} = \lim_{x\to 0^-} x \dfrac{e^{\frac{2}{x}} - 1}{e^{\frac{2}{x}} + 1} = 0 \cdot (-1) = 0.$

Hence, $\lim_{x\to 0} f(x)$ exists. As, $\lim_{x\to 0} f(x) = f(0)$, $f(x)$ is continuous at $x = 0$.

19. $\text{LHL} = \lim_{x\to 0^-} (1 + |\sin x|)^{\frac{a}{\sin x}} = e^{\lim_{x\to 0^-} (1+|\sin x|-1)\frac{a}{|\sin x|}}$

$\text{LHL} = e^a$

$\text{RHL} = \lim_{x\to 0} e^{\frac{\tan 2x}{\tan 3x}} = e^{\lim_{x\to 0}\frac{\tan 2x}{\tan 3x}} = e^{\frac{2}{3}}$

Given, $f(0) = b \Rightarrow e^a = e^{\frac{2}{3}} = b \Rightarrow a = \dfrac{2}{3}, b = e^{\frac{2}{3}}.$

20. [i) Continuity at x = 1

LHL $= \lim_{x\to 1^-} f(x) = \lim_{x\to 1}(x+1) = 1+1 = 2$

RHL $= \lim_{x\to 1^+} f(x) = \lim_{x\to 1} 2x = 2(1) = 2$

$\therefore$ LHL = RHL = 2.

So, $f(x)$ is continuous at $x = 1$.

ii) $\lim_{x\to 2^-} f(x) = \lim_{x\to 2} 2x = 2(2) = 4$; $\lim_{x\to 2^+} f(x) = \lim_{x\to 2}(1+x^2) = 1+4 = 5$

$LHL \neq RHL$ $\therefore f(x)$ is a discontinuous at $x = 2$

21. Both are true statements and R is the correct explanation of A.

22. a) $f(x) = [x]$

The greatest integer function is continuous on $\mathrm{R} - Z$.

b) $\lim_{x\to 0^-} \operatorname{sgn}(x) = -1$ and $\lim_{x\to 0^+} \operatorname{sgn}(x) = 1$. Therefore, $f(x) = \operatorname{sgn}(x)$ is continuous on $R - \{0\}$.

c) $f(x) = \log|x|$ is continuous on $R - \{0\}$.

d) At x = 1,2,3, $f(x)$ is 0 .

23. $f(x) = \begin{cases} \frac{\sqrt{1+kx}-\sqrt{1-kx}}{x} \text{ for } -1 \le x < 0 \\ 2x^2 + 3x - 2 \text{ for } 0 \le x \le 1 \end{cases}$

As, $f(x)$ is continuous at $x = 0, \lim_{x\to 0^-} f(x) = f(0) \Rightarrow \lim_{x\to 0} \frac{\sqrt{1+kx}-\sqrt{1-kx}}{x} = -2$

$\lim_{x\to 0} \frac{2k}{\sqrt{1+kx}+\sqrt{1-kx}} = -2 \Rightarrow k = -2$ $\therefore \left(\frac{1}{3}\right)^k = 9.$

24. $f(0) = \lim_{x\to 0} \frac{1-\cos(1-\cos x)}{x^4} = \lim_{x\to 0} \frac{1-\cos(1-\cos x)}{(1-\cos x)^2} \times \left(\frac{1-\cos x}{x^2}\right)^2 = \frac{1}{2} \times \left(\frac{1}{2}\right)^2 = \frac{1}{8}.$

25. As, $f(x)$ is continuous at $x = 0$,

$\lim_{x\to 0^-} f(x) = f(0) \Rightarrow \lim_{x\to 0} \frac{1-\cos 4x}{x^2} = a \Rightarrow 16 \times \frac{1}{2} = a \Rightarrow a = 8.$

26. $f(x) = \frac{1}{x^2-7x+10} = \frac{1}{(x-2)(x-5)}$

$\therefore$ At $x = 2$ and $x = 5, f(x)$ is not continuous.

27. $f(x) = \begin{cases} [x] & -2 \le x \le -\frac{1}{2} \\ 2x^2 - 1 & -\frac{1}{2} < x \le 2 \end{cases}$

At $x = -2$

R.H.L $= \lim_{x\to -2^+} f(x) = \lim_{x\to -2^+} [x] = -2 = f(-2)$. Therefore, $f(x)$ is continuous at $x = -2$.

Since, [x] is not continuous at integers, at $x = -1, f(x)$ is not continuous.

At $x = -\frac{1}{2}$,

L.H.L $= \lim_{x\to-\frac{1}{2}^-} f(x) = \lim_{x\to-\frac{1}{2}^-}[x] = -1$; R.H.L $= \lim_{x\to-\frac{1}{2}^+} f(x) = \lim_{x\to-\frac{1}{2}^+} 2x^2 - 1 = \frac{-1}{2}$

As, L.H.L ≠ R.H.L, $f(x)$ is not continuous at $x = -\frac{1}{2}$.

At $x = 2, \lim_{x\to2^-} f(x) = 7 = f(2)$. Therefore, $f(x)$ is continuous at $x = 2$.

Hence, $f(x)$ is not continuous at two points -1 and $-\frac{1}{2}$.

28. $$f(x-1) = \begin{cases} [x-1] & -2 \le x-1 \le -\frac{1}{2} \\ 2(x-1)^2 - 1 & -\frac{1}{2} < x-1 \le 2 \end{cases}$$

$$f(x-1) = \begin{cases} [x]-1 & -1 \le x \le \frac{1}{2} \\ 2(x-1)^2 - 1 & \frac{1}{2} < x \le 3 \end{cases}$$

At $x = 0, f(x-1)$ is not continuous.

At $x = \frac{1}{2}$, L.H.L $= \lim_{x\to\frac{1}{2}^-} f(1-x) = \lim_{x\to\frac{1}{2}^-}\{[x]-1\} = -1$

R.H.L $= \lim_{x\to\frac{1}{2}^+} f(1-x) = \lim_{x\to\frac{1}{2}^+} 2(x-1)^2 - 1 = -\frac{1}{2}$

As, L.H.L ≠ R.H.L, $f(1-x)$ is not continuous at $x = \frac{1}{2}$.

Hence, $f(1-x)$ is not continuous at two points 0 and $\frac{1}{2}$.

29. [3] $$|f(x)| = \begin{cases} 2 & -2 \le x < -1 \\ 1 & -1 \le x \le -\frac{1}{2} \\ 1-2x^2 & -\frac{1}{2} < x \le \frac{1}{\sqrt{2}} \\ 2x^2 - 1 & \frac{1}{\sqrt{2}} < x \le 2 \end{cases}$$

$$\frac{d}{dx}(|f(x)|) = \begin{cases} 0 & -2 \le x < -1 \\ 0 & -1 \le x \le -\frac{1}{2} \\ -4x & -\frac{1}{2} < x \le \frac{1}{\sqrt{2}} \\ 4x & \frac{1}{\sqrt{2}} < x \le 2 \end{cases}$$

$|f(x)|$ is not continuous at $x = -1$ and $x = -\frac{1}{2}$

$|f(x)|$ is not differentiable at $x = -1, \frac{-1}{2}$ and $x = \frac{1}{\sqrt{2}}$. [at $x = -1$, LHD = RHD, but the function is not continuous] .

30. [1,3] As, $f(x)$ is continuous at R, $\lim_{x\to1^+} f(x) = f(1) \Rightarrow k^2 - k = 2 \Rightarrow k^2 - k - 2 = 0 \Rightarrow k = -1$ and $k = 2$.

31. $a \to q; b \to r; c \to q; d \to s$.

a) $\lim_{x\to 0} f(x) = \lim_{x\to 0} x\cos\left(\frac{1}{x^2}\right) = 0 \neq f(0). \therefore f(x)$ is discontinuous at $x = 0$.

b) $\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} [x] = 2 = f(2)$. Hence, $f(x)$ is right continuous at $x = 2$.

c) $f(x) = x\sin\left(\frac{1}{x}\right) - [x]\sin\left(\frac{1}{x}\right) \cdot \lim_{x\to 0} x\sin\left(\frac{1}{x}\right) = 0$ and $\lim_{x\to 0} [x]\sin\left(\frac{1}{x}\right)$ doesn't exist.

$\therefore f(x)$ is discontinuous at $x = 0$.

d) $L.H.L = \lim_{x\to -2^-} \frac{\tan^{-1}(x+2)}{|x+2|} = -1$ and R.H.L $= \lim_{x\to -2^+} \frac{\tan^{-1}(x+2)}{|x+2|} = 1$

Given, $f(-2) = -1$. So, $f(x)$ is left continuous.

32. $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \frac{1}{1+e^{1/x}} = \frac{1}{\infty} = 0$

$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} \frac{1}{1+e^{1/x}} = \lim_{h\to 0} \frac{1}{1+e^{1/-\mathrm{h}}} = \frac{1}{1} = 1$$

We have $\lim_{x\to 0^+} f(x) = f(0) \neq \lim_{x\to 0^-} f(x)$.

$\therefore f(x)$ is right continuous at $x = 0$.

14. Differentiation

WORKSHEET – 1

Level – I

1) A 2) B 3) C 4) C 5) A 6) A

7) A 8) D 9) A 10) D

Level – II

1) B 2) C 3) B 4) A 5) B 6) B 7) 0

8) A 9) A 10) A 11) A 12) B 13) A 14) B

15) D 16) B 17) B 18) A 19) C 20) 0 21) A,B

22) A 23) C 24) D 25) C 26) a-r; b-q; c-p; d-s,t

27) B 28) A,B,C,D 29) B 30) A 31) B

32) a-r; b-s; c-q; d-p

33) 0 34) A,B,C 35) A

1. $\frac{d}{dx}\left[3\sqrt[3]{x}-\frac{4}{\sqrt[4]{x}}\right]=3\frac{1}{3}x^{-\frac{2}{3}}-4\left(-\frac{1}{4}\right)x^{\frac{-5}{4}}=x^{\frac{-2}{3}}+x^{\frac{-5}{4}}$

2. $y=3^{\log_9(1+\tan^2 x)}\Rightarrow y=3^{\log_3\sec^2 x}=\sec x\Rightarrow\frac{dy}{dx}=\sec x\tan x$

3. $\frac{d}{dx}\left(\frac{a-b\cos x}{a+b\cos x}\right)=\frac{(a+b\cos x)(b\sin x)-(a-b\cos x)(-b\sin x)}{(a+b\cos x)^2}=\frac{2ab\sin x}{(a+b\cos x)^2}$

4. $y=\cos(1-x^2)^2\Rightarrow\frac{dy}{dx}=-\sin(1-x^2)^2\cdot 2(1-x^2)\cdot(-2x)=4x(1-x^2)\sin(1-x^2)^2$

5. $y=\log\frac{1-x^2}{1+x^2}\Rightarrow y=\log(1-x^2)-\log(1+x^2)\Rightarrow\frac{dy}{dx}=\frac{1}{1-x^2}(-2x)-\frac{1}{1+x^2}(2x)=\frac{-4x}{1-x^4}$

6. $f^1(x)=\begin{cases}1, 0\le x\le 2\\ 0, x\ge 2\end{cases}\Rightarrow f^1(2^+)=0, f^1(2^-)=1\Rightarrow f^1(2^+)\ne f^1(2^-)$

7. $f(-x)=f(x)\Rightarrow f^1(x)+f^1(-x)=0\Rightarrow f^1(0)+f^1(0)=0\Rightarrow f^1(0)=0$

8. $f(x)=\begin{cases}x^2, x>0\\ -x^2, x<0\end{cases}\Rightarrow f'(x)=\begin{cases}2x, x>0\\ -2x, x<0\end{cases}\Rightarrow f'(0^+)=f'(0^-)$

9. $y=(\log x)^3\Rightarrow\frac{dy}{dx}=3(\log x)^2\frac{1}{x}=\frac{3(\log x)^2}{x}$

10. $\frac{d}{dx}(\sqrt{\sin\sqrt{x}})=\frac{1}{2\sqrt{\sin\sqrt{x}}}\frac{d}{dx}(\sin\sqrt{x})=\frac{\cos\sqrt{x}}{2\sqrt{\sin\sqrt{x}}}\frac{1}{2\sqrt{x}}=\frac{\cos\sqrt{x}}{4\sqrt{x\sin\sqrt{x}}}$

11. $\frac{d}{dx}(\sin(\sqrt{2x-3}))=\cos\sqrt{2x-3}\frac{1}{2\sqrt{2x-3}}\cdot 2=\frac{\cos\sqrt{2x-3}}{\sqrt{2x-3}}$

12. $y=2^{ax}\Rightarrow\frac{dy}{dx}=2^{ax}\cdot\log 2\cdot a\Rightarrow\log 256=a\cdot 2^{ax}\cdot\log 2\Rightarrow\log 2^8=a\cdot 2^{ax}\cdot\log 2$

    $\Rightarrow 8\log 2=a\cdot 2^{ax}\log 2\ 8=a\cdot 2^a\Rightarrow a=2$

13. $\frac{d}{dx}(\log(\log(\log x)))=\frac{1}{\log(\log x)}\frac{1}{\log x}\cdot\frac{1}{x}$

14. $f(x)=x-x^2+x^3-x^4+\cdots\ldots\propto$

$f(x)=\frac{x}{1+x}\Rightarrow f'(x)=\frac{1}{(1+x)^2}\left[r=\frac{-x^2}{x}=-x,|r|<1 s_\infty=\frac{a}{1-r}\right]$

15. $\mathrm{Lt}_{x\to 2}\frac{xf(x)-2f(x)}{x-2}$

$$=\underset{x\to 2}{Lt}\frac{xf^1(x)+f(x)-2f^1(x)}{1}=2f^1(2)+f^1(2)-2f^1(2)=2(1)+4-2(1)=4$$

16. $f(x)=\begin{cases}ax^2+1, x<1\\ x^2+ax+b, x>1\end{cases}$ is differentiable $x=1$

Then $f(x)$ is continuous at $x=1$

$f(1^-)=f(1^+)$or $a+1=1+a+b$ or $b=0$

Also ${}_f f^1(x)=\begin{cases}2ax, & x<1\\ 2x+a, & x>1\end{cases}$

We must have $f^1(1^+)$or $2a=2+a$ or $a=2$

17. $f(e^+)=\lim_{h\to 0}(e+h-e)2^{-2^{\frac{1}{e-(e+h)}}}$

$=\lim_{h\to 0}(h)2^{-2^{-\frac{1}{h}}}$

$=0\times 1=0($ As forh $\to 0,-\frac{1}{h}\to-\infty\Rightarrow 2^{\frac{1}{h}}\to 0\big)$

$f(e^-)=\lim_{h\to 0}(-h)2^{-2^{\frac{1}{h}}}=0\times 0=0$

Hence, $f(x)$ is continuous at $x=e$

$$f^1(e^+)=\lim_{h\to 0}\frac{f(e-h)-f(0)}{-h}=\lim_{h\to 0}\frac{(-h)2^{-2^{\frac{1}{h}}}-0}{-h}$$

$$=\lim_{h\to 0}2^{-2^{\frac{1}{h}}}=1$$

$$f^1(e^-)=\lim_{h\to 0}\frac{f(e-h)-f(0)}{-h}=\lim_{h\to 0}\frac{(-h)2^{-2^{\frac{1}{h}}}-0}{-h}$$

$$=\lim_{h\to 0}2^{-2^{\frac{1}{h}}}=0$$

Hence, $f(x)$ is non-differentiable at $x=0$

18. $\sqrt{\tan y}=e^{\cos 2x}\cdot\sin x$

$\log(\tan y)^{1/2}=\log e^{\cos 2x}\cdot\sin x\Rightarrow\frac{1}{2}\log(\tan y)=\cos 2x\log_e^e+\log\sin x$

$\Rightarrow\frac{1}{2}\cdot\frac{1}{\tan y}\cdot\sec^2 y\frac{dy}{dx}=-\sin 2x\cdot 2+\frac{1}{\sin x}\cdot\cos x$

$\Rightarrow\frac{1}{2}\frac{1}{\mathrm{Tan}\,y}\sec^2 y\frac{dy}{dx}=\cot x-2\sin 2x\Rightarrow\frac{1}{\sin 2y}\frac{dy}{dx}=\cot x-2\sin 2x$

$\Rightarrow\frac{dy}{dx}=\sin 2y(\cot x-2\sin 2x).$

19. $\frac{d}{dx}\left(\frac{1+x^2+x^4}{1+x+x^2}\right) = ax + b \Rightarrow \frac{d}{dx}\frac{\left((x^2+1)-x\right)\left((x^2+1)+x\right)}{(x^2+x+1)} = ax + b.$

$\Rightarrow \frac{d}{dx}(x^2 + 1 - x) = ax + b \Rightarrow 2x - 1 = ax + b \Rightarrow a = 2, b = -1$

20.

$f(x) = |x - 1| + |x - 3| \Rightarrow f(x) = x - 1 - (x - 3)$ Since, $1 < x < 3$
$= x - 1 - x + 3 \Rightarrow f(x) = 2 \Rightarrow f^1(x) = 0.$

21. Conceptual

22. Conceptual

23. $f^1(x) = 10$

24. $D[x^2] = 2x.$

25. $f(x) = k \Rightarrow f^1(x) = 0.$

26. a) $\frac{d}{dx}(2x + 1) = 2$

b) $\frac{d}{dx}(3x^2) = 6x$

c) $\frac{d}{dx}(5) = 0$

d) $\frac{d}{dx}[f(x)] = f^1(x) = Lt\frac{f(x+h)-f(x)}{h}$

27. $\mathrm{Lt}_{h\to 0}\frac{2010(x+h)-2010x}{h} = 2010 \Rightarrow \frac{2010}{1005} = 2$

28. Conceptual.

29. $f(x) = \log(x^2 + 1) \Rightarrow f^1(x) = \frac{1}{x^2+1}(2x).$

30. $f(x) = \sin(\sqrt{2x - 3}) \Rightarrow f^1(x) = \cos\sqrt{2x - 3} \cdot \frac{1}{2\sqrt{2x-3}} \cdot 2 = \frac{\cos\sqrt{2x-3}}{\sqrt{2x-3}}$

31. $\frac{d}{dx}\left(\frac{\sin x}{1+\cos x}\right) = \frac{d}{dx}\left(\frac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}\right) = \frac{d}{dx}\mathrm{Tan}\frac{x}{2} = \sec^2\frac{x}{2} \cdot \frac{1}{2} = \frac{1}{2\cos^2\frac{x}{2}} = \frac{1}{1+\cos x}$

32. $f(x) = 2^x$

$g(x) = 3^x$

$f^1(x) = 2^x \log 2$

$g^1(x) = 3^x \log 3$

$f^1(0) = \log 2,$

$g^1(0) = \log 3$

$a \to r, b \to s, c \to q, d \to p$

33. $f^1(x) = \frac{\sqrt{a}}{2\sqrt{x}} - \frac{a^2}{2\sqrt{a}} \times \frac{1}{x\sqrt{x}} \Rightarrow f^1(a) = 0$

34. $f^1(x) = \lim_{x\to 0}\frac{x^p\cos\left(\frac{1}{x}\right)-0}{x-0} = \lim_{x\to 0} x^{p-1}\cos\left(\frac{1}{x}\right).$ $\therefore p > 1$

35. $f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \ldots \Rightarrow f^1(x) = 1 + x + \frac{x^2}{2!} + \cdots \ldots \Rightarrow f^1(0) = 1$

WORKSHEET-2

Level – I

1) B	2) C	3) D	4) A	5) B	6) B	7) A

Level – II

1) A	2) A	3) C	4) B	5) B	6) D	7) A
8) B	9) C	10) A	11) D	12) A	13) C	14) A
15) D	16) 1	17) A	18) A	19) B	20) D	21) A
22) 375	23) C, D	24) B	25) C	26) C	27) C	
28) a-r; b-t; c-p; d-q		29) A	30) A,C	31) a-q; b-r; c-p; d-s		
32) A, B, C	33) B	34) a-q,r; b-p,r,s; c-q, r; d-q,r				

1. (3] Put x = $\sin\theta \Rightarrow \theta = \sin^{-1}$x, then $\frac{d}{dx}\left(\sin^{-1}x + \cos^{-1}\sqrt{1-x^2}\right) = 2\frac{d}{dx}\sin^{-1}x = \frac{2}{\sqrt{1-x^2}}$.
2. [1] $\frac{\mathrm{d}}{\mathrm{dx}}\left(\tan^{-1}\frac{\mathrm{x}}{1+\mathrm{x}^2} + \tan^{-1}\frac{1+\mathrm{x}^2}{\mathrm{x}}\right) = \frac{\mathrm{d}}{\mathrm{dx}}(\pi/2) = 0$
3. [3] $y = Tan^{-1}\left(\tan\left(\frac{\pi}{4} - \frac{x}{2}\right)\right) = \frac{\pi}{4} - \frac{x}{2} \Rightarrow \frac{dy}{dx} = -\frac{1}{2}$
4. [2] $\sec x + \tan x = \tan\left(\frac{\pi}{4} + \frac{x}{2}\right) \Rightarrow \frac{dy}{dx} = \frac{d}{dx}\left(\frac{\pi}{4} + \frac{x}{2}\right) = \frac{1}{2}$
5. [2] Put x = $\sin\theta; \frac{d}{dx}$cosec $^{-1}\frac{1}{2x\sqrt{1-x^2}} = \frac{d}{dx}2\sin^{-1}x = \frac{2}{\sqrt{1-x^2}}$.
6. [4] Put x = $\tan\theta; \frac{d}{dx}\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \frac{d}{dx}2\tan^{-1}x = \frac{2}{1+x^2}$
7. [1] $\tan^{-1}\left(\frac{2+3\tan x}{3-2\tan x}\right) = \tan^{-1}\left(\frac{2/3+\tan x}{1-2/3\cdot\tan x}\right) = \tan^{-1}(2/3) + \tan^{-1}(\tan x)$
8. [2] Put $x = \sin\theta; y = -3\sin^{-1}x \Rightarrow \frac{dy}{dx} = \frac{-3}{\sqrt{1-x^2}}$
9. [3] Put $x/3 = \cos\theta; \frac{d}{dx}\cos^{-1}\left(\frac{4x^3}{27} - x\right) = \frac{d}{dx}3\cos^{-1}\left(\frac{x}{3}\right) = \frac{-3}{\sqrt{9-x^2}}$.
10. [1] Put $x^p = \tan\theta$
11. [4] $y = \tan^{-1}\left(\frac{2(2\sqrt{x})}{1-(2\sqrt{x})^2}\right)$, Put $2\sqrt{x} = \tan\theta$
12. [1] $y = (\sin^{-1}2x + \mathrm{Cos}^{-1}2x) + (\mathrm{Tan}^{-1}3x + \mathrm{Cot}^{-1}3x); y = \frac{\pi}{2} + \frac{\pi}{2} = \pi \Rightarrow \frac{dy}{dx} = 0$
13. [3] Put x = $\tan\theta; y = \tan^{-1}\frac{x}{2} \Rightarrow \frac{dy}{dx} = \frac{1}{2(1+x^2)}$.
14. [1] $\frac{\mathrm{d}}{\mathrm{dx}}(\tan^{-1}\mathrm{x}) = \frac{1}{1+\mathrm{x}^2}$
15. [4] Put x = asin θ
16. [1] $y = \cot^{-1}\left[\frac{2+2\cos x}{2\sin x}\right] = \cot^{-1}\left[\frac{1+\cos x}{\sin x}\right] = \cot^{-1}\left[\cot\frac{x}{2}\right] = \frac{x}{2};\ \frac{dy}{dx} = \frac{1}{2}$
17. [1] Put x = asin α, y = asin β

18. [1] Put $x = a\tan\theta$

19. [2] Put $x = \cos\theta$

20. [4] $\frac{dy}{dx} = \frac{1}{\log(x+\sqrt{1+x^2})} \cdot \frac{1}{x+\sqrt{1+x^2}} \cdot \left(1 + \frac{1}{}.\right) = \frac{1}{\sqrt{1+x^2}\log(x+\sqrt{1+x^2})}$.

21. [1] Taking logarithm on both sides & differentiating

22. [375] $^{y} = \log_3(\log_5 x) \Rightarrow 3^{2^y} = \frac{\log x}{\log 5}$

23. [3,4] $$f(x) = \begin{cases} \pi - x, & in\left(\frac{\pi}{2}, \frac{3\pi}{2}\right) \\ x - 2\pi, & in\left(\frac{3\pi}{2}, \frac{5\pi}{2}\right) \end{cases}$$

24. [2] Both the statements are true, but statement 2 is not correct explanation of statement 1.

    Statement 1 is true as period of $\sin x$ is 2π

    Or, in general if for y = f(x), f(a) = f(b) we cannot say $f'(a) = f'(b)$

25. [3] $y = \cos^{-1}\left(\frac{1-\left(\frac{x}{a}\right)^2}{1+\left(\frac{x}{a}\right)^2}\right) + \sin^{-1}\left(\frac{2\cdot\frac{x}{a}}{1+\left(\frac{x}{a}\right)^2}\right) = 4\tan^{-1}\frac{x}{a}$

26. [3] Put $t = \tan\theta$; $x = \cos^{-1}(\cos\theta) = \theta = \tan^{-1} x$

    $y = \sin^{-1}\left(\frac{\tan\theta}{\sec\theta}\right) = \theta = \tan^{-1} x \; y = x \Rightarrow \frac{dy}{dx} = 1$

27. [3] $\frac{d}{dx}[a\tan^{-1} x + b\{\log(x-1) - \log(x+1)\}] = \frac{1}{x^4-1}$

28. a-r; b-t; c-p; d-q

    a) Put $x = \sin\theta$

    b) $\frac{d}{dx}\left(\cot^{-1}\sqrt{\frac{1+\cos 3x}{1-\cos 3x}}\right)$

    $$= \frac{-1}{1+\frac{1+\cos 3x}{1-\cos 3x}} \times \frac{1}{2\sqrt{\frac{1+\cos 3x}{1-\cos 3x}}} \times \frac{(1-\cos 3x)(-3\sin 3x) - (1+\cos 3x)(3\sin 3x)}{(1-\cos 3x)^2}$$

    $$= -\frac{1}{4} \times \frac{\sqrt{1-\cos 3x}}{\sqrt{1+\cos 3x}} \times \frac{-6\sin 3x}{(1-\cos 3x)}$$

    $$= \frac{6}{4} \times \frac{1-\cos 3x}{\sin 3x} \times \frac{\sin 3x}{1-\cos 3x} = \frac{3}{2}.$$

    c) Put $x = \sin\theta$

    d) Put $x = \tan\theta$

29. [1] $y = \tan^{-1}\left(\frac{1+x}{1-x}\right)$ Put $x = \tan\theta\ x = \tan\theta \Rightarrow \theta = \tan^{-1}x$

$\Rightarrow y = \tan^{-1}\left(\tan\left(\frac{\pi}{4}+\theta\right)\right) = \frac{\pi}{4} + \tan^{-1}x$

$\Rightarrow \frac{dy}{dx} = \frac{1}{1+x^2}$ at $x = 0, \frac{dy}{dx} = 1.$

30. [1,3] $f(x-y), f(x)f(y) \& f(x+y)$ are in A.P

$\Rightarrow f(x+y) + f(x-y) = 2f(x)f(y) \rightarrow (1) \forall x, y \in R$

putting x = 0, y = 0 in (1)

we get f(0) + f(0) = 2f(0)f(0) $\Rightarrow f(0) = 1 (\because f(0) \neq 0)$

Putting x = 0, y = x We get $f(x) + f(-x) = 2f(0)f(x)$

$\Rightarrow f(x) = f(-x) \rightarrow (2) \Rightarrow f(4) = f(-4), f(3) = f(-3)$

Diff. (2) w.r.t x, $f'(x) + f'(-x) = 0 \Rightarrow f'(4) + f'(-4) = 0 \& f'(2) = -f'(-2)$

31. a-q ; b-r ; c-p; d-s

(a) $\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}$

(b) $\frac{d}{dx}\tan^{-1}x = \frac{1}{1+x^2}$

(c) $\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}$

(d) $\frac{d}{dx}\cos^{-1}x = \frac{-1}{\sqrt{1-x^2}}$

32. [1,2,3];

$f(x) = |x^2 - 3|x| + 2|$

$$= \begin{cases} |x^2-3x+2|, & x \geq 0 \\ |x^2+3x+2|, & x < 0 \end{cases} = \begin{cases} x^2-3x+2, & x^2-3x+2 \geq 0, \quad x \geq 0 \\ -x^2+3x-2, & x^2-3x+2 < 0, \quad x \geq 0 \\ x^2+3x+2, & x^2+3x+2 \geq 0, \quad x < 0 \\ -x^2-3x-2, & x^2+3x+2 < 0, \quad x < 0 \end{cases}$$

$$= \begin{cases} x^2-3x+2, & x \in [0,1] \cup [2,\infty) \\ -x^2+3x-2, & x \in (1,2) \\ x^2+3x+2, & x \in (-\infty,-2] \cup [-1,0) \\ -x^2-3x-2, & x \in (-2,-1) \end{cases} \Rightarrow f'(x)$$

$$= \begin{cases} 2x-3, & x \in (0,1) \cup (2,\infty) \\ -2x+3, & x \in (1,2) \\ 2x+3, & x \in (-\infty,-2) \cup (-1,0) \\ -2x-3, & x \in (-2,-1) \end{cases}$$

33. [2] St - I & St - II : $\frac{dy}{dx} = \frac{(dy/d\theta)}{(dx/d\theta)}$

34. a-q,r; b-p,r,s ; c- q,r; d-q,r

a) We know that $2\tan^{-1}x =$

$$\begin{cases}\sin^{-1}\left(\frac{2x}{1+x^2}\right), \text{ if } -1 \le x \le 1\\ \pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right), \text{ if } x > 1 \Rightarrow \frac{dy}{dx} = -\frac{2}{1+x^2} \text{ if } x < -1 \text{ or } x > 1\\ -\pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right), \text{ if } x < -1\end{cases}$$

b) $\cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right) = \begin{cases}\tan^{-1}x, x \ge 0\\ -\tan^{-1}x, x < 0\end{cases} \Rightarrow \frac{dy}{dx} = -\frac{1}{1+x^2}$ if $x < 0$

c) $y = \left|e^{|x|} - e\right| = \begin{cases}|e^x - e|, x \ge 0\\ |e^{-x} - e|, x < 0\end{cases} = \begin{cases}e^x - e, & x \ge 1\\ e - e^x, & 0 \le x < 1\\ e^{-x} - e, & x < -1\end{cases} \Rightarrow \frac{dy}{dx} > 0$ if $x > 1$ or $-1 < x < 0$

d) $u = \log|2x|, v = |\tan^{-1}x| \Rightarrow \frac{du}{dx} = -\frac{1}{x}$, and $\frac{dv}{dx} = \begin{cases}\frac{1}{1+x^2}, x > 0\\ -\frac{1}{1+x^2}, x < 0\end{cases}$

$$\Rightarrow \frac{du}{dv} = \begin{cases}\frac{1+x^2}{x}, x > 0\\ -\frac{1+x^2}{x}, x < 0\end{cases}$$

Now we known that $\frac{1+x^2}{x} = x + \frac{1}{x} > 2$ if $x > 1$ and < -2 if $x < -1$

$\Rightarrow \frac{du}{dv} > 2$ if $x < -1$ or $x > 1$

WORKSHEET – 3
Level – I

1) C 2) A 3) D 4) A 5) A 6) D 7) A
8) B 9) C

1. $x^2 + y^2 = a^2 \Rightarrow 2x + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-x}{y}$

2. $3x^2 + 4xy + 2y^2 = 0 \Rightarrow 6x + 4x\frac{dy}{dx} + 4y + 4y\frac{dy}{dx} = 0$

$\frac{dy}{dx}(4x + 4y) = -(6x + 4y) \Rightarrow \frac{dy}{dx}$ at $(-1,3) = \frac{-(-6+12)}{-4+12} = \frac{-3}{4}$

3. $y = x^x \Rightarrow \log y = x\log x \Rightarrow \frac{1}{y}\frac{dy}{dx} = \frac{x}{x} + \log x \Rightarrow \frac{dy}{dx} = y(1 + \log x)$

4. $y = at^2, x = 5at^3 \Rightarrow \frac{dy}{dt} = 2at, \frac{dx}{dt} = 15at^2 \Rightarrow \frac{dy}{dx} = \frac{2at}{15at^2} = \frac{2}{15t}$

5. $x = at^2, y = 2at \Rightarrow \frac{dx}{dt} = 2at\ \frac{dy}{dt} = 2a \Rightarrow \frac{dy}{dx} = \frac{2a}{2at} = \frac{1}{t}$

6. $x = at^2, y = 2at \Rightarrow \frac{dx}{dt} = 2at, \frac{\mathrm{dy}}{\mathrm{dt}} = 2\mathrm{a} \Rightarrow \frac{dy}{dx} = \frac{1}{t} \Rightarrow \frac{d^2y}{dx^2} = \frac{-1}{2at^3}$

7. $f(x) = \frac{a^x}{x^a} \Rightarrow f'(x) = \frac{1}{x^a}a^x\log a + a^x(-ax^{-a-1})$

$f'(a) = \frac{a^a \log a}{a^a} - a^a a \cdot a^{-a-1} = \log a - 1$

8. $u = a^x; v = \sin^{-1}(x) \Rightarrow \frac{du}{dx} = a^x \log a; \frac{dv}{dx} = \frac{1}{\sqrt{1-x^2}} \Rightarrow \frac{du}{dv} = \frac{a^x \log a}{\sqrt{1-x^2}}$

9. $x = \sin^{-1} t; y = \log(1-t^2)$

$\frac{dx}{dt} = \frac{1}{\sqrt{1-t^2}}; \frac{dy}{dt} = \frac{1}{1-t^2}(-2t) \Rightarrow \frac{dy}{dx} = \frac{-2t}{1-t^2} \times \sqrt{1-t^2}$

$\frac{dy}{dx} = \frac{-2t}{\sqrt{1-t^2}}$

$\Rightarrow \frac{dy}{dx}$ at $t = \frac{1}{2} = \frac{-2\frac{1}{2}}{\sqrt{1-\frac{1}{4}}} = \frac{-2}{\sqrt{3}}$

Level – II

1) C	2) C	3) C	4) B	5) D	6) C	7) A
8) D	9) C	10) C	11) A	12) A	13) A	14) A
15) B	16) C	17) D	18) B	19) B	20) A	21) D
22) D	23) B	24) A	25) A,B	26) A	27) A	28) B
29)a-q; b-p; c-s; d-t		30) C	31) B,C	32) A	33) B,D	34) A

1. $3x^2 - 4xy + 1 = 0 \Rightarrow 6x - 4x\frac{dy}{dx} - 4y = 0$

$\Rightarrow 4x\frac{dy}{dx} = 6x - 4y \Rightarrow \frac{dy}{dx} = \frac{3x-2y}{2x} \Rightarrow \frac{dy}{dx}(1,1) = \frac{1}{2}$

2. $x = a(t + \sin t)\ y = a(1 - \cos t)$

$\frac{dx}{dt} = a(1 + \cos t)\ \frac{dy}{dt} = a(\sin t)$

$\frac{dy}{dx} = \frac{a\sin t}{a(1+\cos t)} = \frac{2\sin\frac{t}{2}\cos\frac{t}{2}}{2\cos^2\frac{t}{2}} = \tan\frac{t}{2}$

$\frac{dx}{dy} = \cot\frac{t}{2}, \therefore \mathrm{p} = \frac{\mathrm{t}}{2}$

3. $x = a\left(\cos t + \log\left(\tan\frac{t}{2}\right)\right), y = a\sin t$

$\Rightarrow \frac{dx}{dt} = a\left(-\sin t + \frac{1}{\mathrm{Tan}\frac{t}{2}}\sec^2\frac{t}{2}\cdot\frac{1}{2}\right) = a\left(-\sin t + \frac{1}{\sin t}\right)$

$= a\left(\frac{1-\sin^2 t}{\sin t}\right) = a\frac{\cos^2 t}{\sin t} \cdot \Rightarrow \frac{dy}{dt} = a\cos t$

$\frac{dy}{dx} = a\cos t \times \frac{\sin t}{a\cos^2 t} = \tan t$

4. $u = e^{\sin^{-1}x}\ v = \sin^{-1}x$

$\frac{du}{dx} = e^{\sin^{-1}x}\frac{1}{\sqrt{1-x^2}}\ \frac{dv}{dx} = \frac{1}{\sqrt{1-x^2}}$

$\frac{du}{dv} = \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}} \times \sqrt{1-x^2} = e^{\sin^{-1}x}$

5. $x^y = \log x$ then ; $y\log x = \log(\log x) \Rightarrow y \cdot \frac{1}{x} + \log x \cdot \frac{dy}{dx} = \frac{1}{\log x} \cdot \frac{1}{x}$

$\Rightarrow \frac{dy}{dx}$ at $x = e = \frac{y}{e} + \frac{dy}{dx} = \frac{1}{e} \Rightarrow \frac{dy}{dx} = \frac{1}{e}$

6. $y = 1 + \sin x$

$\Rightarrow \frac{dy}{dx} = \cos x \Rightarrow \frac{d^2y}{dx^2} = -\sin x$

7. $u = \sin^{-1}x, v = \cos^{-1}\sqrt{1-x^2}$

Let $x = \sin\theta$

$\Rightarrow u = \theta \Rightarrow \frac{du}{d\theta} = 1; v = \cos^{-1}(\cos\theta) \Rightarrow v = \theta \Rightarrow \frac{dv}{d\theta} = 1$

$\Rightarrow \frac{du}{dv} = 1$

8. $\frac{d}{dx}\left(\cos^{-1}x + \sin^{-1}\sqrt{1-x^2}\right)$ (Put $x = \cos\theta \Rightarrow \theta = \cos^{-1}x$)

$= \frac{d}{dx}(\theta + \theta) = \frac{d}{dx}(2\cos^{-1}x) = \frac{-2}{\sqrt{1-x^2}}$

9. $y = x^y \Rightarrow \log y = y\log x$

$\Rightarrow \frac{1}{y}\frac{dy}{dx} = y \cdot \frac{1}{x} + \log x\frac{dy}{dx} \Rightarrow \left(\frac{1}{y} - \log x\right)\frac{dy}{dx} = \frac{y}{x} \Rightarrow \frac{dy}{dx} = \frac{y^2}{x(1-y\log x)}$

10. Let $y = (x^x)^x \Rightarrow \log y = x\log x^x$

$\log y = x^2\log x \Rightarrow \frac{1}{y}\frac{dy}{dx} = x^2 \cdot \frac{1}{x} + \log x(2x)$

$\frac{dy}{dx} = y(x + 2x\log x) = (x^x)^x x[1 + 2\log x]$

11. $e^y + xy = e^x \Rightarrow e^y\frac{dy}{dx} + x\frac{dy}{dx} + y = e^x$

$\Rightarrow \frac{dy}{dx}(e^y + x) = e^x - y \Rightarrow \frac{dy}{dx} = \frac{e^x - y}{e^y + x} \Rightarrow \frac{dy}{dx}(0,0) = 1$

12. $\frac{d}{dx}\left(\cos^{-1}x + \cos^{-1}\sqrt{1-x^2}\right)$. Let $x = \sin\theta \Rightarrow \theta = \sin^{-1}x$

$\frac{d}{dx}(\cos^{-1}(\sin\theta) + \cos^{-1}(\cos\theta)) = \frac{d}{dx}(\cos^{-1}(\cos(90-\theta)) + \theta) = \frac{d}{dx}(90 - \theta + \theta) = 0$

13. $x = a\cos^4 t, y = b\sin^4 t \Rightarrow \frac{dx}{dt} = 4a\cos^3 t(-\sin t), \frac{dy}{dt} = 4b\sin^3 t\cos t$

$\frac{dy}{dx} = \frac{4b\sin^3 t\cos t}{-4a\cos^3 t\sin t} = \frac{-b}{a}\tan^2 t \Rightarrow \frac{dy}{dx}att = \frac{3\pi}{4} = \frac{-b}{a}\text{Tan}^2\left(\pi - \frac{\pi}{4}\right) = \frac{-b}{a}$

14. $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \Rightarrow 2ax + 2by\frac{dy}{dx} + 2hy + 2hx\frac{dy}{dx} + 2g + 2f\frac{dy}{dx} = 0$

$\frac{dy}{dx}(hx + by + f) = -(ax + hy + g) \Rightarrow \frac{dy}{dx} = \frac{-(ax + hy + g)}{(hx + by + f)}$

15. Let $u = \text{Tan}^{-1}\frac{\sqrt{1+x^2}-1}{x}; v = \text{Tan}^{-1}x$. Put $x = \tan\theta \Rightarrow \theta = \tan^{-1}x$.

$u = \text{Tan}^{-1}\left(\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right) = \text{Tan}^{-1}\left(\frac{\sec\theta - 1}{\tan\theta}\right) = \text{Tan}^{-1}\left(\tan\frac{\theta}{2}\right) = \frac{\theta}{2}$

$\Rightarrow u = \frac{\theta}{2} \Rightarrow u = \frac{1}{2}\text{Tan}^{-1}x$. Then, $\frac{du}{d\theta} = \frac{1}{2(1+x^2)}$

$v = \text{Tan}^{-1}(\tan\theta) = \tan^{-1}x$. Then, $\frac{dv}{d\theta} = \frac{1}{1+x^2}. \Rightarrow \frac{du}{dv} = \frac{1}{2(1+x^2)} \times (1 + x^2) = \frac{1}{2}$

16. $x = \cos^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right)\ y = \sin^{-1}\left(\frac{t}{\sqrt{1+t^2}}\right)$

Let $t = \tan\theta \Rightarrow x = \cos^{-1}\left(\frac{1}{\sec\theta}\right) \& y = \sin^{-1}\left(\frac{\text{Tan}\,\theta}{\sec\theta}\right)$

$x = \theta \& y = \sin^{-1}(\sin\theta) = \theta$. Then, $\frac{dx}{d\theta} = \frac{dy}{d\theta} = 1 \Rightarrow \frac{dy}{dx} = 1$

17. $x = \cos^3\theta,\ y = \sin^3\theta$

$\frac{dx}{d\theta} = 3\cos^2\theta(-\sin\theta),\ \frac{dy}{d\theta} = 3\sin^2\theta\cos\theta \Rightarrow \frac{dy}{dx} = \frac{3\sin^2\theta\cos\theta}{-3\cos^2\theta\sin\theta} = -\text{Tan}\,\theta$

$\left(\frac{dy}{dx}\right)^2 = \text{Tan}^2\theta \Rightarrow \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \text{Tan}^2\theta} = |\sec\theta|$

18. $x^y = e^{x-y} \Rightarrow \log x^y = \log e^{x-y} \Rightarrow y\log x = (x - y)\log_e^e \Rightarrow y\frac{1}{x} + \log x \cdot \frac{dy}{dx} = 1 - \frac{dy}{dx}$

$(1 + \log x)\frac{dy}{dx} = 1 - \frac{y}{x} = \frac{x - y}{x}$

$\frac{dy}{dx} = \frac{x - y}{x(1 + \log x)} = \frac{y\log x}{x(1 + \log x)} = \frac{y\log x}{y(1 + \log x)^2}$ $(y\log x + y = x; y(1 + \log y) = x)$

$= \frac{\log x}{(1 + \log x)^2}$

19. $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \cdots}}}\ \infty$

$y^2 = \sin x + \sqrt{\sin x + \sqrt{\sin x} + \cdots \ldots} \Rightarrow\ y^2 = \sin x + y$

$\Rightarrow 2y\frac{dy}{dx} = \cos x + \frac{dy}{dx} \Rightarrow \frac{dy}{dx}(2y - 1) = \cos x \Rightarrow \frac{dy}{dx} = \frac{\cos x}{2y - 1}$

20. $x + y = \sin(x + y)$

$\Rightarrow 1 + \frac{dy}{dx} = \cos(x + y)\left(1 + \frac{dy}{dx}\right) \Rightarrow 1 + \frac{dy}{dx} = \cos(x + y) + \cos(x + y)\frac{dy}{dx}$

$\Rightarrow (1-\cos(x+y))\frac{dy}{dx} = \cos(x+y)-1 \Rightarrow \frac{dy}{dx} = \frac{\cos(x+y)-1}{-[\cos(x+y)-1]} = -1$

21. If $x = \cos\theta, y = \sin 5\theta \Rightarrow \frac{dx}{d\theta} = -\sin\theta \ \frac{dy}{d\theta} = 5\cos 5\theta$

$\frac{dy}{dx} = \frac{5\cos 5\theta}{-\sin\theta} = \frac{5\sqrt{1-\sin^2 5\theta}}{-\sqrt{1-\cos^2\theta}} \Rightarrow \frac{dy}{dx} = \frac{5\sqrt{1-y^2}}{-\sqrt{1-x^2}} \Rightarrow (1-x^2)\left(\frac{dy}{dx}\right)^2 = 25(1-y^2)$

$\Rightarrow (1-x^2)2\frac{dy}{dx}\cdot\frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^2 2x = -50y\frac{dy}{dx} \Rightarrow (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = -25y$

22. $\frac{x^n}{a^n} + \frac{y^n}{b^n} = 6 \Rightarrow\Rightarrow \frac{1}{a^n}n\cdot x^{n-1} + \frac{1}{b^n}ny^{n-1}\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x^{n-1}}{a^n}\times\frac{b^n}{y^{n-1}}$

Then, $\frac{dy}{dx}(a,b) = \frac{-a^{n-1}}{a^n}\times\frac{b^n}{b^{n-1}} = \frac{-b}{a}$

23. $y = \sqrt{\text{Tan}\, x + \sqrt{\tan x +}}$.

$\Rightarrow y^2 = \text{Tan}\, x + y \Rightarrow 2y\frac{dy}{dx} = \sec^2 x + \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{\sec^2 x}{2y-1}$

24. $x^y = y^x \Rightarrow \log x^y = \log y^x \Rightarrow y\log x = x\log y$

$\Rightarrow y\frac{1}{x} + \log x\frac{dy}{dx} = x\cdot\frac{1}{y}\frac{dy}{dx} + \log y \Rightarrow \left(\frac{x}{y} - \log x\right)\frac{dy}{dx} = -\log y + \frac{y}{x}$

$\Rightarrow \frac{(x - y\log x)}{y}\frac{dy}{dx} = \frac{-x\log y + y}{x}$

$\Rightarrow x(x - y\log x)\frac{dy}{dx} = y(-x\log y + y)$

25

1) $y = (\log x)^x \Rightarrow \log y = x\log(\log x)$
$\Rightarrow \frac{1}{y}\frac{dy}{dx} = x\frac{1}{\log x}\frac{1}{x} + \log(\log x) \Rightarrow \frac{dy}{dx} = y\frac{[1+\log x\cdot\log(\log x)]}{\log x}$
$= (\log x)^{x-1}(1+\log x\cdot\log(\log x))$

2. $y = x^x \Rightarrow \log y = x\log x \Rightarrow \frac{1}{y}\frac{dy}{dx} = x\cdot\frac{1}{x} + \log x$

$\frac{dy}{dx} = x^x(1+\log x) = x^x(\log ex)$

3. $y = x^y \Rightarrow \log y = y\log x \Rightarrow \frac{1}{y}\frac{dy}{dx} = y\cdot\frac{1}{x} + \log x\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{y^2}{x(1-y\log y)}$

26. Let $y = t^{10}+1$ and $x = t^8+1$ then $\frac{d^2y}{dx^2}$ is

$\frac{dy}{dx} = 10t^9,\ \frac{dx}{dt} = 8t^7 \Rightarrow \frac{dy}{dx} = \frac{10t^9}{8t^7} \Rightarrow \frac{dy}{dx} = \frac{10t^9}{8t^7}$
$\frac{dy}{dx} = \frac{5}{4}t^2 \Rightarrow \frac{dy}{dx} = \frac{5}{4}(x-1)^{1/4} \Rightarrow \frac{d^2y}{dx^2} = \frac{5}{16t^6}$

27. $y = \sin x + e^x \Rightarrow y^1 = \cos x + e^x \Rightarrow y^{11} = -\sin x + e^x \Rightarrow \frac{d^2x}{dy^2} = (-\sin x + e^x)^{-1}$

28. $y = x + e^x \Rightarrow 1 = \frac{dx}{dy} + e^x \frac{dx}{dy} \Rightarrow 1 = \frac{dx}{dy}(1 + e^x) \Rightarrow \frac{dx}{dy} = \frac{1}{1+e^x}$

$\frac{dx}{dy} = (1 + e^x)^{-1} \Rightarrow \frac{d^2x}{dy^2} = -(1 + e^x)^{-2}\left(e^x \frac{dx}{dy}\right) \Rightarrow \frac{d^2x}{dy^2} = \frac{-e^x}{(1 + e^x)^3}$

29. a) $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2} \Rightarrow \frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-y^2}}\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-\sqrt{1-y^2}}{\sqrt{1-x^2}}$

b) $\mathrm{Tan}^{-1} x + \mathrm{Tan}^{-1} y = \frac{\pi}{2} \Rightarrow \frac{1}{1+x^2} + \frac{1}{1+y^2}\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{(1+y^2)}{1+x^2}$

c) $\log(x^m y^n) = \log(x + y)^{m+n} \Rightarrow m\log x + n\log y = (m + n)\log(x + y) \Rightarrow \frac{dy}{dx} = \frac{y}{x}$

d) $x = at^2\ y = 2at$

$\frac{dx}{dt} = 2at\ \frac{dy}{dt} = 2a \Rightarrow \frac{dy}{dx} = \frac{2a}{2at} = \frac{1}{t} \Rightarrow \frac{d^2y}{dx^2} = \frac{-1}{t^2}$

30. $\sqrt{x} + \sqrt{y} = 4 \Rightarrow \frac{dx}{dy} = \frac{\sqrt{y}-4}{\sqrt{y}} \Rightarrow \frac{1}{2\sqrt{x}}\frac{dx}{dy} + \frac{1}{2\sqrt{y}} = 0$

$\frac{dx}{dy} = \frac{-\sqrt{x}}{\sqrt{y}} \Rightarrow \frac{dx}{dy}$ at $y = 1 = -3$

31. $f(x) = ax^2 + bx + c \Rightarrow f^1(x) = 2ax + b \Rightarrow f^1(5) = 10a + b$

32. $x = \mathrm{Tan}\,\theta$

$u = \mathrm{Tan}^{-1}\left(\frac{2\mathrm{Tan}\,\theta}{1-\mathrm{Tan}^2\,\theta}\right) = 2\theta \Rightarrow \frac{du}{d\theta} = 2.$

$. v = \sin^{-1}(\sin 2\theta) \Rightarrow v = 2\theta \Rightarrow \frac{dv}{d\theta} = 2.$ Then $\frac{du}{dv} = 1.$

33. $f(x) = \sin^{-1}(\sin x) \Rightarrow f(x) = \begin{cases} \pi - x, & \left(\frac{\pi}{2}, \frac{3\pi}{2}\right) \\ x - 2\pi, & \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right) \end{cases} \Rightarrow f^1(x) = \begin{cases} -1, & \left(\frac{\pi}{2}, \frac{3\pi}{2}\right) \\ 1, & \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right) \end{cases}$

34. $f(x) = \cos x \cdot \cos 2x \cdot \cos 4x \cdot \cos 8x \cdot \cos 16x$

$$f(x) = \frac{2\sin x \cos x \cdot \cos 2x \cdot \cos 4x \cdot \cos 8x \cdot \cos 16x}{2\sin x} \Rightarrow f(x) = \frac{\sin 32x}{2^5 \sin x}$$

$$f^1(x) = \frac{1}{32}\left[\frac{32\cos 32x \cdot \sin x - \cos x \cdot \sin 32x}{\sin^2 x}\right]$$

$$\Rightarrow f^1\left(\frac{\pi}{4}\right) = \frac{32 \cdot \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}(0)}{32 \times \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{2}.$$

15. Application of Derivatives

WORKSHEET – 1
Level – I

1) B 2) A 3) A 4)B 5) A 6) A
7) A 8) A 9) D

1. $f(x) = \frac{x^2}{2} + 3x, x = 2, \delta x = 0.05$

   $\delta f = f^1(x)\delta x = (x+3)\delta x = (2+3)0.05 = 0.25.$

2. Let $f(x) = \sqrt{x}, x = 1, \delta x = 0.02$

   Approximate value of $\sqrt{1.02}$

   $= f(x) + f^1(x)\delta x = \sqrt{x} + \frac{1}{2\sqrt{x}}dx = \sqrt{1} + \frac{1}{2\sqrt{1}} \times 0.02 = 1 + 0.01 = 1.01.$

3. $f(x) = \sqrt[5]{x}, x = 32, \delta x = 1$

   Approximate value of $\sqrt[5]{38} = f(x) + f^1(x)\delta x = x^{\frac{1}{5}} + \frac{1}{5}x^{-\frac{4}{5}}dx$

   $= (32)^{\frac{1}{5}} + \frac{1}{5}\frac{1}{(32)^{\frac{4}{5}}}(1) = 2 + \frac{1}{5}\cdot\frac{1}{16} = 2 + \frac{1}{80} = 2.0125.$

4. Let $f(x) = \cos x, x = 60°, \delta x = 1^0 = 0.01745$

   Approximate value of $\cos 61^0$

   $= f(x) + f^1(x)\delta x = \cos 60° - \sin 60° \times 0.01745$

   $= \frac{1}{2} - \frac{\sqrt{3}}{2} \times 0.01745 = 0.5 - 0.86603 \times 0.01745 = 0.5 - 0.01511 = 0.48489.$

5. $s = t^3 - 9t^2 + 24t$

   $v = \frac{ds}{dt} = 3t^2 - 18t + 24$

   If V < 0 ⇒ s decreases

   $3t^2 - 18t + 24 < 0 \Rightarrow t^2 - 6t + 8 < 0 \Rightarrow (t-2)(t-4) < 0 \Rightarrow t \in (2,4).$

6. Let side of square plate be x

   Area, $A = x^2$, perimeter , $p = 4x$

   $\frac{dA}{dt} = 2x\frac{dx}{dt}, \frac{dp}{dt} = 4\frac{dx}{dt}$

   $\frac{dA}{dt} = \frac{dp}{dt} \Rightarrow 2x\frac{dx}{dt} = 4\frac{dx}{dt}$

   $x = 2$ units .

7. $f(x) = x^2 + 4x, x = 2, \delta x = 0.1$

   $df = f^1(x)dx = (2x+4)dx = (4+4)(0.1) = 0.8.$

8. Let $f(x) = \log(x), x = 2, \delta x = 0.01$

   Approximate value of $\log_e(2.01)$

   $= f(x) + f^1(x)\delta x = \log x + \frac{1}{x}.dx = \log 2 + \frac{1}{2} \times 0.01 = 0.6934 + 0.005 = 0.6984.$

9. $y = \frac{1}{x^2}, x = 1, \delta x = -0.01$

   $dy = \frac{dy}{dx}.dx = \frac{-2}{x^3}dx = \frac{-2}{1}(-0.01) = 0.02.$

Level – II

1) C	2) A	3) A	4) B	5) A	6) A	7) B
8) C	9) C	10) C	11) B	12) C	13) A	14) B
15) 5	16) D	17) B	18) C	19) 54	20) C	21) C
22) 96	23) B,D	24) D	25) B	26) A	27) a-r; b-p; c-s; d-q	
28) 0	29) A,C	30) A	31) D	32) D	33) C	
34) a-s; b-p; c-q; d-r		35) A,B,C				

1. Radius of sphere, $r = 3$ cm, $\delta r = -0.02$

   volume, $V = \frac{4}{3}\pi r^3$

   $dv = \frac{4}{3}\pi \times 3r^2\delta r = -4\pi \times 3 \times 3 \times 0.02 = -0.72\pi cm^3.$

2. Surface area of sphere, $A = 4\pi r^2 = A\% = 2 \times r\%$

   $K = 2r\% \Rightarrow r\% = \frac{K}{2}$

   Volume, $V = \frac{4}{3}\pi r^3$

   $V\% = 3r\% = \frac{3K}{2}.$

3. Percentage error is radius, $r\% = 0.1$

   Volume of sphere, $V = \frac{4}{3}\pi r^3$

   $V\% = 3r\% = 3 \times 0.1 = 0.3\%.$

4. Mass of the body, $m = 100$ kg

   $s = 2t^2 + 3t + 1$

   Velocity, $V = \frac{ds}{dt} = 4t + 3$; Velocity after 5sec = 23 m/sec

   $K.E = \frac{1}{2}mv^2 = \frac{1}{2} \times 100 \times 23 \times 23 = 26450.$

5. Equation of ellipse is $16x^2 + 9y^2 = 400$

   diff. w.r.t t

   $32x\frac{dx}{dt} + 18y\frac{dy}{dt} = 0 \Rightarrow 18y\frac{dy}{dt} = -32x\frac{dx}{dt}$

   $18y = 32x \left(\because \frac{dy}{dt} = -\frac{dx}{dt}\right)$

$y = \frac{32}{18}x = \frac{16}{9}x \ \therefore$ Point $= \left(3, \frac{16}{3}\right)$.

6. Let $f(x) = \frac{1}{\sqrt{x}}, x = 25, \delta x = 0.25$

Approximate value of $\frac{1}{\sqrt{25.25}} = f(x) + f^1(x)\delta x$

$= \frac{1}{\sqrt{x}} - \frac{1}{2}x^{-3/2}\delta x = \frac{1}{\sqrt{25}} - \frac{1}{2} \cdot \frac{1}{(25)^{3/2}}(0.25) = \frac{1}{5} - \frac{1}{2} \cdot \frac{1}{125} \times 0.25 = 0.2 - 0.001 = 0.199.$

7. Let $f(x) = \sqrt[4]{x}, x = 81, \sqrt{x} = -1$

Approximate value of $\sqrt[4]{80}$ is $= f(x) + f^1(x)\delta x$

$$= x^{1/4} + \frac{1}{4}x^{-3/4} \cdot \delta x \Rightarrow (81)^{1/4} + 1/4 \cdot \frac{1}{(81)^{3/4}}(-1)$$

$$= 3 - \frac{1}{4} \cdot \frac{1}{27} \Rightarrow 3 - 0.009259 = 2.9907 \Rightarrow [2.9907] = 2$$

8. Period of simple pendulum is

$$T = 4\pi\sqrt{\frac{l}{g}} \Rightarrow T\% = \frac{1}{2}l\% \Rightarrow a = \frac{1}{2}l\% \ \therefore l\% = 2a$$

9. $PV^{1/4} = C \Rightarrow P\% + \frac{1}{4}V\% = 0 \Rightarrow P\% = -\frac{1}{4}V\% \Rightarrow -\frac{1}{4}\left(-\frac{1}{2}\right) = \frac{1}{8}.$

10. Equation of curve is $y = x^2 + 2x \Rightarrow \frac{dy}{dt} = (2x + 2)\frac{dx}{dt}$

$\Rightarrow 2x + 2 = 1 \Rightarrow x = -1/2 \Rightarrow y = \frac{1}{4} - 1 = -3/4 \ \therefore (x, y) = (-1/2, -3/4).$

11. Let $f(x) = \sin x, x = 45°, \delta x = 1°$

Approximate value of $\sin 46°$

$= f(x) + f^1(x)\delta x = \sin x + \cos x \cdot \delta x = \sin 45 + \cos 45.1° = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \times 0.01745.$

12. Volume of cone is $V = \frac{1}{3}\pi r^2 h \Rightarrow V\% = 2 \times r\% + h\% = 2 \times 1 + 1 = 3.$

13. Area of triangle, $s = \frac{1}{2}bc\sin A$

$$\Rightarrow \delta s = \frac{1}{2}bc\cos A\delta A \Rightarrow \frac{\delta s}{s} = \cot A\delta A \Rightarrow \delta A = 15^1 = \frac{15}{60} \times \frac{\pi}{180}$$

$\Rightarrow \frac{\delta s}{s} \times 100 = \cot 63° \times \frac{15}{60} \times \frac{\pi}{180} \times 100 = \frac{5\pi}{36}\cot 63°.$

14. $a^2 = b^2 + c^2 - 2bc\cos A \Rightarrow 2a\delta a = 2bc\sin A\delta A$

$\Rightarrow \delta a = \frac{bc\sin A\delta A}{a} = \frac{2\Delta\delta A}{a}. \ \left(\because \Delta = \frac{1}{2}bc\sin A\right)$

15. Area of circular plate, $A = \pi r^2 \Rightarrow \delta A = 2\pi r\delta r$

$\Rightarrow \frac{\pi}{10} = 2\pi r \times 0.01 \Rightarrow r = \frac{1}{10 \times 2 \times 0.01} = 5$ cm.

16. $x^2 + y^2 = 100 \Rightarrow 6^2 + y^2 = 100 \Rightarrow y = 8m$

$$\Rightarrow 2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow 6 \times 2 + 8\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = \frac{-12}{8} = \frac{-3}{2} \text{ m/min} \Rightarrow A = \frac{1}{2}xy$$

$\Rightarrow \frac{dA}{dt} = \frac{1}{2}\left(x\frac{dy}{dt} + y\frac{dx}{dt}\right) \Rightarrow \frac{1}{2}\left[6\left(-\frac{3}{2}\right) + 8(2)\right] \Rightarrow \frac{1}{2}(-9+16) = \frac{7}{2} sq \cdot m/\text{min}$

17. Volume of ball, $V = \frac{4}{3}\pi r^3 \Rightarrow 288\pi = \frac{4}{3}\pi r^3 \Rightarrow r^3 = \frac{288\times3}{4} = 72 \times 3 = 216$

$\therefore r = 6$ cm $\Rightarrow \frac{dV}{dt} = \frac{4}{3}\pi \times 3r^2\frac{dr}{dt} \Rightarrow 2\pi = 4\pi \times 6 \times 6 \times \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{72}$ cm/sec.

18. Let radius of sphere be r

and Height of cone be h

$\therefore h = r$

Volume of balloon $= \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3 \Rightarrow V = \pi r^3 \Rightarrow \frac{dV}{dt} = 3\pi r^2\frac{dr}{dt} = 3\pi \times (2)^2\frac{dr}{dt} =$

$12\pi\frac{dr}{dt}$.

19. $y = x^3 \Rightarrow \frac{dy}{dx} = 3x^2 \Rightarrow m = 3x^2 \Rightarrow \frac{dm}{dt} = 6x\frac{dx}{dt} = 6 \times 3 \times 3 = 54$

20. $\theta\alpha t^2$

$\Rightarrow \theta = kt^2 \Rightarrow 2\pi = k \times 8^2 \Rightarrow k = \frac{2\pi}{8\times8} \Rightarrow k = \frac{\pi}{32} \Rightarrow \frac{d\theta}{dt} = k \cdot 2t = \frac{\pi}{32} \times 2 \times 8 = \frac{\pi}{2}$rad/sec.

21. $\frac{h}{r} = \frac{10}{5} \Rightarrow \frac{h}{2} = r$

Volume, $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \times \frac{h^3}{4}$

$\therefore V = \frac{\pi}{12}h^3$.

$\Rightarrow \frac{dV}{dt} = \frac{\pi}{12} \times 3 \times h^2\frac{dh}{dt} \Rightarrow \frac{3}{2} = \frac{\pi}{12} \times 3 \times 6 \times 6\frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{3}{2\times4\pi} = \frac{3}{8\pi}$mt/min.

22. $s = 160t - 16t^2 \Rightarrow v = \frac{ds}{dt} = 160 - 32t$

At $s = 256 \Rightarrow 160t - 16t^2 = 256$

$\Rightarrow 16t^2 - 160t + 256 = 0 \Rightarrow t^2 - 10t + 16 = 0$

$\therefore (t-2)(t-8) = 0 \ \therefore t = 2,8.$

$\Rightarrow V(2) = 160 - 322 = 96 \Rightarrow V(8) = 160 - 32(8) = -96$

Velocity on the way up = 96 m/sec.

23. $s = \frac{t^3}{3} - 3t^2 + 8t + 5 \& V = \frac{ds}{dt} \Rightarrow V = t^2 - 6t + 8$

Direction of particle changes when $V = 0$

$\Rightarrow t^2 - 6t + 8 = 0 \Rightarrow (t-2)(t-4) = 0 \ \therefore t = 2,4.$

24. $\frac{dr}{dt} = 1$ cm/sec $\Rightarrow$ Area of circle, $A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r\frac{dr}{dt} = 2\pi \times 5 \times 1 = 10\pi.$

25. $\frac{dA}{dt} = 2\pi r\frac{dr}{dt} = 2\pi \times 3 \times 1 = 6\pi.$

26. Perimeter, $P = 2\pi r \Rightarrow \frac{dP}{dt} = 2\pi\frac{dr}{dt} = 2\pi \times 1 = 2\pi.$

27. a) $x^2 + y^2 = 2 \Rightarrow 2x\frac{dy}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow y\frac{dy}{dt} = -x\frac{dx}{dt} = 0 \Rightarrow y = -x \ \left(\because \frac{dy}{dt} = \frac{dx}{dt}\right)$

By verification, point = $(1,-1)$.

b) $x^2 = 8y \Rightarrow 2x\frac{dx}{dt} = 8\frac{dy}{dt} \Rightarrow x = 4 \& y = 2.$

$\therefore$ point = (4,2).

c) $x^2 - y^2 = 0 \Rightarrow 2x\frac{dx}{dt} - 2y\frac{dy}{dt} = 0 \Rightarrow x\frac{dx}{dt} = y\frac{dy}{dt} \Rightarrow x = y.$

$\therefore$ point = (4,4).

d) $y^2 = 8x \Rightarrow 2t\frac{dy}{dt} = 8\frac{dx}{dt} \Rightarrow y = 4 \& x = 2.$

$\therefore$ point = (2,4).

28. $s = t^3 - 3t^2 + 3t + 12 \Rightarrow v = \frac{ds}{dt} = 3t^2 - 6t + 3$

$$a = \frac{dv}{dt} = 6t - 6; a = 0 \Rightarrow 6t - 6 = 0 \Rightarrow t - 1 = 0 \Rightarrow t = 1 \ \therefore v(1) = 3 - 6 + 3 = 0$$

29. $y = \frac{x^3}{3} - \frac{3}{2}x^2 + 5x + 4 \Rightarrow \frac{dy}{dt} = (x^2 - 3x + 5)\frac{dx}{dt} \Rightarrow 3\frac{dx}{dt} = (x^2 - 3x + 5)\frac{dx}{dt}$

$\Rightarrow x^2 - 3x + 2 = 0 \Rightarrow (x-1)(x-2) = 0 \ \therefore x = 1,2.$

30. $s = t^3 - 2t^2 - 3t + 5 \Rightarrow v = \frac{ds}{dt} = st^2 - 4t - 3 \Rightarrow \therefore v(2) = 3(2)^2 - 4(2) - 3 = 1.$

31. Let $f(x) = \sin x, x = 60°, \delta x = 6^1$

Approximation value of $\sin 60° 6^1$

$$= f(x) + f^1(x)\delta x = \sin x + \cos x \cdot \delta x = \sin 60° + \cos 60° \cdot \frac{6}{60} \times 0.0175$$

$$= \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{10} \times 0.0175 = 0.86603 + 0.00087 = 0.8669.$$

32. Let $f(x) = \tan x, x = 60°, \delta x = 6^1$

Approximate value of $\tan 60° 6^1$

$$= \tan x + \sec^2 x\delta x = \tan 60 + \sec^2 60 \times \frac{6}{60} \times 0.0175 = \sqrt{3} + 2 \times \frac{1}{10} \times 0.0175$$

$$= 1.732 + 0.0035 = 1.7355.$$

33. Let $f(x) = \sin x, x = 60°, \delta x = 1^1$

Approximate value of $\sin 60° 1^1$

$$= \sin 60° + \cos 60° \times \frac{1}{60} \times 0.01745 = \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{60} \times 0.01745 = 0.86603 + 0.00015.$$

$\therefore 0.86618$

34. a) Let $f(x) = \log x, x = 4, \delta x = 0.01$

Approximate value of $\log 4.01 = \log x + \frac{1}{x}\delta x = \log 4 + \frac{1}{4} \times 0.01 = 1.3868 + 0.0025 =$ 1.3893.

b) Let $f(x) = \tan x, x = 45°, \delta x = -1°$

Approximate value of $\tan 44° = \tan 45 - \sec^2 45 \times 0.01745$

$= 1 - 2 \times 0.01745 = 1 - 0.0349 = 0.9651.$

c) Let $f(x) = \frac{1}{\sqrt{x}}, x = 25, \delta x = 0.25$

Approximate value of $\frac{1}{\sqrt{25.25}} = \frac{1}{\sqrt{x}} - \frac{1}{2}x^{-3/2}\delta x$

$= \frac{1}{25} - \frac{1}{2}\frac{1}{(25)^{3/2}}(0.25) \Rightarrow \frac{1}{5} - \frac{1}{2} \cdot \frac{1}{125} \times 0.25 = 0.2 - 0.001 = 0.199.$

d) Let $f(x) = \sqrt[6]{x}, x = 729, \delta x = -1$

Approximate value of $\sqrt[6]{728}$

$= x^{1/6} + \frac{1}{6}x^{-5/6} \cdot \delta x = (729)^{1/6} + \frac{1}{6}\frac{1}{(729)^{5/6}}(-1) = 3 - \frac{1}{6} \times \frac{1}{243} \Rightarrow 3 - 0.0007 = 2.9993.$

35.

$s = at^2 + bt + e \Rightarrow s(1) = a + b + c = 16$

$V = \frac{ds}{dt} = 2at + b \Rightarrow V(2) = 4a + b = 24$

$a = \frac{dV}{dt} = 2a \Rightarrow 2a = 8 \Rightarrow a = 4 \Rightarrow 16 + b = 24 \Rightarrow b = 8 \Rightarrow 4 + 8 + c = 16 \Rightarrow c = 4$

WORKSHEET – 2
Level – I

1) 3	2) 2	3) 3	4) 1	5) 1	6) 2
7) 2	8) 3	9) 4	10) 1	11) 2	

1. Slope of line $m = -1$.
2. Equation of curve is $y = 4x \Rightarrow \frac{dy}{dx} = 4$

   slope of tangent,$m = 4$

   point = (1,4) = (x_1, y_1)

   length of sub normal = $|y_1 m| = |4 \times 4| = 16$.
3. Equation of the lines are $x + y = 4 \& x - y = 10$.

   They are perpendicular lines.
4. Equation of lines are $2x + 3y = 5 \& 4x + 6y = 10$.

   They are parallel lines.
5. Conceptual 6. Conceptual
6. Equation of curve is $x^2 = 4by \Rightarrow 2x = 4b\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{x}{2b}$

   point = (0,0) = (x_1, y_1)

   slope of Tangent, m = 0.

   Eq. of tangent is $y - 0 = 0 \Rightarrow y = 0$.

8. Equation of curve is $y = x^3 - 3x^2 - 2x + 7$

$\Rightarrow \frac{dy}{dx} = 3x^2 - 6x - 2$

Gradient of curve at (1,3) is $m = 3 - 6 - 2 = -5$.

9. $f(x) = 2x^3 + x^2 - 4x - 2$

$[a, b] = [-\sqrt{2}, \sqrt{2}]$
$f^1(x) = 6x^2 + 2x - 4$
$f^1(x) = 0 \Rightarrow 6x^2 + 2x - 4 = 0 \Rightarrow x = -1, \frac{2}{3}$.

10. $y = \log(\log_e x)$

$\Rightarrow \frac{dy}{dx} = \frac{1}{\log_e^x} \cdot \frac{1}{x} = \frac{1}{x\log_e^x}$

slope of tangent at $x = e = \frac{1}{e}$

$\Rightarrow$ slope of normal $= -e$.

11. Conceptual.

12. Conceptual.

Level – II

1) B	2) D	3) B	4) D	5) D	6) B
7) B	8) B	9) A	10) B	11) A	12) C
13) C	14) D	15) B	16) 9	17) A	18) C
19) D	20) A	21) B	22) A	23) 0	24) A,D
25) A	26) A	27) C	28) B	29) a-q, b-s, c-p, d-r	
30) C	31) C,D	32) a-r; b-p; c-s; d-r		33) C	34) A,B
35) a-r; b-p; c-s; d-q					

1. Equation of curve is $y = 16 - x^2 \Rightarrow \frac{dy}{dx} = -2x$.

Slope of tangent at (0,16) is $m = 0$

Equations tangent is $y = 16$.

2. Equation of curve is $\left(\frac{x}{a}\right)^4 + \left(\frac{y}{b}\right)^4 = 2$

On differentiating both the sides, w.r.t.x,

$\Rightarrow \frac{4x^3}{a^4} + \frac{4y^3}{b^4}\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{4x^3}{a^4} \times \frac{b^4}{4y^3}$

slope of tangent at (a, b) is

$\Rightarrow m = \frac{-4a^3}{a^4} \cdot \frac{b^4}{4b^3} = -\frac{b}{a}$

slope of normal $= \frac{-1}{m} = \frac{a}{b}$

Equation of normal is $y-b=\frac{a}{b}(x-a) \Rightarrow by-b^2=ax-a^2 \ \therefore ax-by=a^2-b^2$.

3. Equation of curves are $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and $\frac{x^2}{l^2}+\frac{y^2}{m^2}=1$

$\Rightarrow a^2x^2+b^2y^2=1$ and $a_1^2x^2+b_1^2y^2=1$

cut orthogonally, then $\frac{1}{a}-\frac{1}{a_1}=\frac{1}{b}-\frac{1}{b_1} \Rightarrow a^2-l^2=b^2-m^2 \Rightarrow a^2-b^2=l^2-m^2$.

4. Equation of curve is $f(x)=x^3-xy^2-4x^2-xy+5x+3y+1=0$

$\frac{dy}{dx}=-\frac{\partial f}{\partial x}/\frac{\partial f}{\partial y}=\frac{-[3x^2-y^2-8x-y+5]}{-2xy-x+3}$ slope of tangent at $(1,-1)$ is

$m=\left(\frac{3-1-8+1+5}{2-1+3}\right)$. Equation of tangent is $y=y_1 \Rightarrow y=-1 \Rightarrow y+1=0$.

5. Equation of curve is $xy=a^2 \Rightarrow y=\frac{a^2}{x} \Rightarrow \frac{dy}{dx}=-\frac{a^2}{x^2}$

slope of tangent at (x,y) is $m=-\frac{a^2}{x^2}$

Length of sub normal $=|y\times m|=\left|y\cdot\left(-\frac{a^2}{x^2}\right)\right|=y\cdot\frac{y^2}{a^2}=\frac{y^3}{a^2}$.

6. $f(x)=(x-a)(x-b) \Rightarrow f(x)=x^2-(a+b)x+ab \Rightarrow f^1(x)=2x-(a+b)$

By Rolle's theorem, $f^1(x)=0 \Rightarrow 2x-(a+b)=0 \Rightarrow x=\frac{a+b}{2}$.

7. $f(x)=x(x+3)(x-2)=(x^2+3x)(x-2)=x^3+x^2-6x$.

$f^1(x)=3x^2+2x-6=10$ [$\because f^1(x)=10$]

$\Rightarrow 3x^2+2x-6=10 \Rightarrow 3x^2+2x-16=0$

$\Rightarrow 3x^2-6x+8x-16=0 \Rightarrow 3x(x-2)+8(x-2)=0$

$\Rightarrow (x-2)(3x+8)=0 \ \therefore x=2$ or $-\frac{8}{3}$.

8. Equation of curve is $x^3+y^3=6xy \Rightarrow f(x,y)=x^3+y^3-6xy$

$\Rightarrow \frac{dy}{dx}=-\frac{\partial f}{\partial x}/\frac{\partial f}{\partial y}=-\frac{(3x^2-6y)}{3y^2-6x}$

Gradient of the curve at (3,3) is $m=-\frac{(27-18)}{27-18}=-1$.

9. Equation of curves are

$y=x^3+x+1;\ 2y=x^3+5x \Rightarrow 2x^3+2x+2=x^3+5x \Rightarrow x^3-3x+2=0 \ \therefore x=1; y=3$.

The curve touch each other at (1,3).

10. $x=2\sin t\ ;\ y=2\cos t$

On differentiating with respect x,

$\Rightarrow \frac{dx}{dt}=2\cos t\ \frac{dy}{dt}=-2\sin t$

$\Rightarrow \frac{dy}{dx}=\frac{dy}{dt}/\frac{dx}{dt}=\frac{-2\sin t}{2\cos t}=-\tan t$

Slope of tangent at $t = \frac{\pi}{2}$ is $m = \infty$

slope of normal = 0

Equation of normal $y - 2\cos\frac{\pi}{2} = 0 \Rightarrow y = 0.$

11. $f(x) = 1 - \sqrt[3]{x^4} = 1 - x^{\frac{4}{3}} \Rightarrow f^1(x) = -\frac{4}{3}x^{1/3}$

$\Rightarrow -\frac{4}{3}x^{1/3} = 0[\because f^1(x) = 0] \Rightarrow x = 0.$

12. Equation of curve is $y = x + \frac{4}{x^2} \Rightarrow \frac{dy}{dx} = 1 + 4\left(\frac{-2}{x^3}\right) = 1 - \frac{8}{x^3}$

slope of tangent $= \frac{x^3-8}{x^3}$

equation of tangent is $y = k$. it is parallel to x-axis

slope $= 0 \Rightarrow x^3 - 8 = 0 \Rightarrow x^3 = 8 \Rightarrow x = 2$

$\Rightarrow y = 2 + \frac{4}{4} = 3 \ \therefore k = 3.$

13. Let $f(x) = \frac{4ax^4}{4} + \frac{3bx^3}{3} + \frac{2cx^2}{2} + dx$

$f(x) = ax^4 + bx^3 + cx^2 + dx \Rightarrow f^1(x) = 4ax^3 + 3bx^2 + 2cx + d = 0 \ \therefore f(0) = 0$

$f(3) = 81a + 27b + 9c + 3d = 3(27a + 9b + 3c + d) = 0 \ \therefore f(0) = 0 = f(3)$

$\exists c \in (0,3)$ such that $f^1(c) = 0.$

14. Equation of curve is $y^2x^2 = a^2(x^2 - a^2) \Rightarrow y^2 = a^2 - \frac{a^4}{x^2} \Rightarrow 2yy^1 = \frac{2a^4}{x^3}$

$\Rightarrow y^1 = \frac{a^4}{x^3y} \Rightarrow$ slope of tangent, $m = \frac{a^4}{x^3y}$

length of subnormal $= |ym| = \left|y\frac{a^4}{x^3y}\right| = \frac{a^4}{x^3}\alpha\frac{1}{x^3}.$

15. Equation of curve is

$x = e^{\sin y} \Rightarrow \log x = \sin y \log e \Rightarrow \log x = \sin y \Rightarrow \frac{1}{x} = \cos y\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{x\cos y}$

slope of tangent at (1,0) is $m = 1$

slope of normal $= -1$

Equation normal is $y - 0 = -1(x - 1)$

$\therefore x + y = 1.$

Area of triangle formed by the normal with co-ordinate axes $= \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$ sq.units.

16. $y = ax^2 + bx + c \Rightarrow \frac{dy}{dx} = 2ax + b$

Where $x = 1, y = 0 \Rightarrow a + b + c = 0$

$y = 3x - 3 \ y = x + 1$

$\Rightarrow \frac{dy}{dx} = 3$ at $x = 1 \ \frac{dy}{dx} = 1$ at $x = 3$

$\Rightarrow 2a+b=3 \Rightarrow 6a+b=1 \Rightarrow 4a=-2 \Rightarrow a=-\frac{1}{2};\ b=4$

$\Rightarrow -\frac{1}{2}+4+c=0\ \therefore c=-\frac{7}{2}$

$2a-b-4c=2\left(-\frac{1}{2}\right)-4-4\left(-\frac{7}{2}\right)\Rightarrow -1-4+14=9.$

17. Equation of curves are

$\Rightarrow x=a\left[\cos t+\log\left(\tan\frac{t}{2}\right)\right], y=a\sin t \Rightarrow \frac{dx}{dt}=a\left[-\sin t+\frac{1}{\tan t/2}\sec^2 t/2\cdot\frac{1}{2}\right]$

$=a\left[-\sin t+\frac{\cos t/2}{\sin t/2}\cdot\frac{1}{\cos^2 t/2}\cdot\frac{1}{2}\right]=a\left[-\sin t+\frac{1}{\sin t}\right]\Rightarrow a\left[\frac{1-\sin^2 t}{\sin t}\right]=a\frac{\cos^2 t}{\sin t}$

$\therefore \frac{dy}{dt}=a\cos t$

$\Rightarrow \frac{dy}{dx}=\frac{dy/dt}{dx/dt}=\frac{a\cos t}{a\cos^2 t}\cdot\sin t=\tan t$

slope of tangent at $t=\frac{\pi}{3}$ is $\Rightarrow m=\tan\frac{\pi}{3}=\sqrt{3}$

length of subtangent $=\left|\frac{y_1}{m}\right|=\left|\frac{a\sin\pi/3}{\sqrt{3}}\right|=\frac{a\sqrt{3}}{\sqrt{3}\times 2}=\frac{a}{2}.$

18. $f(0)=2, g(0)=0, f(1)=6, g(1)=2$

From cauchys' mean value theorem

$\Rightarrow \frac{f(b)-f(a)}{g(b)-g(a)}=\frac{f^1(c)}{g^1(c)}\Rightarrow \frac{f(1)-f(0)}{g(1)-g(0)}=\frac{f^1(c)}{g^1(c)}\Rightarrow \frac{6-2}{2-0}=\frac{f^1(c)}{g^1(c)}\ \therefore f^1(c)=2g^1(c)$

19. $f(x)=4-(6-x)^{2/3}$ is $[5,7]\Rightarrow f(a)=f(5)=4-(6-5)^{2/3}=4-1=3$

$f(b)=f(7)=4-(6-7)^{2/3}=4-(-1)^{2/3}\Rightarrow f(a)\neq f(b)$

Rolle's theorem is not applicable.

similarly LMVT also not applicable.

20. Equation of curve is $\Rightarrow \sqrt{x}+\sqrt{y}=2\Rightarrow \frac{1}{2\sqrt{x}}+\frac{1}{2\sqrt{y}}\frac{dy}{dx}=0\Rightarrow \frac{dy}{dx}=\frac{-2\sqrt{y}}{2\sqrt{x}}=\frac{-\sqrt{y}}{\sqrt{x}}$

slope of tangent at (1,1) is $m=-1$

Area of triangle formed by tangent, normal and y-axis =

$\frac{x_1^2(1+m^2)}{2|m|}=\frac{1(1+1)}{2(1)}=1$ sq.units .

21. Equation of curve is $y=be^{-x/a}\Rightarrow \frac{dy}{dx}=be^{-x/a}\left(-\frac{1}{a}\right)=-\frac{b}{a}e^{-x/a}$

for $x=0\Rightarrow y=b\Rightarrow$ slope of tangent $=-\frac{b}{a}$

Equation of tangent is $\Rightarrow y-b=-\frac{b}{a}(x-0)\Rightarrow \frac{y}{b}-1=-\frac{x}{a}\Rightarrow \frac{x}{a}+\frac{y}{b}=1.$

22. $h(x) = \frac{f(x)}{g(x)}$ By applying LMVT

$$\Rightarrow \frac{h(p) - h(a)}{b - a} = h^1(c) \Rightarrow \frac{f(b)/g(b)^{-f(a)}/g(a)}{(b-a)g(a)g(b)} = \frac{g(c)f^1(c) - f(c)g^1(c)}{(g(c))^2}$$

$$\Rightarrow \frac{g(a)f(b) - f(a)g(b)}{g(c)f^1(c) - f(c)g^1(c)} = (b-a)\frac{g(a)g(b)}{(g(c))^2} \Rightarrow k = b - a.$$

23. Equation of the curve is, $y = ax^2 + bx + c$

Points are $(p, ap^2 + bp + c), (-p, ap^2 - bp + c)$

Slope of the line joining two points $= \frac{2bp}{2p} = b$

$\frac{dy}{dx} = 2ax + b; \left[\frac{dy}{dx}\right]_{(\alpha,\beta)} = 2a\alpha + b; 2a\alpha + b = b \Rightarrow \alpha = 0.$

24. Given equations to curves are $y = x^2 + ax + b$ and $y = x(c - x) \Rightarrow y = cx - x^2$

$y^1 = 2x + a, y^1 = c - 2x$

$y^1_{(1,0)} = 2 + a, y^1_{(1,0)} = c - 2$

Curves touch each other at (1,0). Then the slopes of tangents to curves at (1,0) are equal

i.e., $2 + a = c - 2$

(1,0) is point on $y = x^2 + ax + b$ and $y = x(c - x)$

i.e., $0 = 1 + a + b$

$1(c - 1) = 0 = c = 1$

sub (3) in (1),

$2 + a = 1 - 2 \Rightarrow a = -3$

from (2), $a + b + 1 = 0 \Rightarrow -3 + b + 1 = 0 \Rightarrow b = 2.$

25. $m = \left(\frac{dy}{dx}\right)_{(2,3)} = 4$. Equation of normal is $x + 4y - 14 = 0.$

26. $y = x^3 - 2x + 7;$

$y^1 = 3x^2 - 2 \Rightarrow [y^1]_{(1,6)} = 1$

equation of tangent is $y - 6 = 1(x - 1) \Rightarrow x - y + 5 = 0.$

27. $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2;\ (a, b)$

$$n\left(\frac{x}{a}\right)^{n-1} \times \frac{1}{a} + n\left(\frac{y}{b}\right)^{n-1} \times \frac{1}{b}y^1 = 0 \Rightarrow \left(\frac{y}{b}\right)^n \cdot \left(\frac{y}{b}\right)^{-1} \times \frac{1}{b}y^1 = -\left(\frac{x}{a}\right)^n \left(\frac{x}{a}\right)^{-1} \times \frac{1}{a}$$

$$y^1 = -\frac{b}{a}\frac{\left(\frac{x}{a}\right)^n \times \left(\frac{x}{a}\right)^{-1}}{\left(\frac{y}{b}\right)^n \cdot \left(\frac{y}{b}\right)^{-1}} \Rightarrow y^1\Big|_{(a,b)} = \frac{-\frac{b}{a}(1)^n(1)^{-1}}{(1)^n(1)^{-1}} = -\frac{b}{a}$$

Equation of tangent is

$y - b = -\frac{b}{a}(x - a) \Rightarrow ay - ab = -bx + ab$

$\therefore bx + ay = 2ab \Rightarrow \frac{bx+ay}{ab} = 2 \Rightarrow \frac{x}{a} + \frac{y}{b} = 2.$

28. $6y = 7 - x^3;\ (1,1) \Rightarrow 6y^1 = 0 - 3x^2 \Rightarrow y^1 = \frac{-3x^2}{6} = y^1|_{(1,1)} = -\frac{1}{2}$

Equation of tangent is $y - 1 = -\frac{1}{2}(x - 1)$

$2y - 2 = -x + 1 \Rightarrow x + 2y - 3 = 0.$

29. Given point (1,1)

a) $6y = 7 - x^3 \Rightarrow 6y^1 = 0 - 3x^2 \Rightarrow y^1 = \frac{-3x^2}{6} = -\frac{x^2}{2}$

$y^1|_{(1,1)} = -\frac{1}{2}$

Equation of tangent is $y - 1 = -1/2(x - 1).$

$2y - 2 = -x + 1 \Rightarrow x + 2y - 3 = 0.$

b) $y = 3x^2 + 4x - 6 \Rightarrow y^1 = 6x + 4 \Rightarrow y^1|_{(1,1)} = 10.$

Equation of tangent is $y - 1 = 10(x - 1) \Rightarrow y - 1 = 10x - 10 \Rightarrow 10x - y - 9 = 0.$

c) $x^2 + y^2 = 2 \Rightarrow 2x + 2yy^1 = 0 \Rightarrow yy^1 = -x \Rightarrow y^1 = -x/y \Rightarrow y^1|_{(1,1)} = -1$

Equation of tangent is $y - 1 = -1(x - 1) \Rightarrow x + y - 2 = 0.$

d) $x^2y = 1 \Rightarrow x^2y^1 + y(2x) = 0$

$x^2y^1 = -2yx \Rightarrow y^1 = \frac{-2y}{x} \Rightarrow y^1\Big|_{(1,1)} = -2$

Equation tangent is $y - 1 = -2(x - 1) \Rightarrow 2x + y - 3 = 0.$

30. Given $y = x^3;$

$y^1 = 3x^2 \Rightarrow y^1|_{(1,1)} = 3$

$LSN = |y_1 \cdot m| = |1.3| = 3.$

31. $\mathrm{x} = 4\mathrm{t}^2 + 3, \mathrm{y} = 8\mathrm{t}^3 - 1 \Rightarrow \frac{\mathrm{dy}}{\mathrm{dx}} = \frac{\left(\frac{\mathrm{dy}}{\mathrm{dt}}\right)}{\left(\frac{\mathrm{dx}}{\mathrm{dt}}\right)} = \frac{24\mathrm{t}^2}{8\mathrm{t}} = 3\mathrm{t}$

Equation of tangent at the point $(4t^2 + 3, 8t^3 - 1)$ is $y - 8t^3 + 1 = 3t(x - 4t^2 - 3)$

The tangent meets the curve again at $\mathrm{P}(4\mathrm{p}^2 + 3, 8\mathrm{p}^3 - 1)$

$\Rightarrow 8\mathrm{p}^3 - 1 - 8\mathrm{t}^3 + 1 = 3\mathrm{t}(4\mathrm{p}^2 + 3 - 4\mathrm{t}^2 - 3)$

$8p^3 - 8t^3 = 3t \cdot (4p^2 - 4t^2) \Rightarrow 8p^3 - 8t^3 = 12t(p^2 - t^2)$

$8(p^3 - t^3) = 12t(p^2 - t^2) \Rightarrow 2(p - t)(p^2 + pt + t^2) = 3t(p - t)(p + t)[p \neq t]$

$2(p^2 + pt + t^2) = 3t(p + t) \Rightarrow 2p^2 + 2pt + 2t^2 = 3pt + 3t^2$

$2\mathrm{p}^2 - \mathrm{pt} - \mathrm{t}^2 = 0 \Rightarrow (\mathrm{p} - \mathrm{t})(2\mathrm{p} + \mathrm{t}) = 0 \Rightarrow 2\mathrm{p} + \mathrm{t} = 0 \Rightarrow \mathrm{p} = \frac{-\mathrm{t}}{2}$

32. $y^2 = x; x^2 = 6y; (2,2)$

$2yy^1 = 1\ 2x = y^1$

$y^1 = \frac{1}{2} y\ y^1 = 2x$

$y^1|_{(1,1)} = \frac{1}{2}\ y^1\Big|_{(1,1)} = 2$

$\tan\theta = \left|\frac{\frac{1}{2} - 2}{1+1}\right| = \left|-\frac{3}{4}\right| \Rightarrow \tan\theta = \frac{3}{4} \Rightarrow \theta = \tan^{-1}\left(\frac{3}{4}\right)$

2. $x^2y = 1, y = e^{2(1-x)};$

$(1,1).$

$x^2y^1 + y(2x) = 0\ y^1 = e^{2(1-x)} \times -2$

$x^2y^1 = -2xy\ y^1 - 2e^{2(1-x)}$

$y^1 = \frac{-2y}{x}\ y^1\Big|_{(1,1)} = -2e^{2(1-1)}$

$y^1|_{(1,1)} = -2\ y^1|_{(1,1)} = -2$

Slopes are equal.

∴ Curves touch each other.

∴ Angle between curves is zero.

3. $xy = 4, x^2 - y^2 = 15, (-4,-1)$

$xy^1 + y(1) = 0\ 2x - 2yy^1 = 0$

$xy^1 = -y\ 2yy^1 = 2x$

$y^1 = -\frac{y}{x}\ y^1 = \frac{x}{y}$

$y^1|_{(-4,1)} = \frac{-(-1)}{-4} = \frac{-1}{4}\ y^1\Big|_{(-4,1)} = \frac{-4}{-1} = 4$

∴ Product of slopes $= -1$

∴ Curves touch each other orthogonally.

4. $x^2 = 4y, y^2 = 4x,$

$2x = 4y^1\ 2yy^1 = 4$

$y^1 = \frac{x}{2}\ y^1 = \frac{2}{y}$

$y^1|_{(4,4)} = \frac{4}{2} = 2\ y^1|_{(4,4)} = \frac{2}{4} = \frac{1}{2}$

$\tan\theta = \left|\frac{2-\frac{1}{2}}{1+1}\right| = \left|\frac{3}{4}\right| \Rightarrow \theta = \tan^{-1}\left(\frac{3}{4}\right).$

33. Given $f(x) = \sqrt{x-2}$; [2,6]

$f^1(x) = \dfrac{1}{2\sqrt{x-2}}$

By LMVT,

$f^1(c) = \dfrac{f(6)-f(2)}{4} \Rightarrow \dfrac{1}{2\sqrt{c-2}} = \dfrac{2-0}{4}$

$\frac{1}{2\sqrt{c-2}} = \frac{1}{2} \Rightarrow \sqrt{c-2} = 1 \Rightarrow c-2 = 1 \Rightarrow c = 3.$

34. $x^3 - y^2 = 0$

$2yy^1 = 3x^2 \Rightarrow y^1 = \dfrac{3x^2}{2y}$

$y^1|_{(4m^2,8m^3)} = 3m$

Equation of tangent at p is $y - 8m^3 = 3m(x - 4m^2)$

$y = 3mx - 4m^3$

it cuts curve again at Q

solving (1) & (2),

We get $x = 4m^2, m^2$

put $x = m^2$ in (2), then

$y = 3m(m^2) - 4m^3 = -m^3$

$\therefore Q \equiv (m^2, -m^3)$

Slope of tangent at Q is $\frac{dy}{dx}$

$\left.\dfrac{dy}{dx}\right|_{(m^2,-m^3)} = \dfrac{3(m^4)}{2 \times (-m^3)} = \dfrac{-3}{2}m$

slope of normal at Q = $\frac{2}{3m}$.

Since tangent at P is normal at Q.

$\frac{2}{3m} = 3m \Rightarrow 9m^2 = 2 \Rightarrow m^2 = \frac{2}{9} \Rightarrow m = \pm\frac{\sqrt{2}}{3}.$

35. a) Given equation of curve is $y = x^3 + 1 \Rightarrow \frac{dy}{dx} = 3x^2$

Length of tangent = $\left|\frac{y_1\sqrt{1+m^2}}{m}\right| = \left|\frac{2\cdot\sqrt{1+3^2}}{3}\right| = \frac{2\sqrt{10}}{3}$.

b) Given equation of curve is $y = x^2 + 1$

$\frac{dy}{dx} = 2x \Rightarrow m = \left[\frac{dy}{dx}\right]_{(1,2)} = 2(1) = 2.$

Length of normal = $\left|y_1 \cdot \sqrt{1+m^2}\right| = \left|2 \cdot \sqrt{1+2^2}\right| = 2\sqrt{5}.$

c) Given equation of curve is $y = b \cdot e^{x/a} \Rightarrow \frac{dy}{dx} = \frac{b}{a}e^{x/a}$

$m = \left(\frac{dy}{dx}\right)_{(x_1,y_1)} = \frac{b}{a}e^{x_1/a}$

Length of sub tangent $= \left|\frac{y_1}{m}\right| = \left|\frac{y_1}{\frac{b}{a}e^{x_1}a}\right| = \left|\frac{be^{x_1/a}}{\frac{b}{a}e^{x/a}}\right| = |a|$.

d) Given equation of curve is $y = x^3 \Rightarrow \frac{dy}{dx} = 3x^2$

$m = \left(\frac{dy}{dx}\right)_{(1,1)} = 3(1)^2 = 3$ Length of sub normal $= |y_1 \times m| = |1 \times 3| = 3$.

WORKSHEET – 3
Level – I

1) A 2) C 3) A 4) C 5) B 6) A
7) D 8) B 9) C

1. $f(x) = \sin x + \cos x$

$= \sqrt{2}\left(\sin x \cdot \frac{1}{\sqrt{2}} + \cos x \cdot \frac{1}{\sqrt{2}}\right) = \sqrt{2}\left(\sin x \cdot \cos\frac{\pi}{4} + \cos x \cdot \sin x\frac{\pi}{4}\right) = \sqrt{2}\sin\left(x + \frac{\pi}{4}\right)$

maximum value of $f(x) = \sqrt{2}.1 = \sqrt{2}$

$\left(\because \text{maximum value of } \sin\left(x + \frac{\pi}{4}\right) = 1\right)$

2. $f(x) = \sin^2 x + \text{cosec }^2 x$

A.M $\geq G.M$

$\frac{\sin^2 x + \text{cosec}^2 x}{2} = \sqrt{\sin^2 x \cdot \text{cosec}\, e^2 x}$

$\sin^2 x + \text{cosec }^2 x \geq 2.1$

$\therefore$ Least value of $\sin^2 x + \text{cosec}^2 x$ is 2 .

3. $f(x) = \frac{\log x}{x} \Rightarrow f^1(x) = \frac{x\cdot\frac{1}{x} - \log x \cdot 1}{x^2}$

For increasing function, $f^1(x) > 0 \Rightarrow \frac{1-\log x}{x^2} > 0$

$\Rightarrow 1 > \log x \Rightarrow \log x < 1 \Rightarrow x < e \Rightarrow x \in (0, e)$.

4. $f(x) = xe^x \Rightarrow f^1(x) = x \cdot e^x + e^x.1$

for increasing function, $f^1(x) > 0 \Rightarrow x \cdot e^x + e^x > 0$

$\Rightarrow e^x(x+1) > 0 \Rightarrow x + 1 > 0 \Rightarrow x > -1 \Rightarrow x \in (-1, \infty)$.

5. Conceptual.

6. Conceptual.

7. $f(x) = 3 + 12x - 9x^2 + 2x^3 \Rightarrow f^1(x) = 12 - 18x + 6x^2$

for increasing function, $f^1(x) > 0 \Rightarrow 6x^2 - 18x + 12 > 0$

$\Rightarrow x^2 - 3x + 2 > 0 \Rightarrow (x-1)(x-2) > 0 \Rightarrow x \in (-\infty, 1) \cup (2, \infty)$

8. $f(x) = \frac{\log x}{x^2}$

$f^1(x) = \frac{x^2 - \frac{1}{x} - \log x \cdot 2x}{x^4}$

For maxima or minima, $f^1(x) = 0$

$\Rightarrow x - 2x \cdot \log x = 0 \Rightarrow x(1 - 2\log x) = 0$

$\Rightarrow 2\log x = 1 \Rightarrow \log x = \frac{1}{2} \Rightarrow x = e^{\frac{1}{2}} = \sqrt{e}$

Maximum value occurs at $x = \sqrt{e}$.

9. $f(x) = x^3 - 3x \Rightarrow f^1(x) = 3x^2 - 3$

For maxima or minima $f^1(x) = 0 \Rightarrow x^2 - 1 = 0$

$\Rightarrow x = \pm 1; f(0) = 0,$

$f(1) = 1 - 3 = -2,\ f(2) = 8 - 6 = 2$

$\therefore$ Maximum value = 2.

Level – II

1) B	2) A	3) A	4) C	5) D	6) D
7) 6	8) B	9) A	10) C	11) A	12) D
13) 128	14) C	15) A	16) B	17) A	18) A
19) A	20) C	21) B	22) A	23) B	24) B
25) C	26) a-r; b-p; c-s; d-q		27) A	28) A	29) A,B,C,D
30) C	31) B	32) D	33) a-q; b-s; c-r; d-p		34) B,D

35) a-r,s; b-r,s; c-p,q,r; d-s

1. $f(x) = x(\log x)^2$

$f^1(x) = x \cdot 2\log x \cdot \frac{1}{x} + (\log x)^2 \cdot 1 = 2\log x + (\log x)^2$

$f^1(x) = 0$

i.e., $2\log x + (\log x)^2 = 0$

$\Rightarrow \log x(2 + \log x) = 0$

$\Rightarrow \log x = -2$

$\therefore x = e^{-2}.$

Stationary value of $f(x) = f(e^{-2}) = e^{-2}(\log_e e^{-2})^2 = e^{-2}(-2)^2 = 4e^{-2}.$

2. $f(x) = \sin^3 x + \cos^3 x$

$\Rightarrow f^1(x) = 3\sin^2 x\cos x - 3\cos^2 x\sin x = 3\sin x\cos x(\sin x - \cos x)$

$f^1(x) = 0$ i.e., $3\sin x\cos x(\sin x - \cos x) = 0$

$\Rightarrow \sin x - \cos x = 0, \sin x = 0, \cos x = 0$

$\Rightarrow \sin x = \cos x, x = 0°, x = \frac{\pi}{2}$

$\therefore x = \frac{\pi}{4}$

$f(0) = \sin^3 0 + \cos^3 0 = 0 + 1 = 1$

$\Rightarrow f\left(\frac{\pi}{4}\right) = \sin^3 \frac{\pi}{4} + \cos^3 \frac{\pi}{4} = \frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$

$\Rightarrow f\left(\frac{\pi}{2}\right) = \sin^3 \frac{\pi}{2} + \cos^3 \frac{\pi}{3} = 1 + 0 = 1$

$\therefore$ greatest value of $f(x) = 1$.

3. $f(x) = x^x$

$f^1(x) = x^x(1 + \log x)$ for decreasing, $f^1(x) < 0$

$\Rightarrow x^x(1 + \log x) < 0 \Rightarrow 1 + \log x < 0 \Rightarrow \log x < -1$

$\Rightarrow x < \frac{1}{e} \Rightarrow x \in \left(0, \frac{1}{e}\right)$

4. Let x and y be the numbers.

$x + y = 6 \Rightarrow y = 6 - x \Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{1}{x} + \frac{1}{6-x} = \frac{6-x-x}{x(6-x)} = \frac{6}{x(6-x)}$

Let $f(x) = \frac{6}{x(6-x)} = \frac{6}{6x-x^2} \Rightarrow f^1(x) = 6\left(\frac{-1}{(6x-x^2)^2}\right)(6-2x)$

$f^1(x) = 0 = \frac{-6(6-2x)}{(6x-x^2)^2} = 0 \Rightarrow 6 - 2x = 0 \Rightarrow x = 3$

Minimum value of $f(x) = f(3) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$.

5. Let area of triangle be $\Delta = \frac{1}{2}bc\sin A$.

where b and c are constants $\Rightarrow \Delta^1 = \frac{1}{2}bc\cos A$

$\Delta^1 = 0 \Rightarrow \frac{1}{2}bc\cos A = 0 \Rightarrow \cos A = 0 \Rightarrow A = \frac{\pi}{2}$.

6. $y = f(x) = \frac{\log x}{x}$

$\Rightarrow f^1(x) = \frac{x \cdot \frac{1}{x} - \log x \cdot 1}{x^2} = 0 \Rightarrow 1 - \log x = 0 \Rightarrow \log x = 1 \ \therefore x = e$

Stationary point $= (x, f(x)) = \left(e, \frac{\log e}{e}\right) = \left(e, \frac{1}{e}\right)$.

7. $f(x) = 9x + \frac{1}{x} \Rightarrow f^1(x) = 9 - \frac{1}{x^2} = 0 \ \because f^1(x) = 0$

$\Rightarrow 9 - \frac{1}{x^2} = 0 \Rightarrow \frac{1}{x^2} = 9$

$\Rightarrow x^2 = \frac{1}{9} \ \therefore x = \pm\frac{1}{3}$

$\Rightarrow f\left(\frac{1}{3}\right) = 9 \times \frac{1}{3} + 3 = 6 \Rightarrow f\left(-\frac{1}{3}\right) = 9\left(-\frac{1}{3}\right) - 3 = -6.$

8. $f(x) = x^3 + ax^2 + bx + c \Rightarrow f^1(x) = 3x^2 + 2ax + b$

for increasing, $f^1(x) > 0$

$\Rightarrow 3x^2 + 2ax + b > 0 \Rightarrow \Delta < 0 \Rightarrow 4a^2 - 12b < 0 \Rightarrow a^2 < 3b.$

9. $y = f(x) = x^3 - 3x + 2$

$\Rightarrow f^1(x) = 3x^2 - 3 = 0$

$\because f^1(x) = 0$

$\Rightarrow x^2 = 1 \Rightarrow x = \pm 1 \Rightarrow f(0) = 2, f(-1) = -1 + 3 + 2 = 4 \Rightarrow f(1) = 1 - 3 + 2 = 0$

$\therefore$ Absolute maximum $= 4.$

10. Equation of ellipse is $\frac{x^2}{64} + \frac{y^2}{49} = 1$

comparing with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow a = 8, b = 7$

The least value of tangents to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is intercepted between coordinate axes is $a + b = 8 + 7 = 15.$

11. $f(x) = (\sin^{-1} x)^2 + (\cos^{-1} x)^2 \Rightarrow f^1(x) = 2\sin^{-1} x \cdot \frac{1}{\sqrt{1-x^2}} + 2\cos^{-1} x \left(-\frac{1}{\sqrt{1-x^2}}\right) = 0$

$\Rightarrow \frac{2}{\sqrt{1-x^2}}(\sin^{-1} x - \text{cis}^{-1} x) = 0 \Rightarrow \sin^{-1} x = \cos^{-1} x$

$\therefore x = \frac{1}{\sqrt{2}}$ $\therefore f(x)$ is stationary at $x = \frac{1}{\sqrt{2}}.$

12. $f(x) = \sin x - \cos x - ax + b \Rightarrow f^1(x) = \cos x + \sin x - a$

for decreasing, $f^1(x) < 0 \Rightarrow \cos x + \sin x - a < 0$

$\Rightarrow a > \sqrt{2}.$ ($\because$ max. value of $\sin x + \cos x = \sqrt{2}$)

13. $f(x) = x^2 + (16 - x)^2 = 2x^2 + 256 - 36x = 4x - 32 = 0$

$\therefore x = 8 \Rightarrow y = 8$

$\therefore$ Minimum value of $x^2 + y^2 = 128.$

14. $f(x) = 2x - \tan^{-1} x - \log\left(x + \sqrt{1 + x^2}\right)$

$f^1(x) = 2 - \frac{1}{1+x^2} - \frac{1}{x+\sqrt{1+x^2}}\left(1 + \frac{1}{2\sqrt{1+x^2}} 2x\right)$

$= 2 - \frac{1}{1+x^2} - \frac{1}{x+\sqrt{1+x^2}} \frac{\sqrt{1+x^2}+x}{\sqrt{1+x^2}} = 2 - \frac{1}{1+x^2} - \frac{1}{\sqrt{1+x^2}}$

$= \frac{2(1+x^2)-1-\sqrt{1+x^2}}{1+x^2} = \frac{1+2x^2-\sqrt{1+x^2}}{1+x^2}$

for increasing $f^1(x) > 0 \Rightarrow (1 + 2x^2)^2 > 1 + x^2$

$\Rightarrow 1 + 4x^4 + 4x^2 > 1 + x^2 \Rightarrow 4x^4 + 3x^2 > 0 \Rightarrow x^2(4x^2 + 3) > 0 \Rightarrow x > 0.$

15. $f(x) = 8x^2 - x^4 - 4$

$\Rightarrow f^1(x) = 16x - 4x^3 \Rightarrow f^1(x) = 0 \Rightarrow 16x - 4x^3 = 0 \Rightarrow 4x(4 - x^2) = 0$

$\therefore x = 0$ or $x^2 = 4 \Rightarrow x = \pm 2$

$f(0) = -4, f(-2) = 12, f(2) = 12$

$\therefore$ stationary points are $(0,-4),(-2,12),(2,12)$.

16. $f(x)=|x-1|$ is not differentiable at $x=1$ critical point is at $x=1$.

17. Let x,y be two numbers. ; $x+y=20 \Rightarrow y=20-x$

Let $f(x)=x^2(20-x)^3$

$f^1(x)=x^2\times 3(20-x)^2(-1)+(20-x)^3 2x=(20-x)^2x[-3x+2(20-x)]$

$=(20-x)^2x(40-5x) \Rightarrow f^1(x)=0 \Rightarrow x=8 \Rightarrow y=12.$

18. Equation of curve is

$x^{2/3}+y^{2/3}=a^{2/3}$.

Let $p(a\cos^3\theta, a\sin^3\theta)bc=$ point on the curve

Distance from origin $=a^2\cos^6\theta+a^2\sin^6\theta$

Let $f(\theta)=\sqrt{a^2\cos^6\theta+a^2\sin^6\theta}$

$=a\sqrt{\cos^6\theta+\sin^6\theta}=a\sqrt{1-3\sin^2\theta\cos^2\theta}=a\sqrt{1-\frac{3}{4}\sin^2 2\theta}$

Max. value of $f(\theta)=a$. ($\because$ min.value of $\sin^2 2\theta=0$)

19. To have minimum area, the point (3,4) is the mid point of potion of line intercepted between coordinate axes $A=(6,0),B=(0,8)$

Equation of line is $\frac{x}{6}+\frac{y}{8}=1=8x+6y=48 \Rightarrow 4x+3y=24.$

20. Equation of curve is

$2x^2+y^2=2x \Rightarrow y^2=2x-2x^2$.

Let $p(x,y)$ be a point on the curve

distance from $A(a,0)$ is

$PA^2=(x-a)^2+y^2=(x-a)^2+2x-2x^2$

Let $f(x)=(x-a)^2+2x-2x^2$

$f^1(x)=2(x-a)+2-4x=2x-2a+2-4x$

$2x=2(1-a)\ \because f^1(x)=0$

$x=1-a$

tangent distance, $\sqrt{f(1-a)}=\sqrt{(1-a-a)^2+2(1-a)-2(1-a)^2}$

$=\sqrt{1+a^2-4a+2-2a-2+4a-2a^2}=\sqrt{2a^2-2a+1}.$

21. $f^1(x)=x^2-1=0$

$\therefore x=1,-1$

for three distinct roots $f(x_1)\cdot f(x_2)<0$ where x_1,x_2 are roots of $f^1(x)=0$

$\left(\frac{1}{3}-1-b\right)\left(\frac{-1}{3}+1-b\right)<0$ or $\left(b+\frac{2}{3}\right)\left(b-\frac{2}{3}\right)<0.\ \therefore b\in\left(-\frac{2}{3},\frac{2}{3}\right)$

22. $R^2 = \frac{h^2}{4} + r^2 \Rightarrow r^2 = R^2 - \frac{h^2}{4}$ volume of cylinder is $V = \pi r^2 h$

$= \pi\left(R^2 - \frac{h^2}{4}\right)h = \pi R^2 h - \frac{\pi h^3}{4} \Rightarrow V^1 = \pi R^2 - \frac{3\pi h^2}{4} = V^1 = 0 \Rightarrow \frac{3\pi h^2}{4} = \pi R^2$

$\Rightarrow h^2 = \frac{4R^2}{3} \ \ \therefore h = \frac{2R}{\sqrt{3}}$

$\therefore$ volume is maximum when $h = \frac{2R}{\sqrt{3}}$.

23. Let x be the side of square base and volume, $V = x^2 h \Rightarrow x^2 h = k \Rightarrow h = \frac{k}{x^2}$ Area of that lining is gives x^2

= Base area + Area of 4 wells $= x^2 + 2h(x + x) = x^2 + 4hx = x^2 + 4x\left(\frac{k}{x^2}\right)$

Let $f(x) = x^2 + 4\frac{k}{x} \Rightarrow f^1(x) = 2x - \frac{4k}{x^2} \Rightarrow f^1(x) = 0$

$\therefore 2x = \frac{4k}{x^2} \Rightarrow 2x = 4 \times \frac{x^2 h}{x^2} \Rightarrow \frac{h}{x} = \frac{2}{4} = 1:2.$

24. $f(x) = \frac{|x-1|}{x^2}$

f is not defined for $x = 0$ and $f^1(x) = 0 \Rightarrow x = 2$

$\therefore f$ is not differentiable for $x = 1$

$\therefore$ Critical points are 1,2 .

25. $f(x) = xe^{-x} \Rightarrow f^1(x) = -xe^{-x} + e^{-x} = e^{-x}(1 - x)$

$\Rightarrow f^1(x) = 0 \Rightarrow 1 - x = 0 \Rightarrow x = 1 \Rightarrow f(0) = 0, f(\infty) = 0, f(1) = e^{-1} = \frac{1}{e}.$

26. a) $f(x) = x + \frac{1}{x} \Rightarrow f^1(x) = 1 - \frac{1}{x^2}$

for decreasing, $f^1(x) < 0$.

$\Rightarrow 1 - \frac{1}{x^2} < 0 \Rightarrow x^2 - 1 < 0 \Rightarrow (x + 1)(x - 1) < 0 \therefore x \in (-1,0) \cup (0,1).$

b) $f(x) = 1 - x^3 \Rightarrow f^1(x) = -3x^2 \Rightarrow -3x^2 < 0 \Rightarrow x^2 > 0 \ \ \therefore x \in R$

c) $f(x) = xe^{-x} \Rightarrow f^1(x) = -xe^{-x} + e^{-x} \Rightarrow e^{-x}(1 - x)$

for decreasing $f^1(x) < 0 \Rightarrow 1 - x < 0 \Rightarrow x > 1 \therefore x \in (1, \infty).$

d) $f(x) = x\log x$

$\Rightarrow f^1(x) = x \cdot \frac{1}{x} + \log x \Rightarrow f^1(x) < 0 \Rightarrow 1 + \log x < 0 \Rightarrow \log x < -1. \Rightarrow x < e^{-1} \therefore x \in \left(0, \frac{1}{e}\right).$

27. $f(x) = 2x^3 - 3x^2 - 12x + 8$

$\Rightarrow f^1(x) = 6x^2 - 6x - 12 \ \Rightarrow f^1(x) = 0 \Rightarrow 6x^2 - 6x - 12 = 0$

$\Rightarrow x^2 - x - 2 = 0 \Rightarrow (x - 2)(x + 1) = 0 \therefore x = 2, -1$

$f^{11}(x) = 12x - 6 \Rightarrow f^{11}(2) = 24 - 6 = 18 > 0$

$\therefore f(x)$ has minimum value at $x = 2$.

28. Let $f(x) = x + \frac{1}{x} \Rightarrow f^1(x) = 1 - \frac{1}{x^2} \Rightarrow f^1(x) = 0 \Rightarrow x^2 - 1 = 0\ x = \pm 1\ f^{11}(x) = \frac{2}{x^3} \Rightarrow$ $f^{11}(1) = 2 > 0 \Rightarrow f(x)$ has minimum value at $x = 1$

29. $f(x) = x^3 - 6x^2 + 9x + 15 \Rightarrow f^1(x) = 3x^2 - 12x + 9 \Rightarrow f^1(x) = 0 \Rightarrow x^2 - 4x + 3 = 0$
$\Rightarrow (x-3)(x-1) = 0\ \therefore x = 1,3$
$f^{11}(x) = 6x - 12 \Rightarrow f^{11}(1) = 6 - 12 < 0 \Rightarrow f^{11}(3) = 18 - 12 > 0 \Rightarrow f^{11}(2) = 12 - 12 = 0.$
$f(x)$ is monotonic.
$f(x)$ increases is $(-\infty, 1) \cup (3, \infty)$.
$f(x)$ decrease is $(1,3)$.
It has point of inflection at $x = 2$.

30. $f(x) = \sin x(1 + \cos x) \Rightarrow f^1(x) = \sin x(-\sin x) + (1 + \cos x)\cos x$
$= -\sin^2 x + \cos x + \cos^2 x\ = \cos 2x + \cos x.$
$f^1(x) = 0 \Rightarrow 2\cos^2 x - 1 + \cos x = 0$
$\Rightarrow 2\cos^2 x + 2\cos x - \cos x - 1 = 0 \Rightarrow 2\cos x(\cos x + 1) - 1(\cos x + 1) = 0$
$\Rightarrow (\cos x + 1)(2\cos x - 1) = 0 \Rightarrow \cos x = -1$ or $\cos x = 1/2 \Rightarrow x = \pi/3.$

31. $y = x^{-x} \Rightarrow \log y = -x\log x \Rightarrow \frac{1}{y}\frac{dy}{dx} = -\left(x \cdot \frac{1}{x} + \log x\right) \Rightarrow \frac{dy}{dx} = -e^{-x}(1 + \log x)$
$\Rightarrow \frac{dy}{dx} = 0 \Rightarrow 1 + \log x = 0 \Rightarrow \log x = -1 \Rightarrow x = e^{-1} = \frac{1}{e} \Rightarrow y = \left(\frac{1}{e}\right)^{-\frac{1}{e}} = e^{1/e}.$

32. Perimeter of sector $= 2r + r\theta = k \Rightarrow r(2 + \theta) = k \Rightarrow r = \frac{k}{2+\theta}$
Area, $A = \frac{1}{2}r^2\theta \Rightarrow \frac{1}{2}\frac{k^2}{(2+\theta)^2}\theta \Rightarrow \frac{k^2}{2}\frac{\theta}{(2+\theta)^2}$
$$\Rightarrow A^1 = \frac{k^2}{2}\left[\frac{(2+\theta)^2 \cdot 1 - 2\theta(2+\theta)}{(2+\theta)4}\right]$$
$$\Rightarrow \frac{k^2}{2}\left[\frac{4 + \theta^2 + 4\theta - 4\theta - 2\theta}{(2+\theta)4}\right] = \frac{4 - \theta^2}{(2+\theta)^4}$$
$\Rightarrow A^1 = 0 \Rightarrow 4 - \theta^2 = 0 \Rightarrow \theta^2 = 4 \Rightarrow \theta = 2.$

34. Equation of ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$
Let $p(a\cos\theta, b\sin\theta)$ be one vertex of rectangle,
Length $= 2a\cos\theta$, Breadth $= 2b\sin\theta$
Area $= 2ab.2\sin\theta\cos\theta = 2ab\sin 2\theta.$
Max. area = 2ab sq.units.

35. a) $g(x) = 2\sin x + \cos 2x \Rightarrow g^1(x) = 2\cos x - 2\sin 2x$
$2\cos x - 4\sin x\cos x \Rightarrow 2\cos x(1 - 2\sin x) > 0$
$\Rightarrow \cos x > 0$ and $1 - 2\sin x > 0$

$\Rightarrow -\frac{\pi}{2} < x < \frac{\pi}{2}$ (or) $1 > 2\sin x \Rightarrow 2\sin x < 1 \Rightarrow \sin x < \frac{1}{2} \Rightarrow x < \frac{\pi}{6}$

(or) $\cos x < 0$ and $1 - 2\sin x < 0 \sin x > 1/2 \Rightarrow x > \pi/6$.

b) $f(x) = \cos^2 x + \sin x \Rightarrow f^1(x) = -2\cos x \sin x + \cos x = \cos x(1 - 2\sin x)$.

$f^1(x) = 0 \Rightarrow \cos x = 0$ or $1 - 2\sin x = 0 \Rightarrow \sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}$.

c) $f(x) = \tan^{-1}(\sin x + \cos x)$

$$\Rightarrow f^1(x) = \frac{1}{1 + (\sin x + \cos x)^2} \times (\cos x - \sin x)$$

for increasing, $f(x) > 0 \Rightarrow \cos x - \sin x > 0 \Rightarrow \cos x > \sin x$.

d) $f(x) = x^{99} + \sin x - 1 \Rightarrow f^1(x) = 99x^{98} + \cos x$

$f^1(x) > 0 \Rightarrow \cos x + 99.x^{98} > 0 \ \therefore x \in \left(0, \frac{\pi}{2}\right)$